

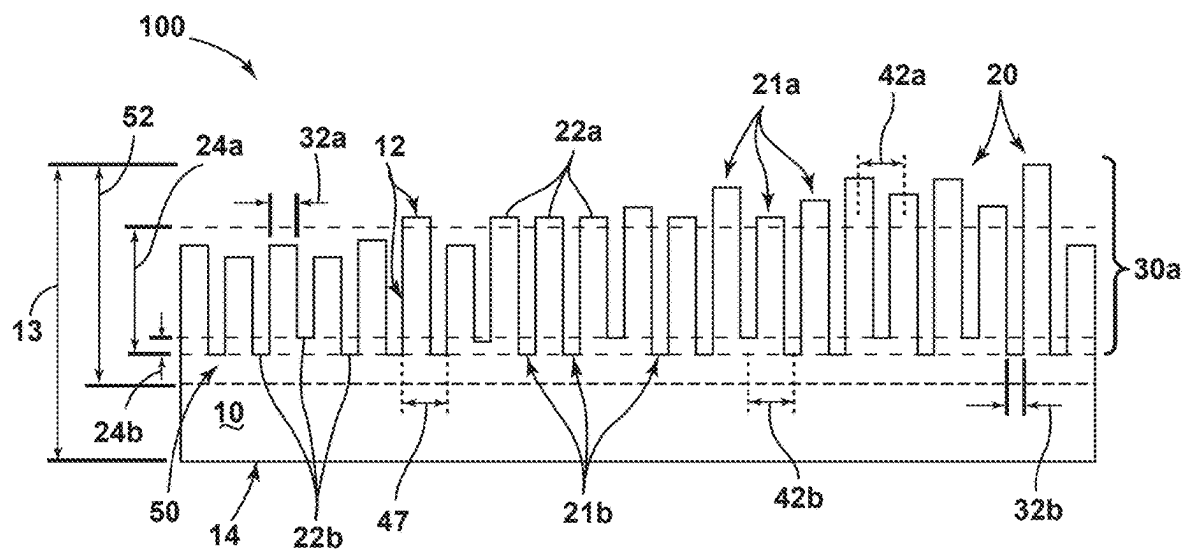


FIG. 1A

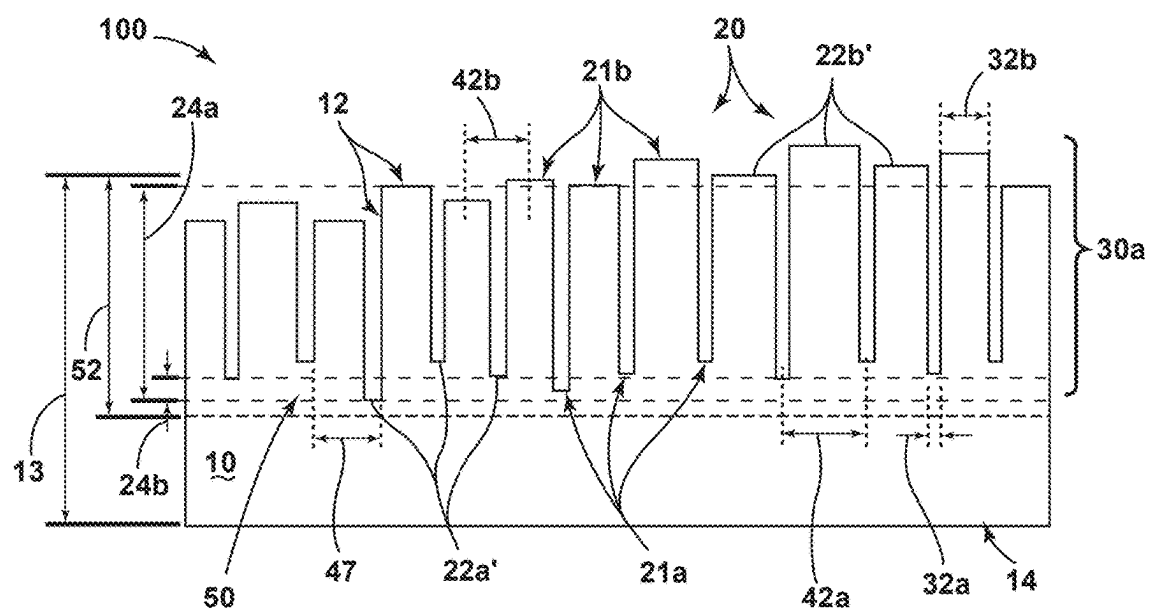


FIG. 1B

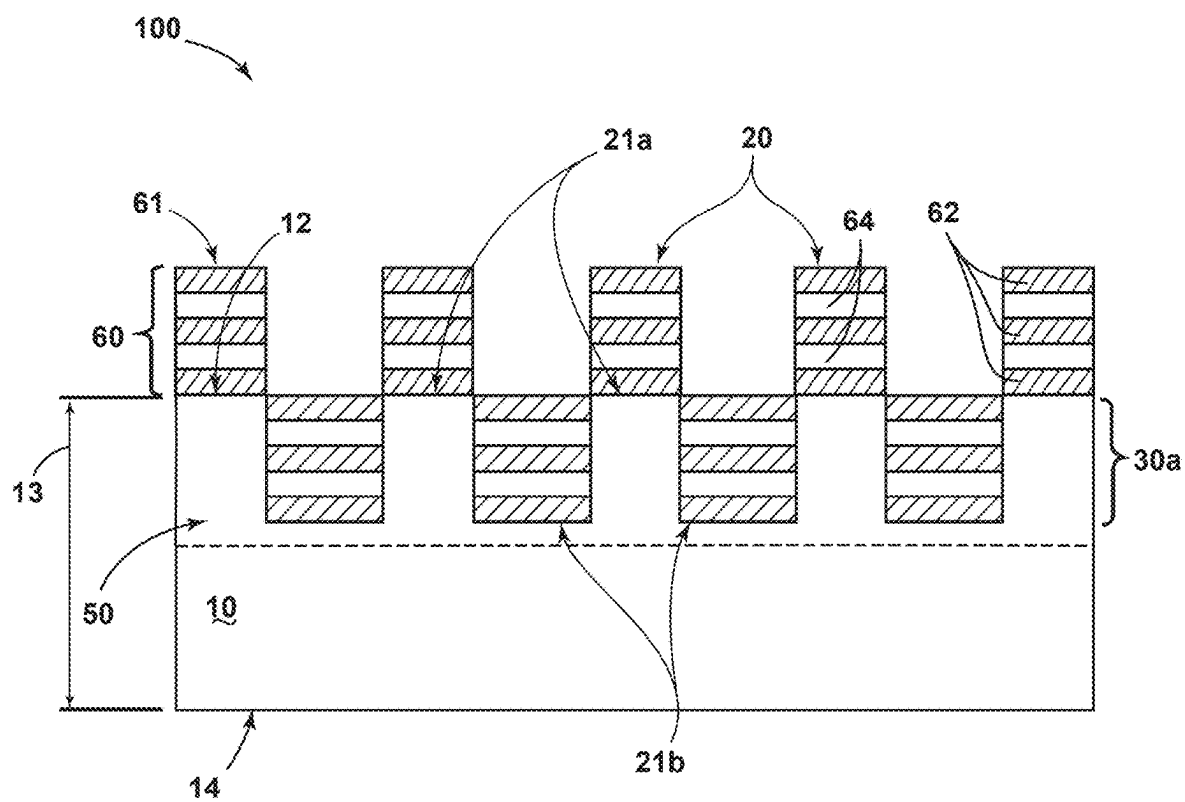


FIG. 1C

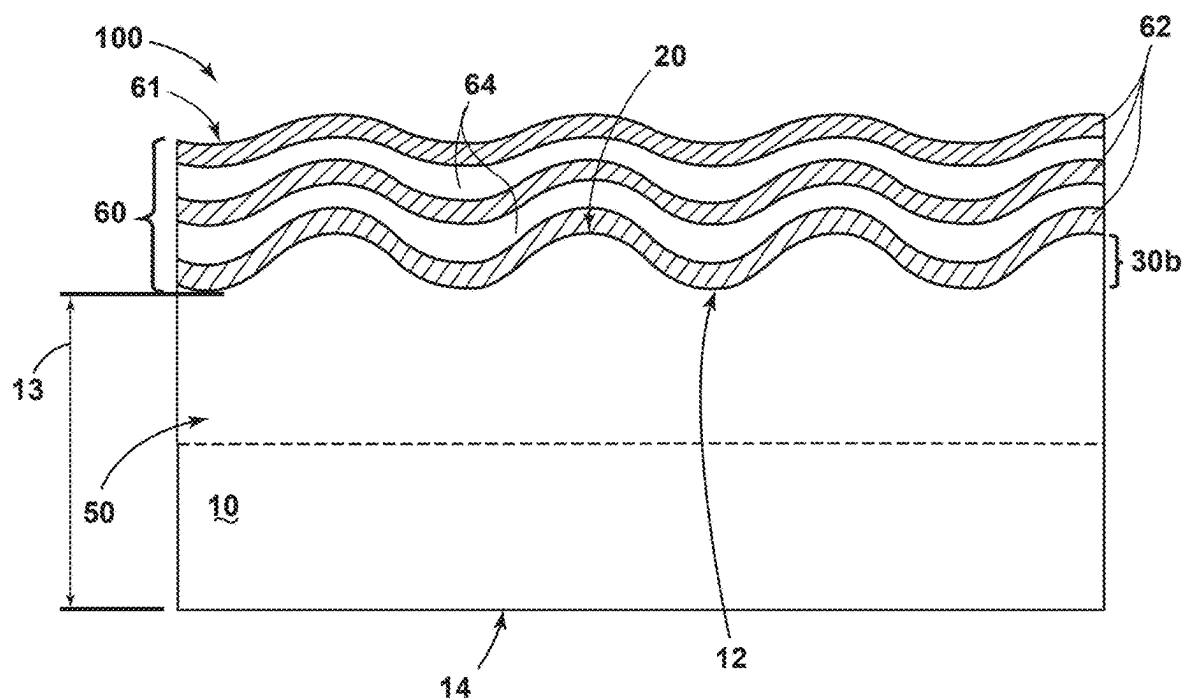


FIG. 1D

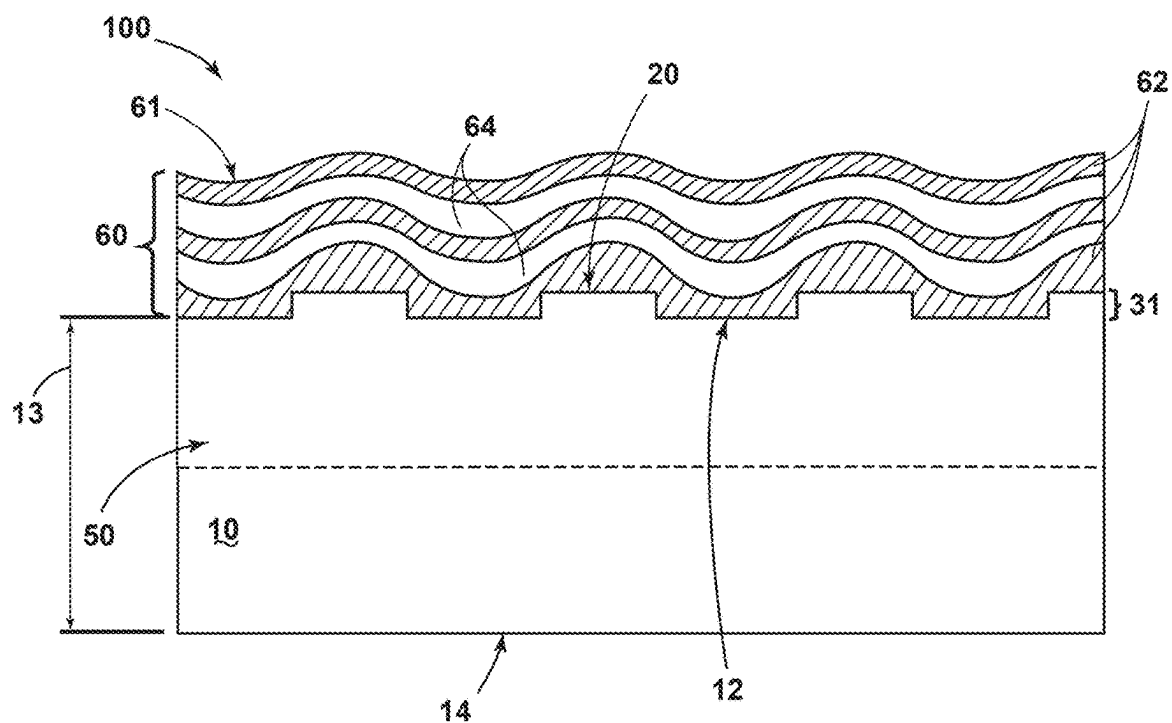
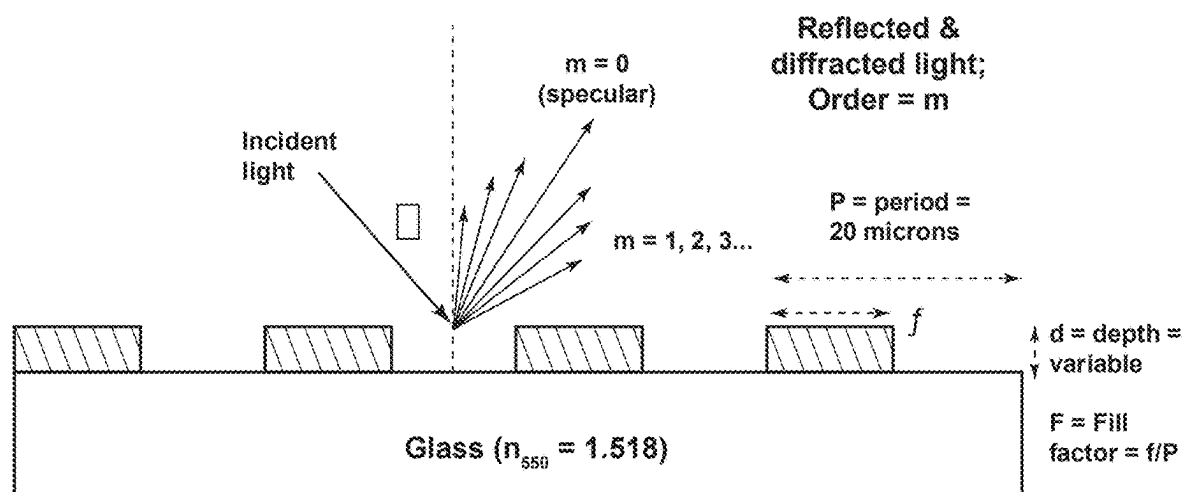


FIG. 1E

**FIG. 2**

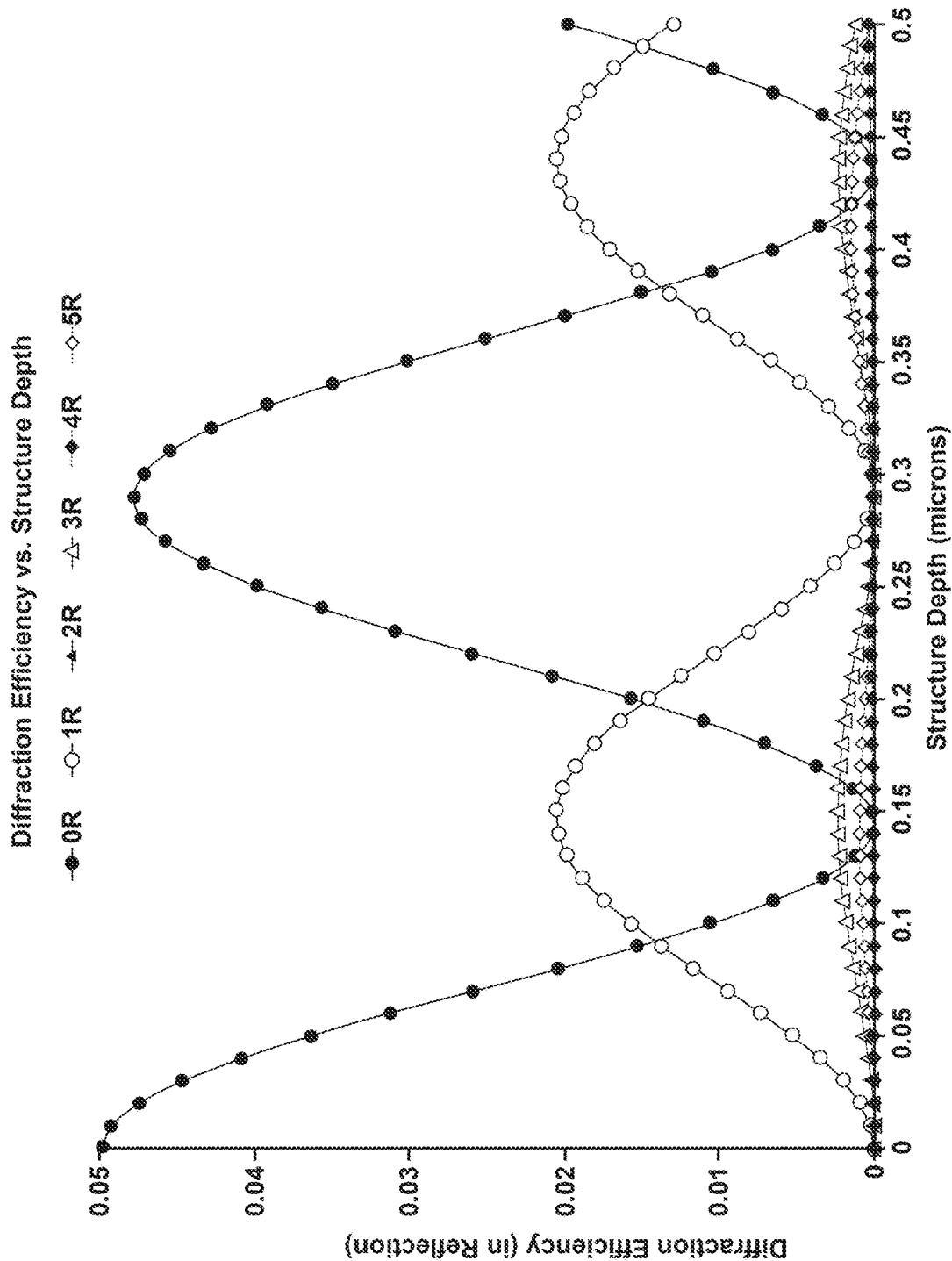


FIG. 3A

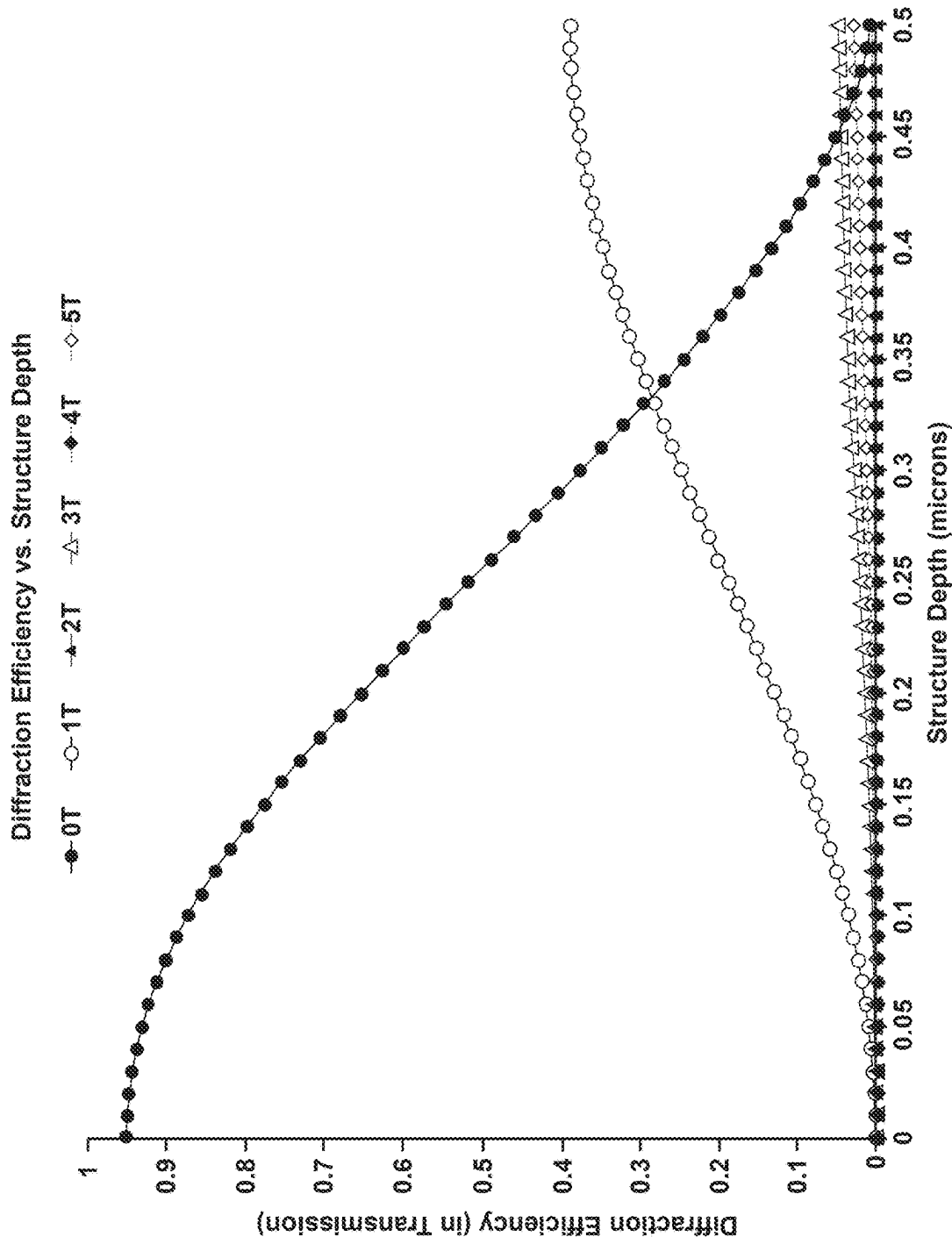


FIG. 3B

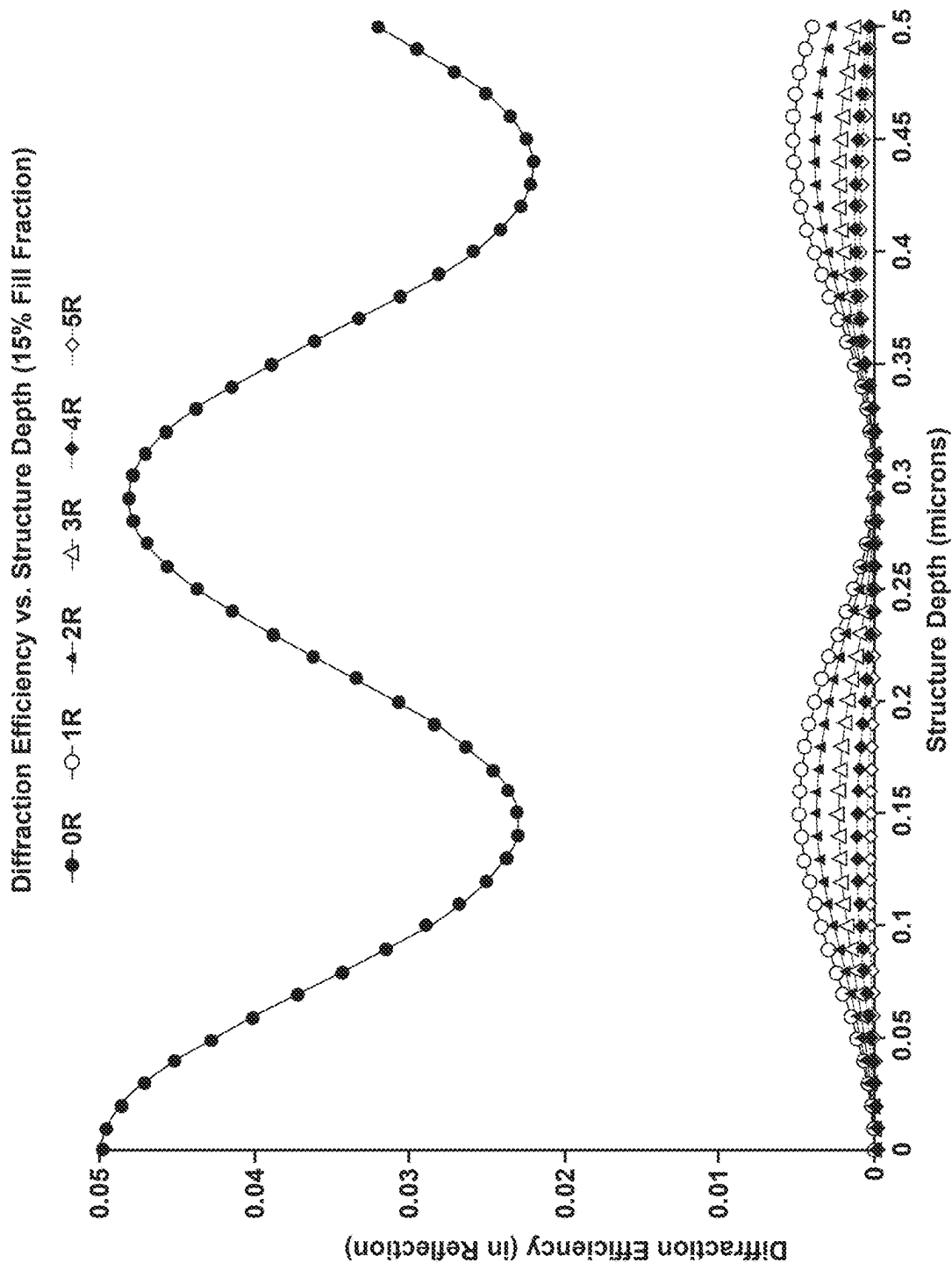


FIG. 4A

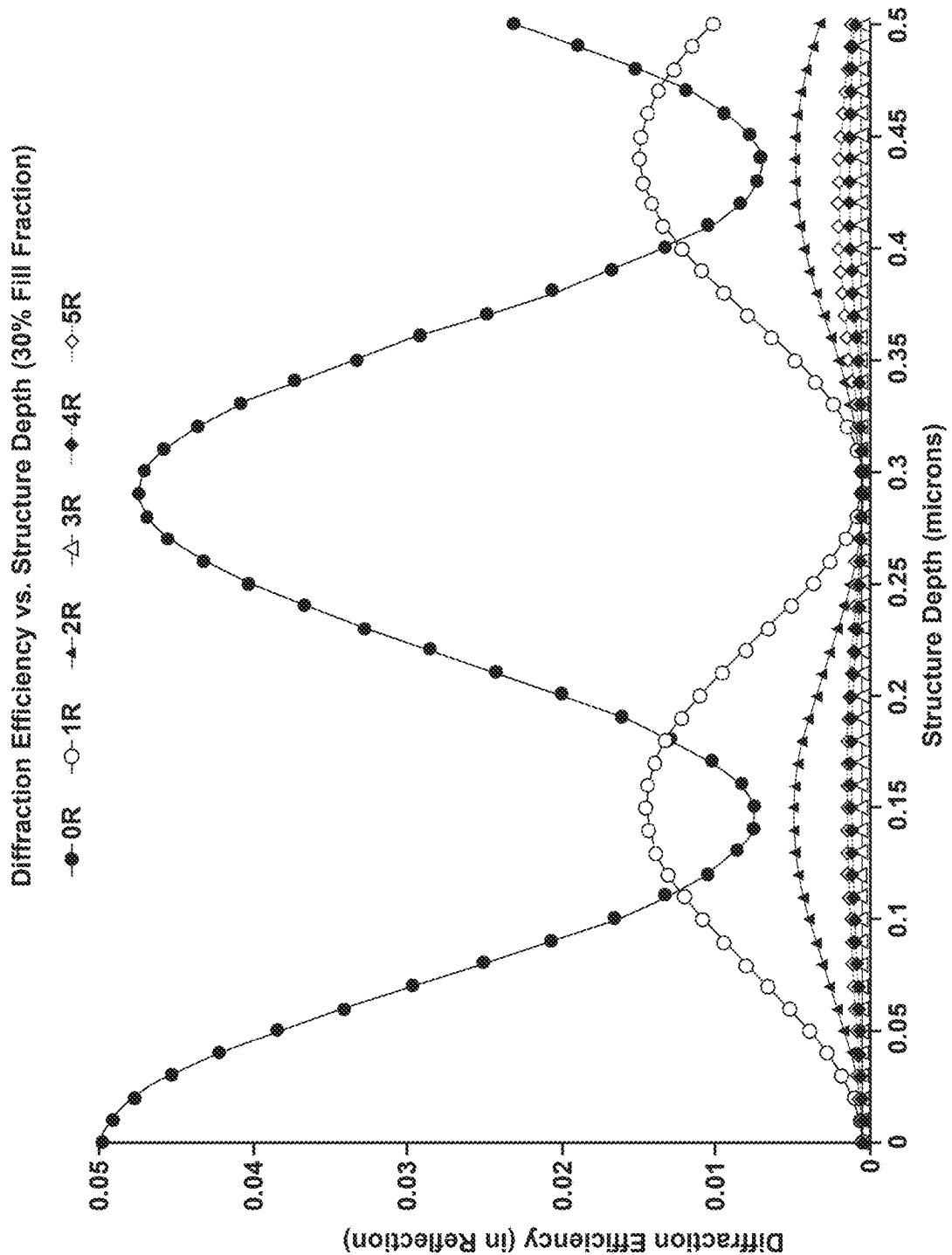


FIG. 4B

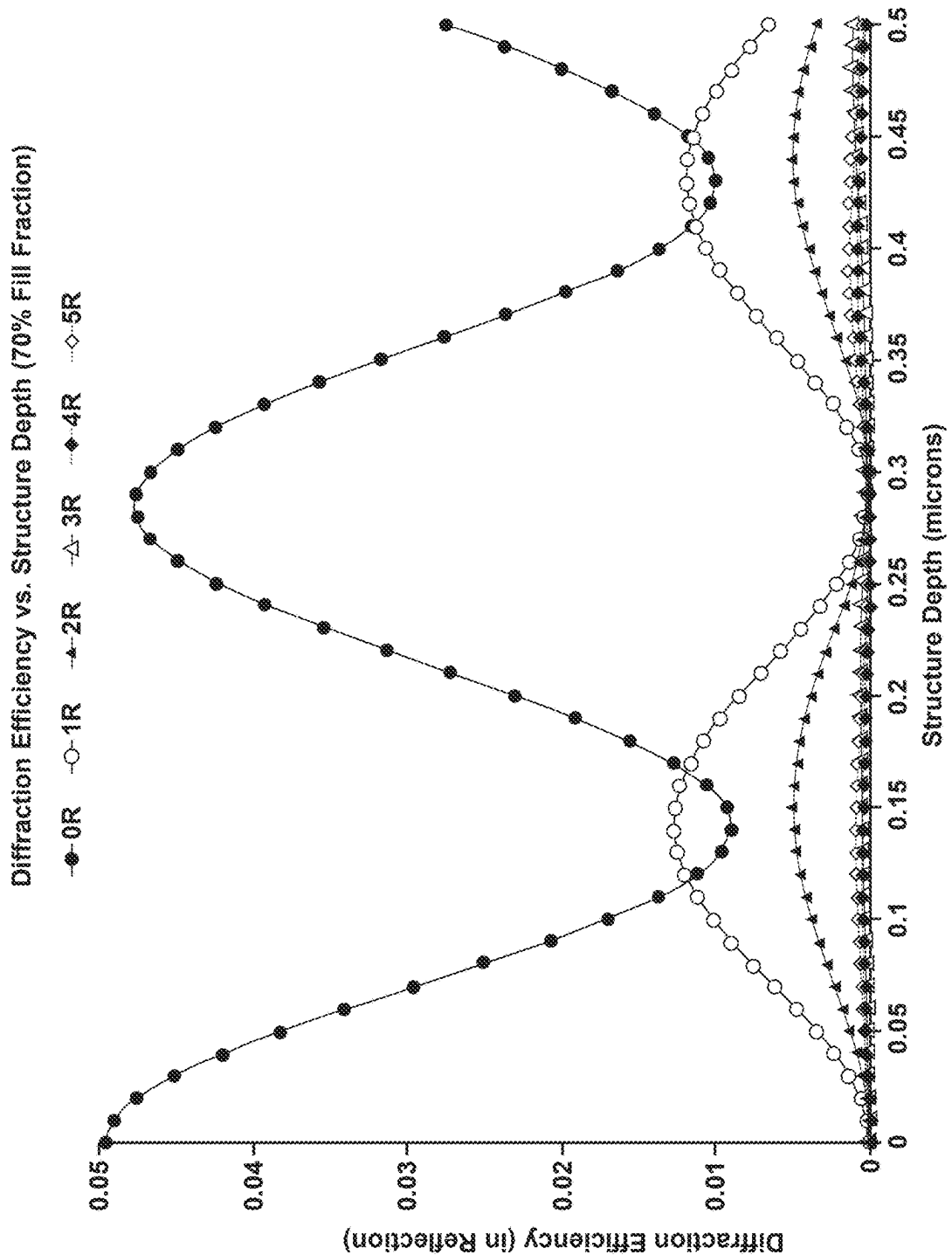


FIG. 4C

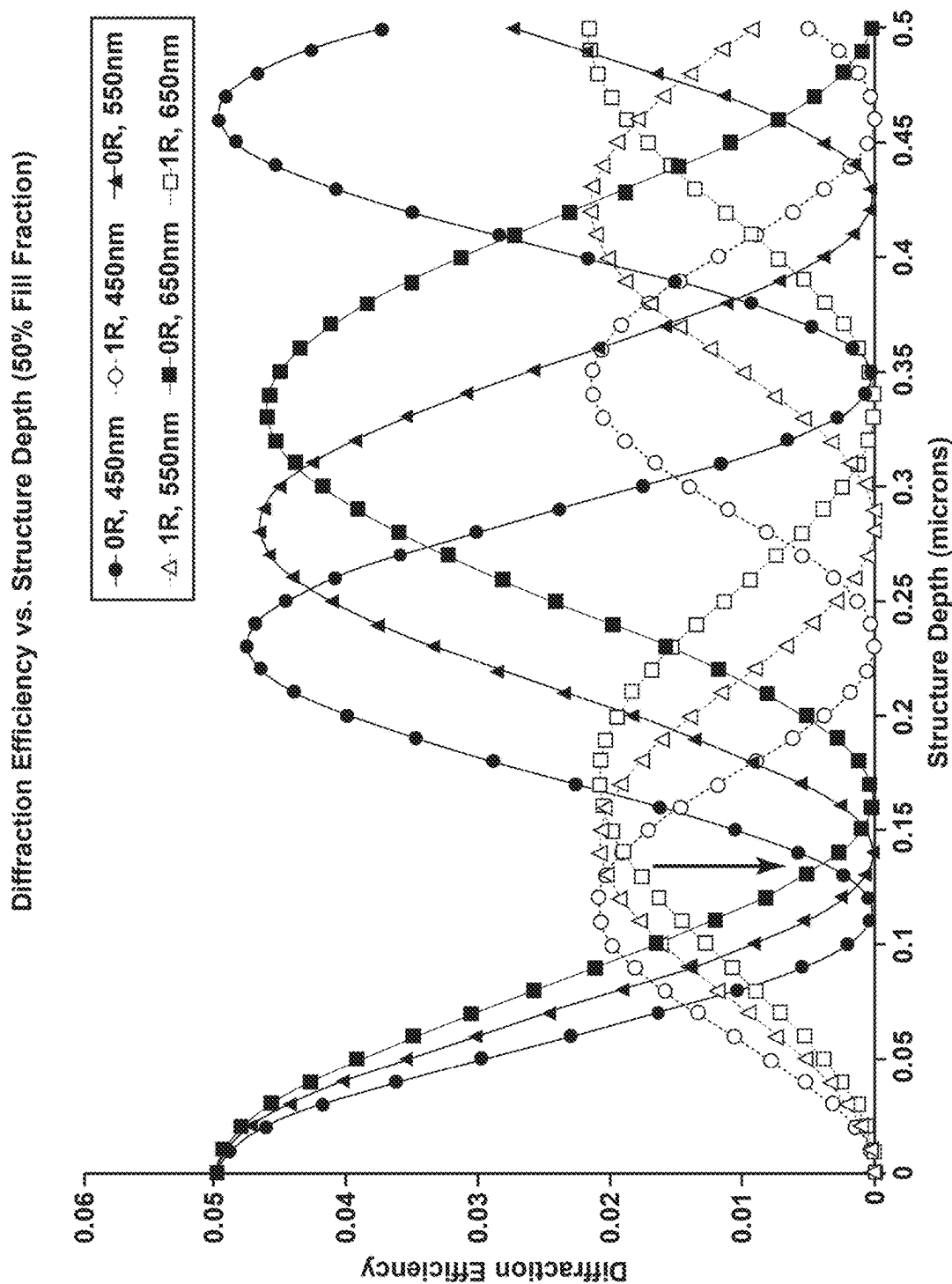


FIG. 5

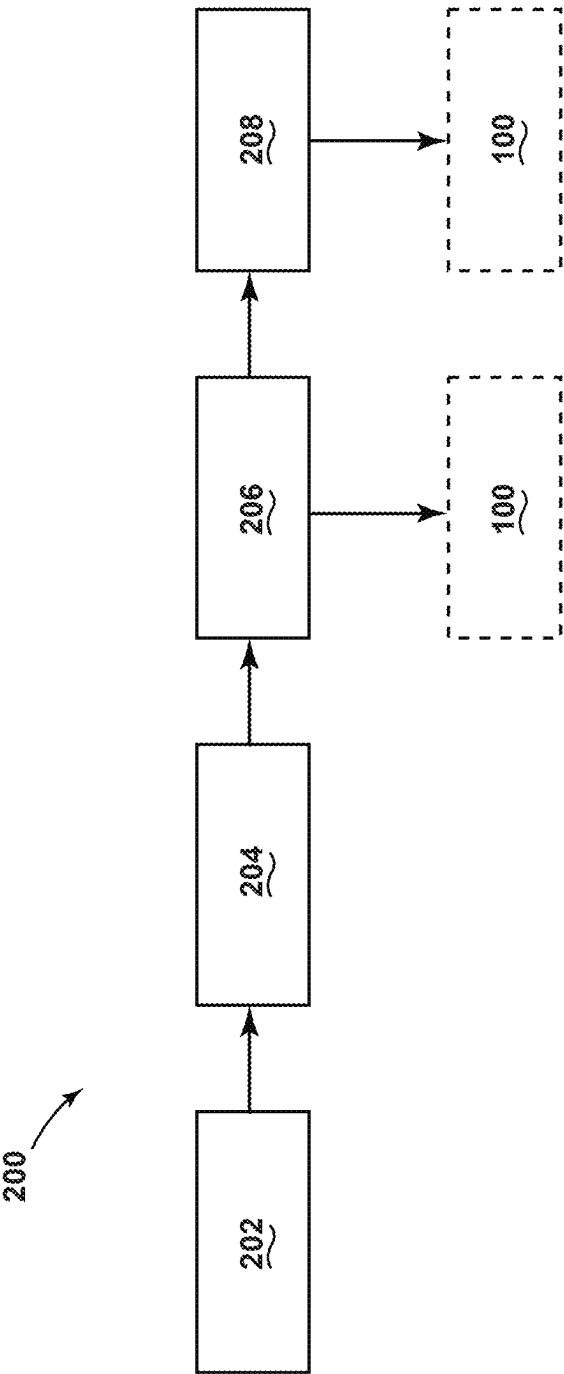


FIG. 6

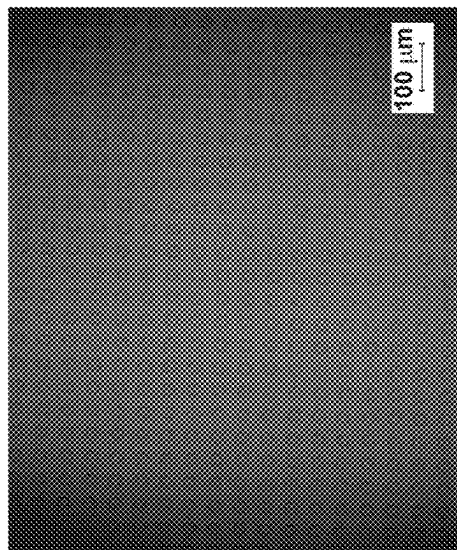


FIG. 7A

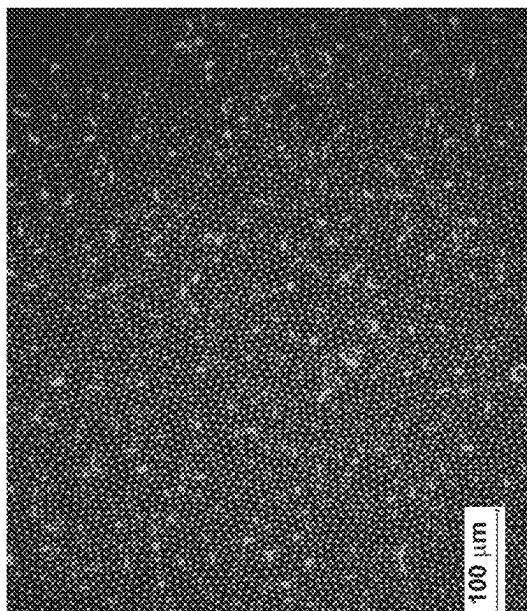


FIG. 7B

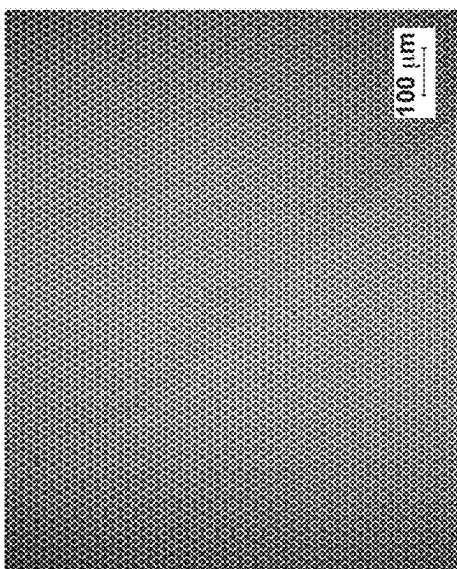


FIG. 7C

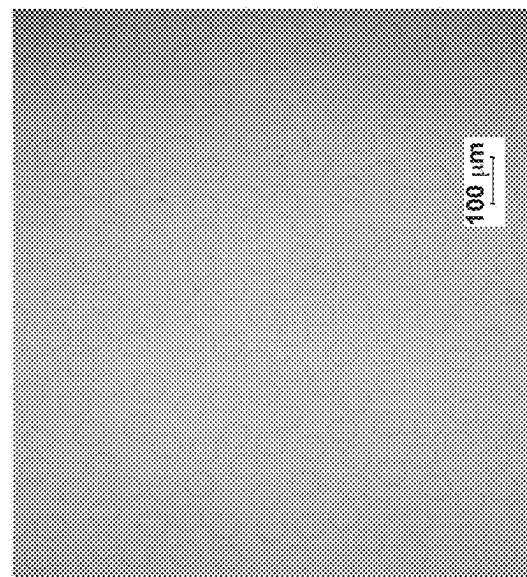


FIG. 7D

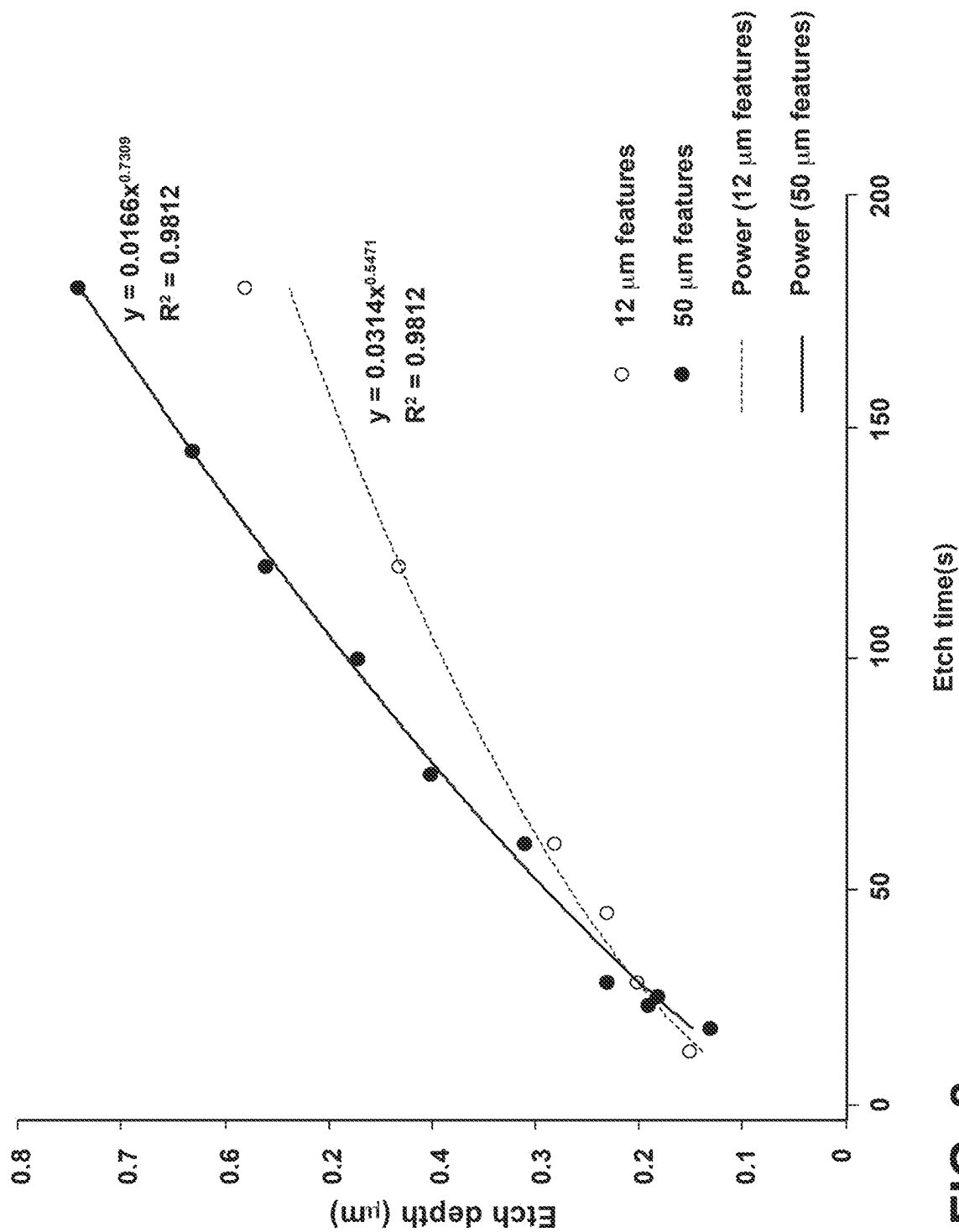


FIG. 8

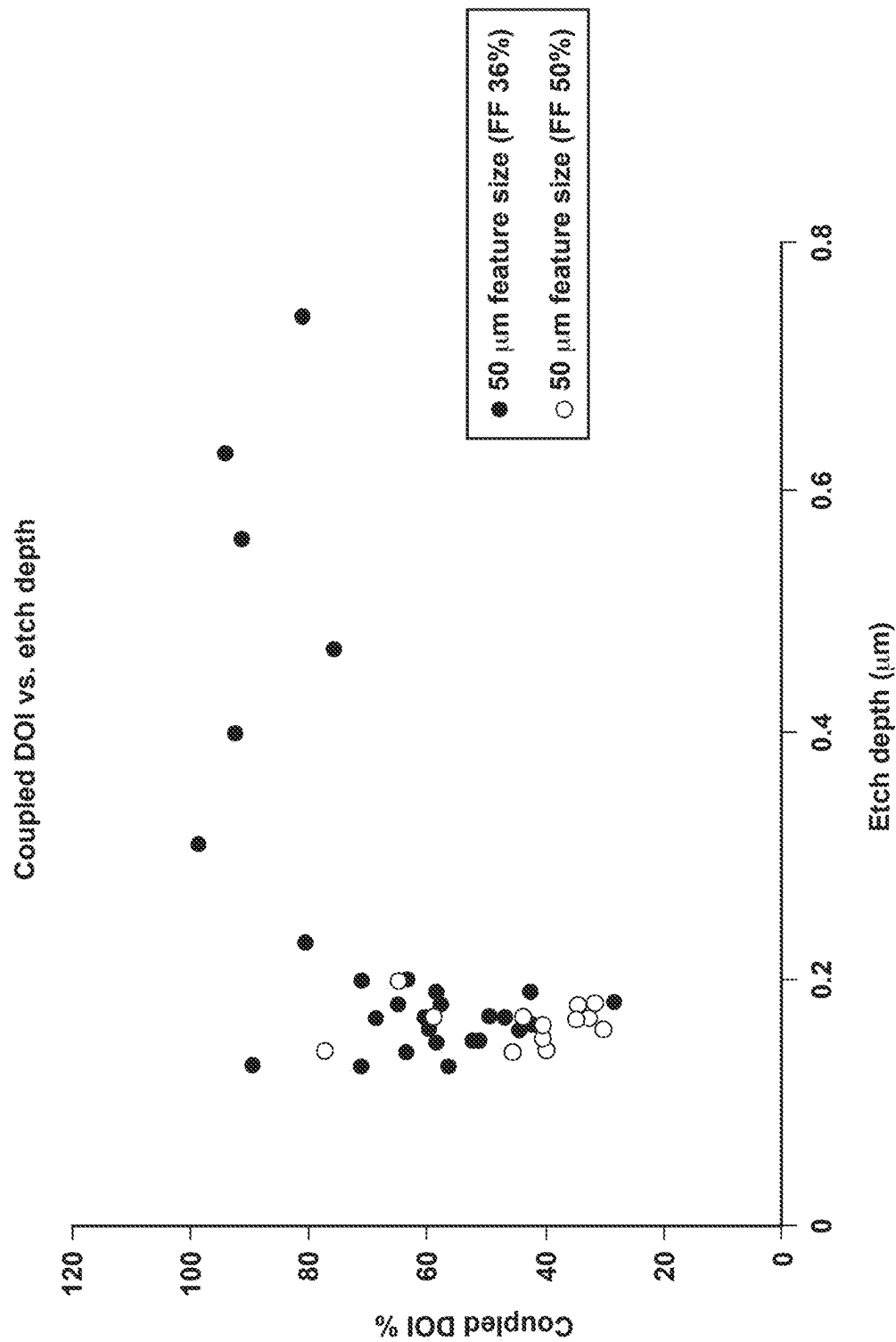


FIG. 9A

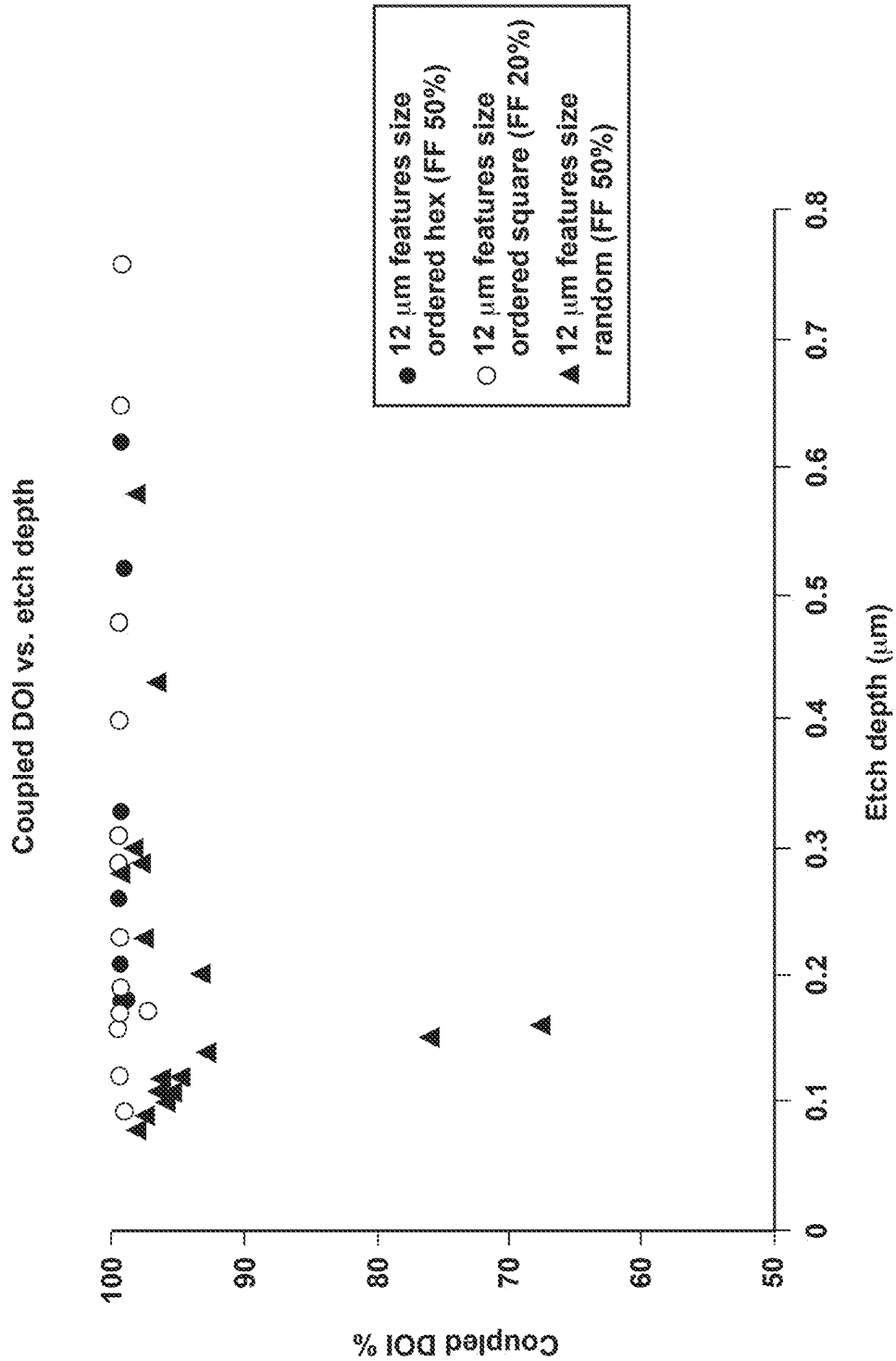


FIG. 9B

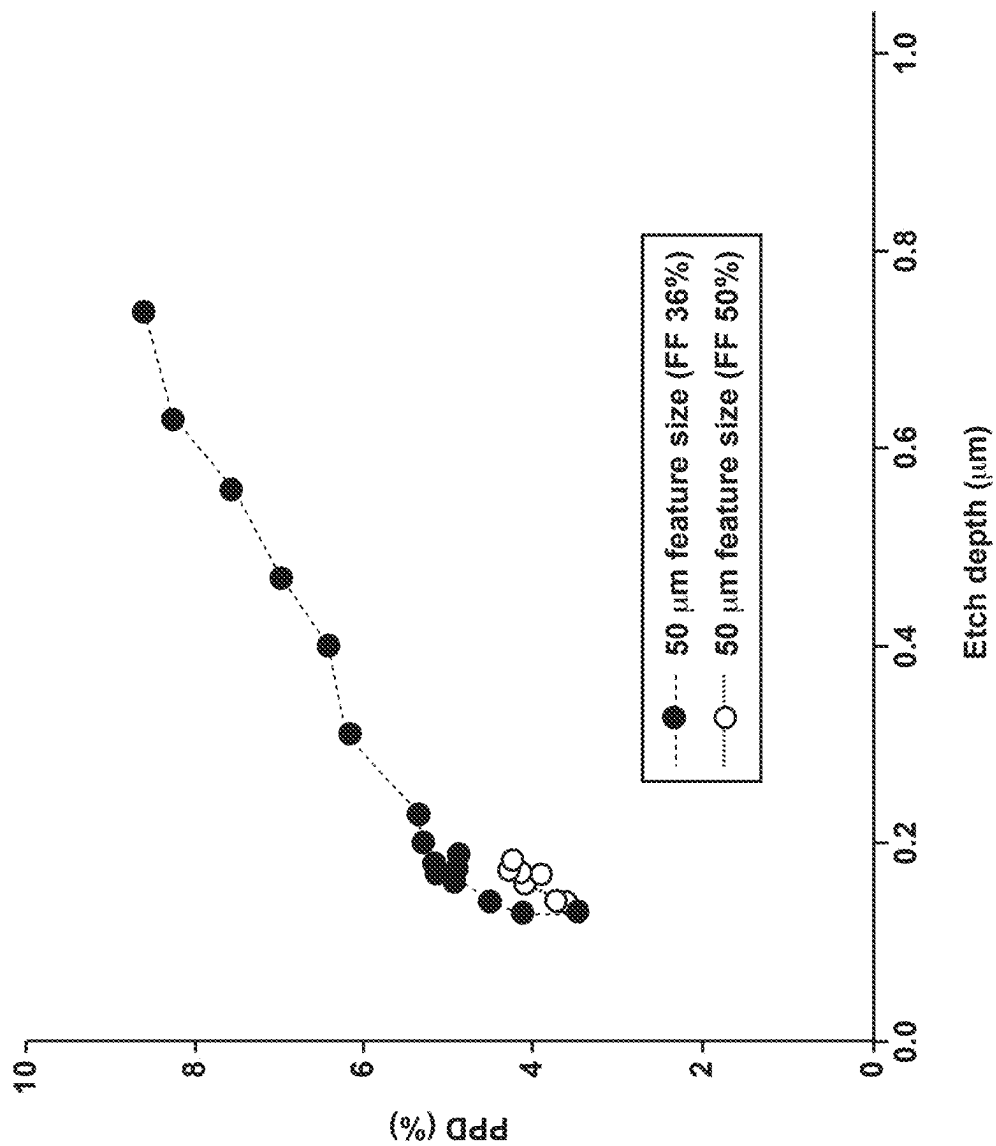


FIG. 10A

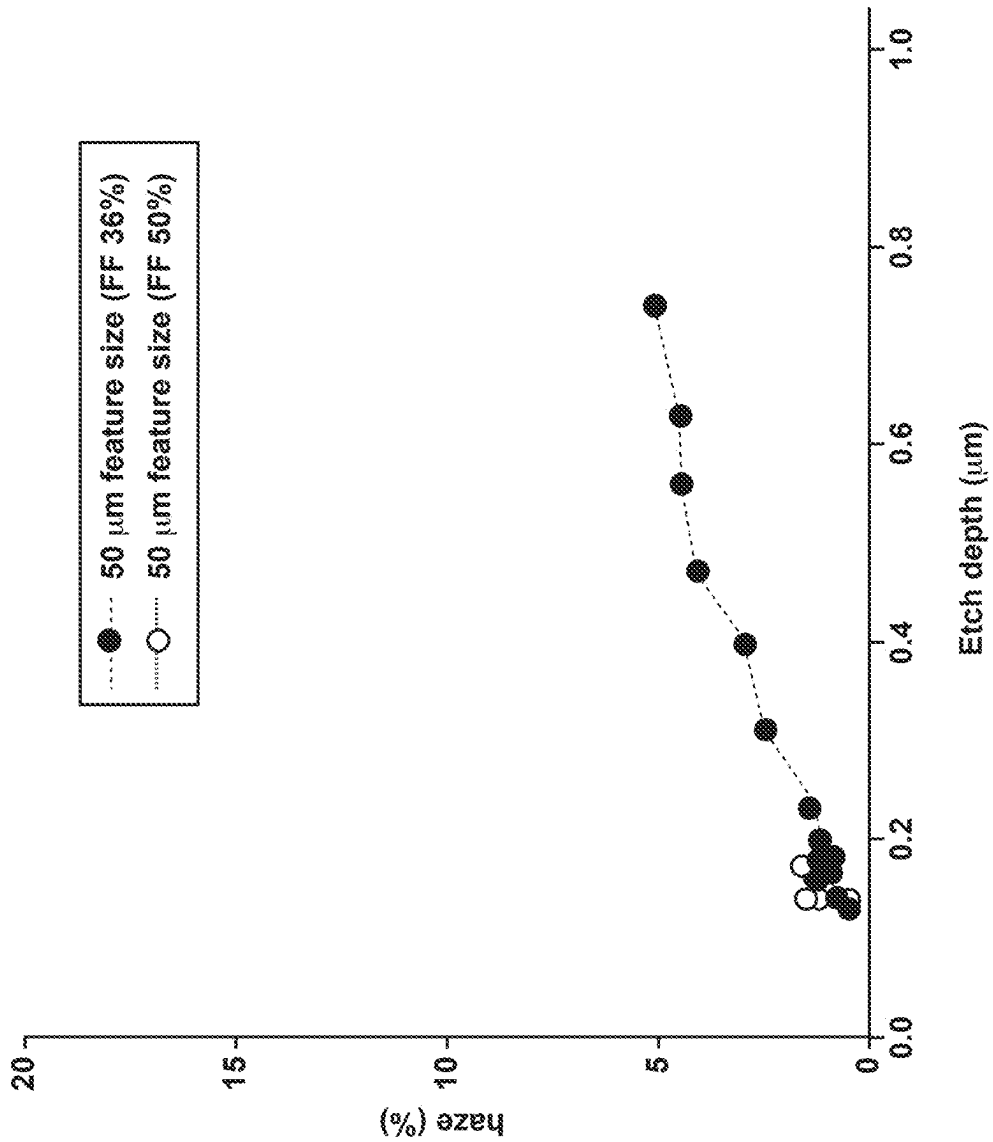


FIG. 10B

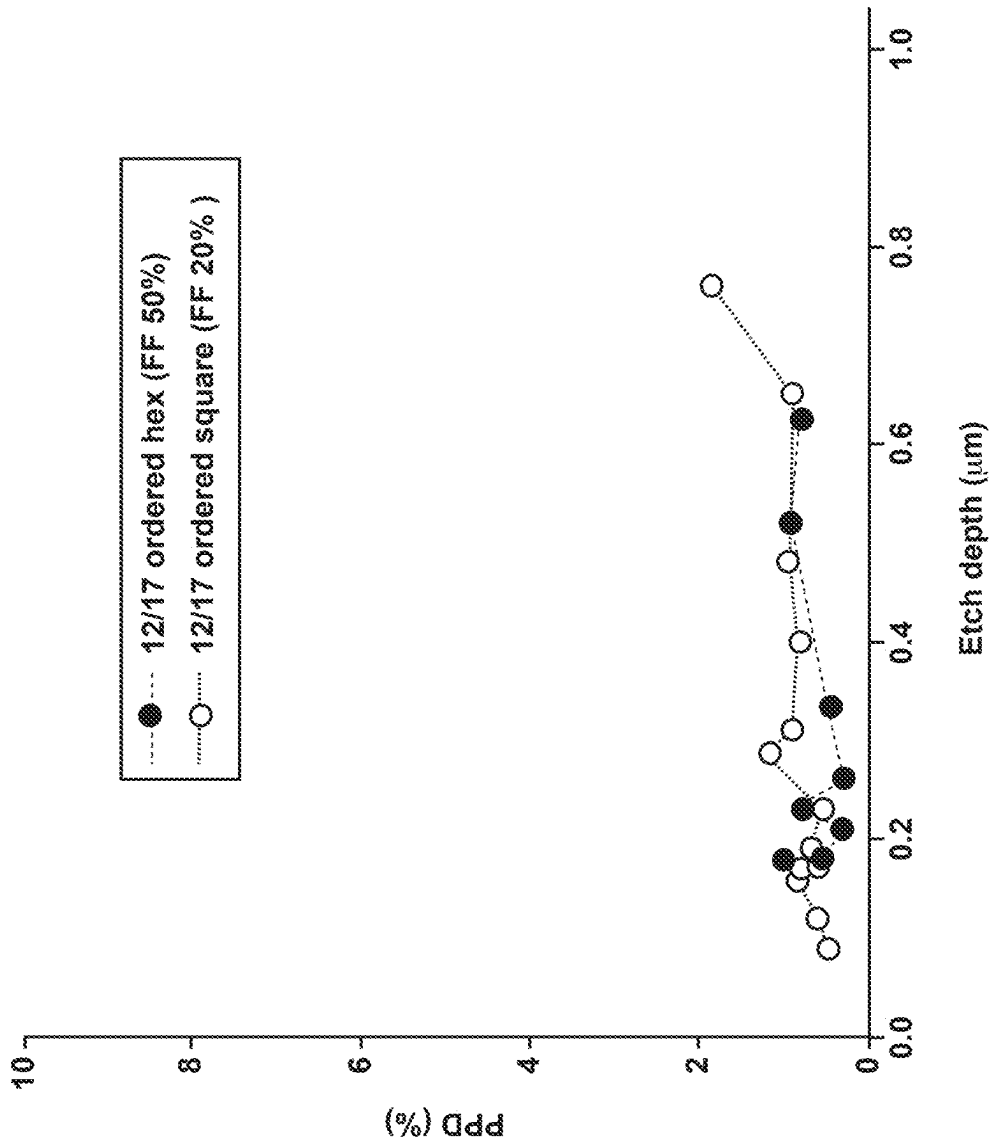


FIG. 10C

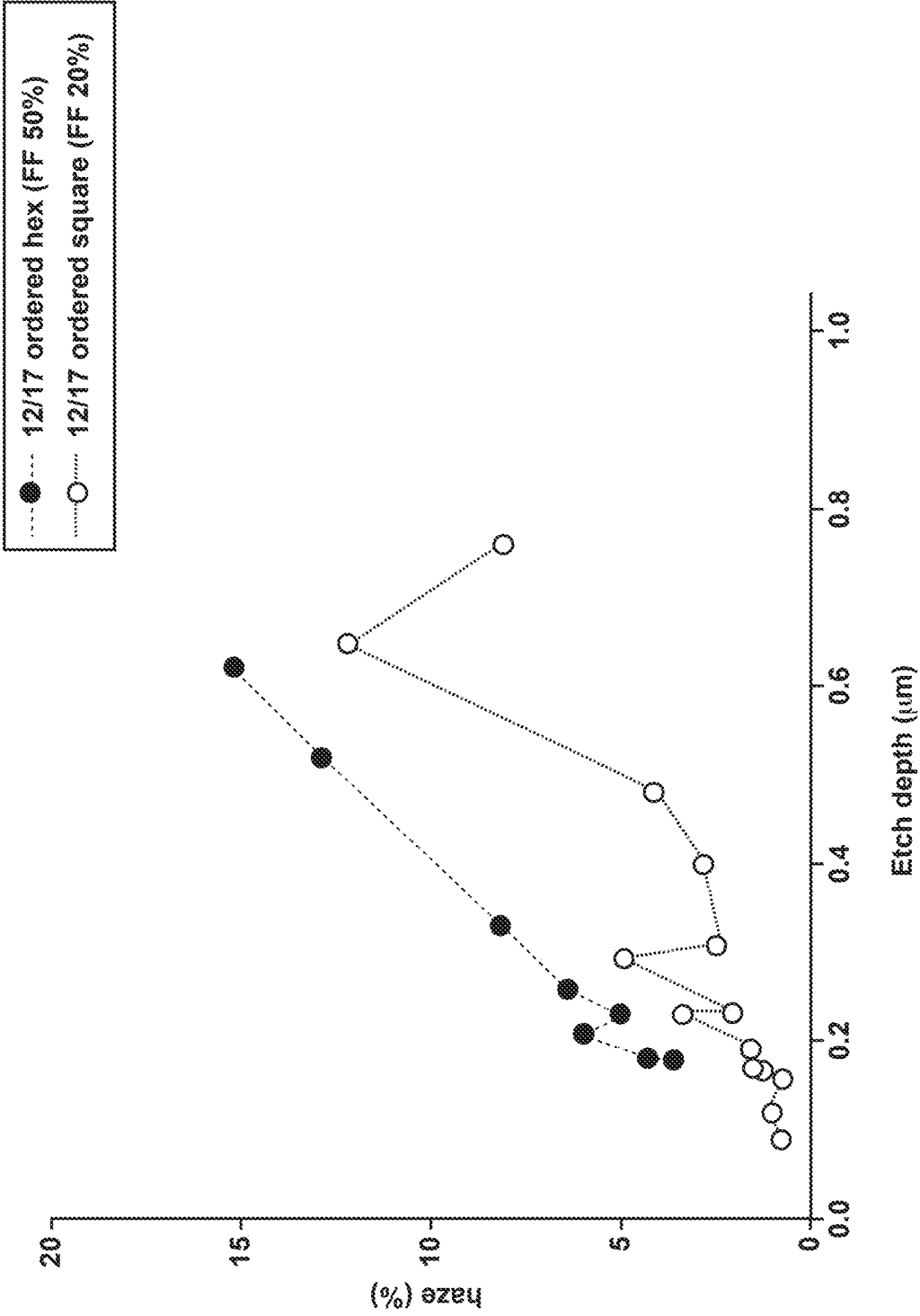


FIG. 10D

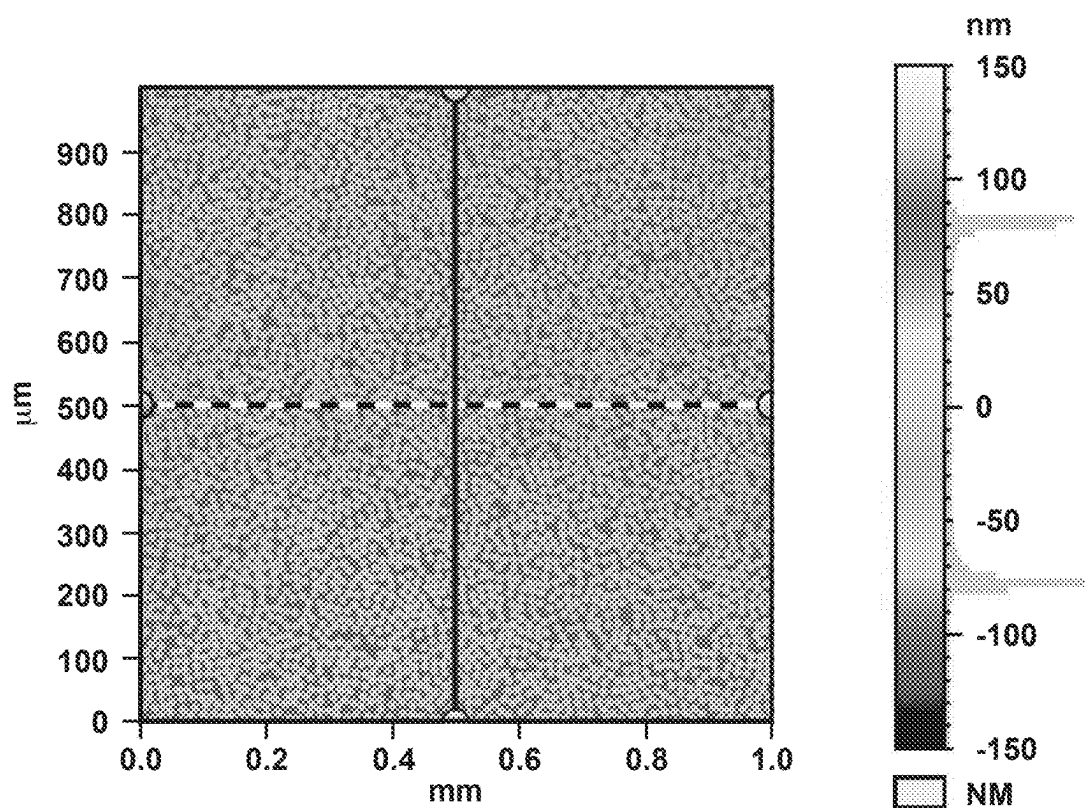


FIG. 11A

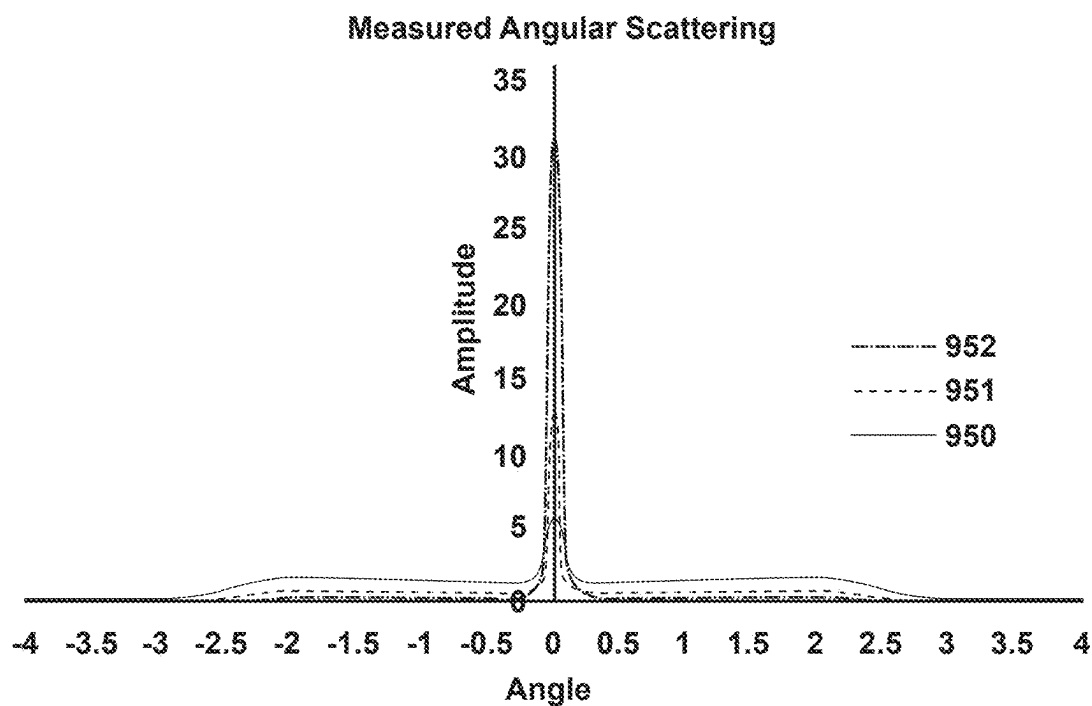


FIG. 11B

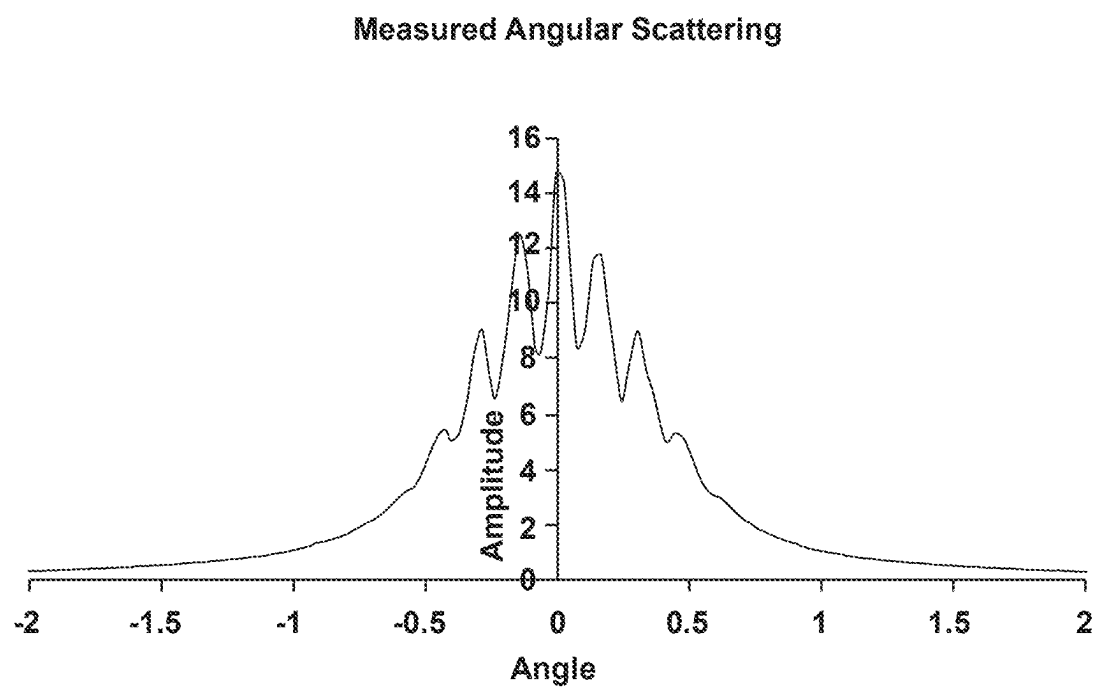


FIG. 12

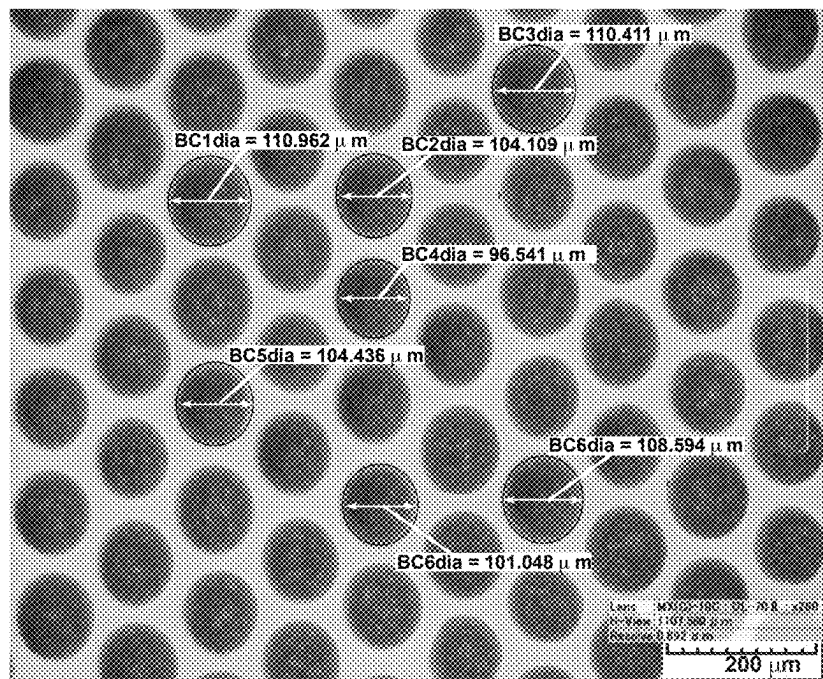


FIG. 13A

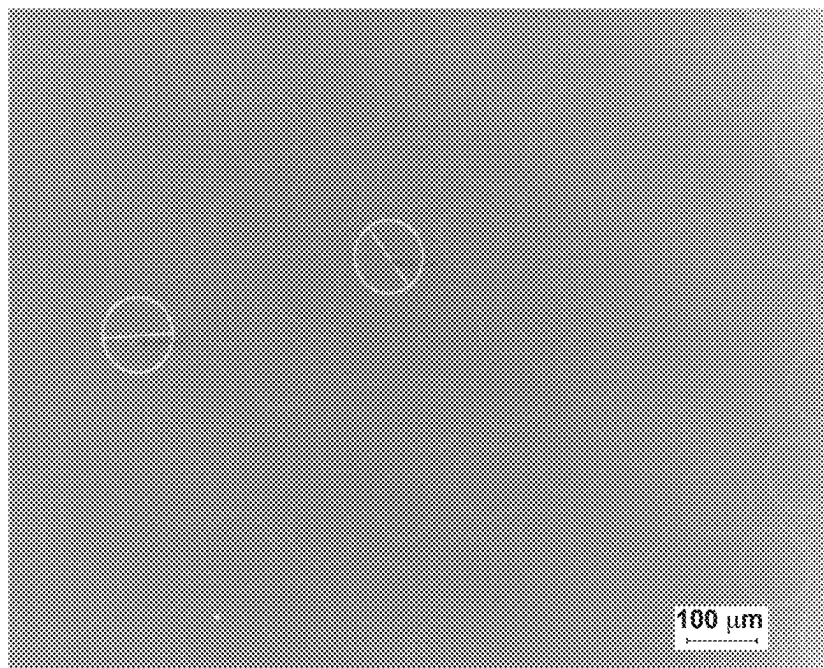


FIG. 13B

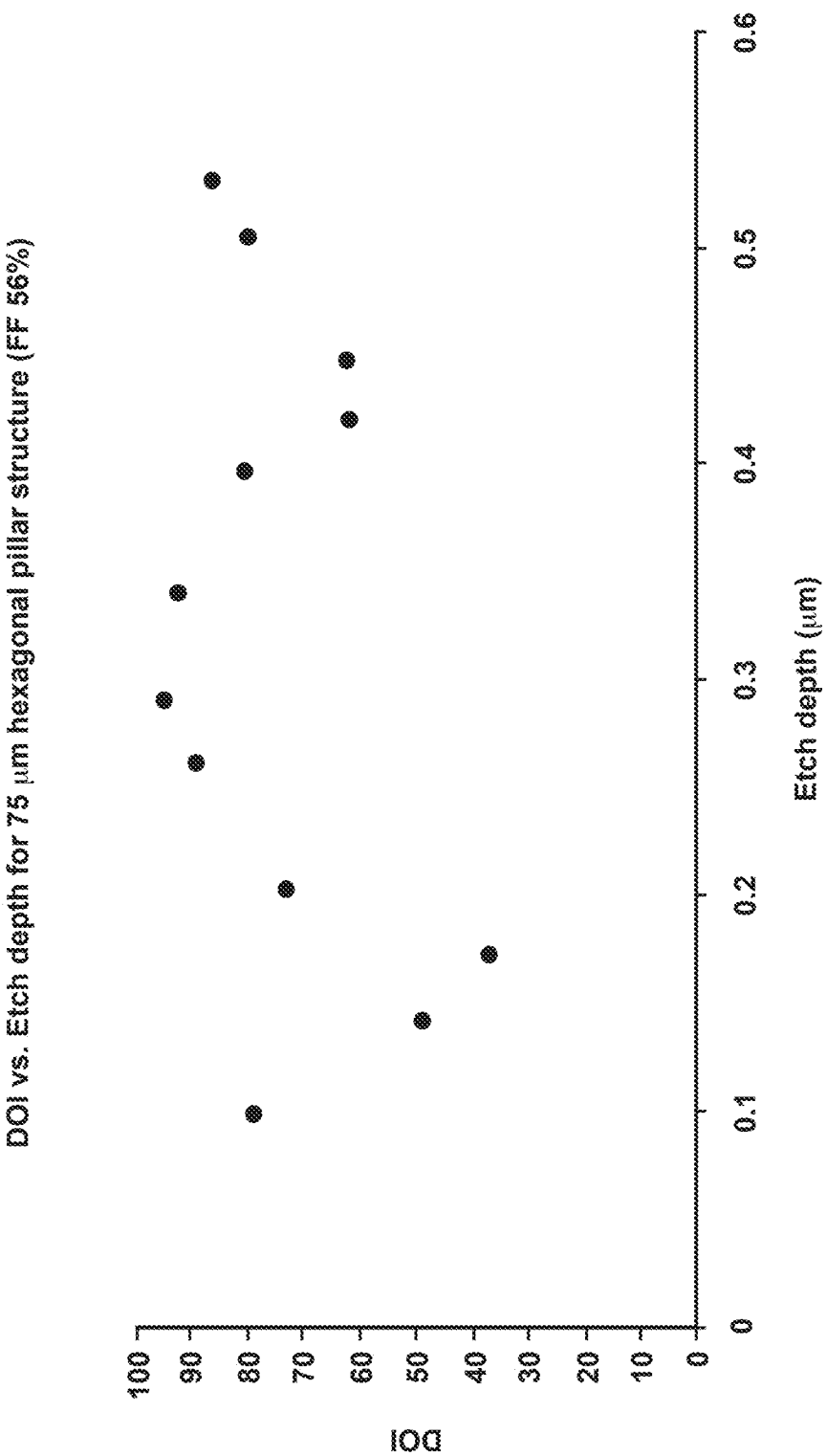


FIG. 14

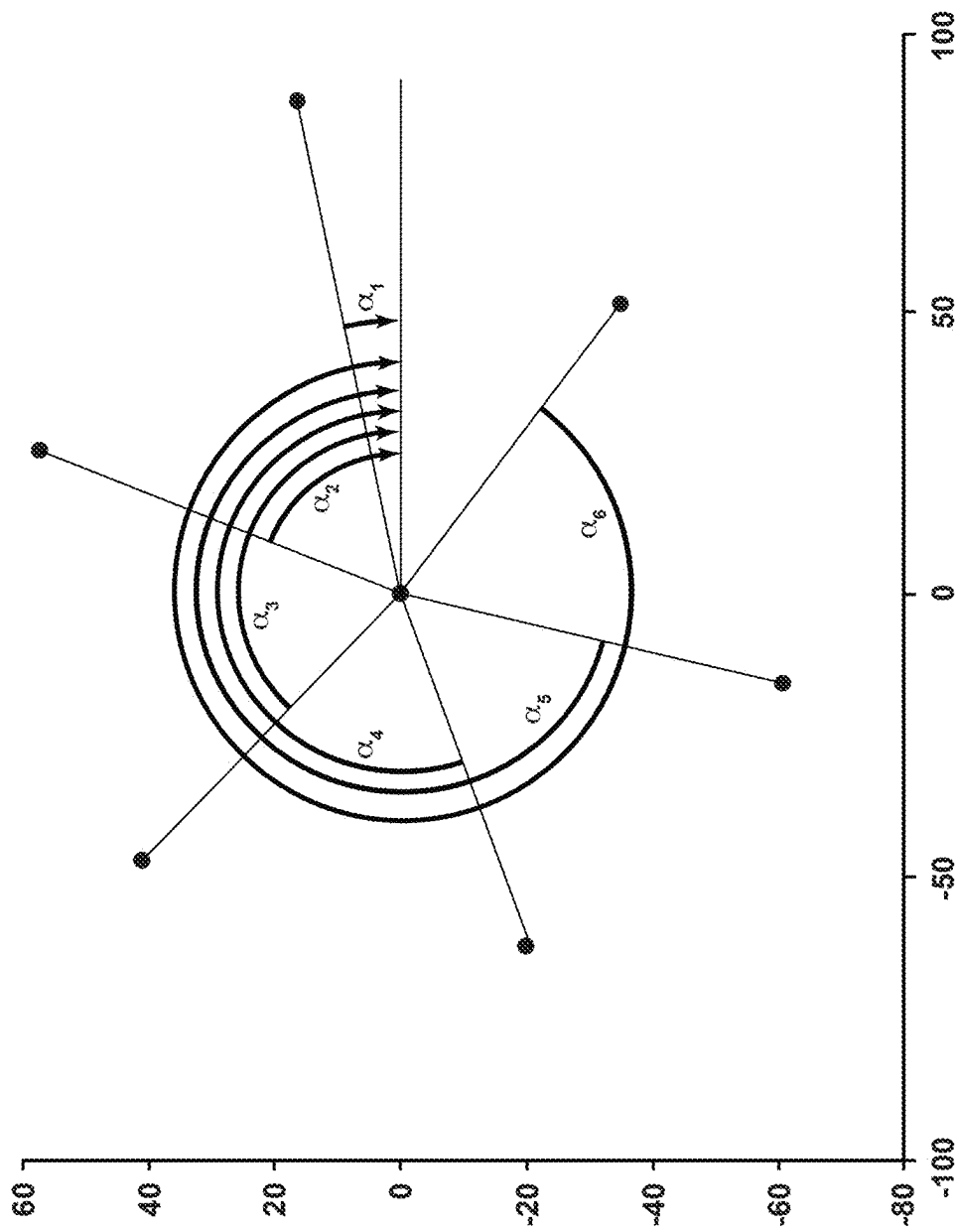


FIG. 15

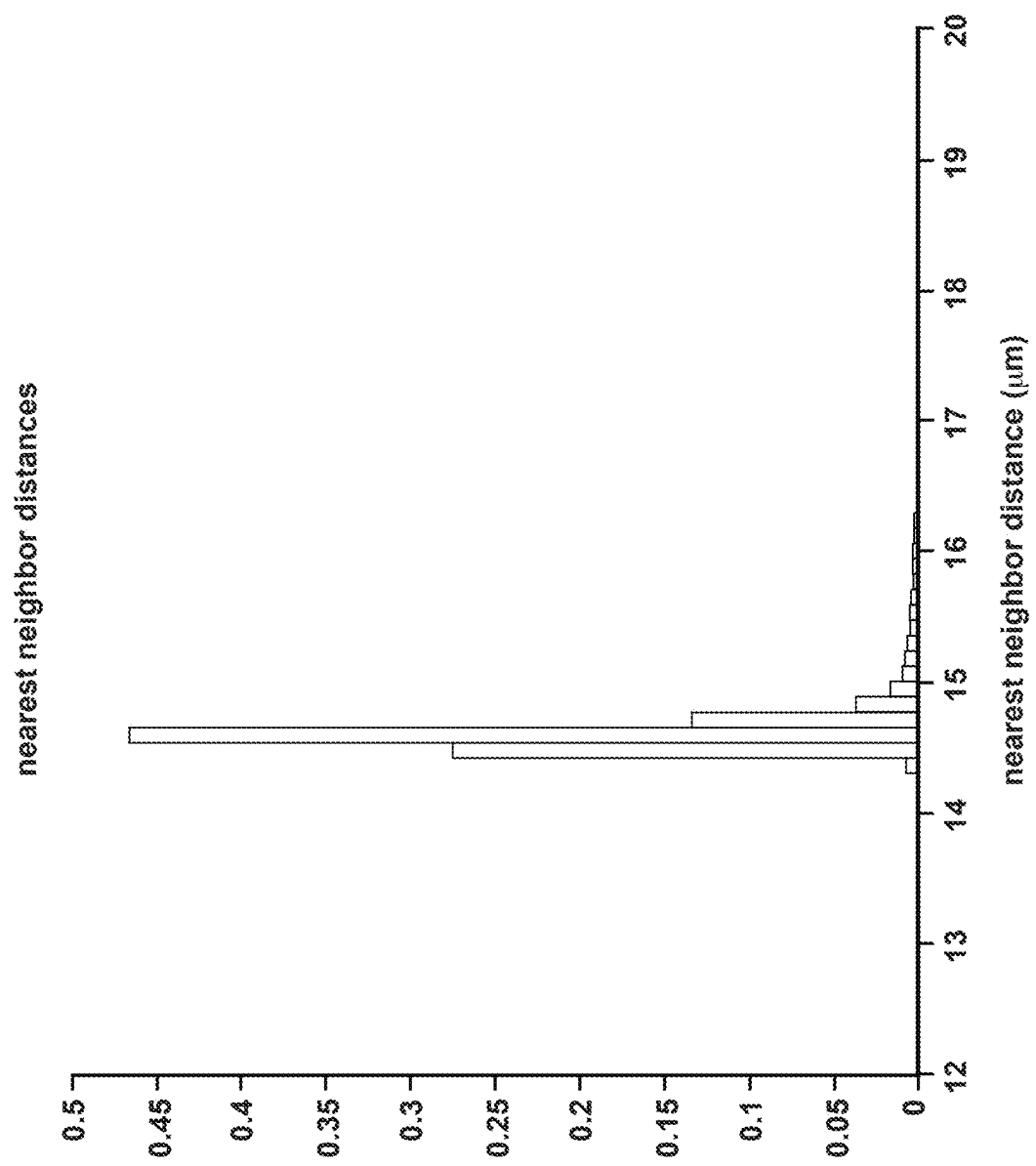


FIG. 16A

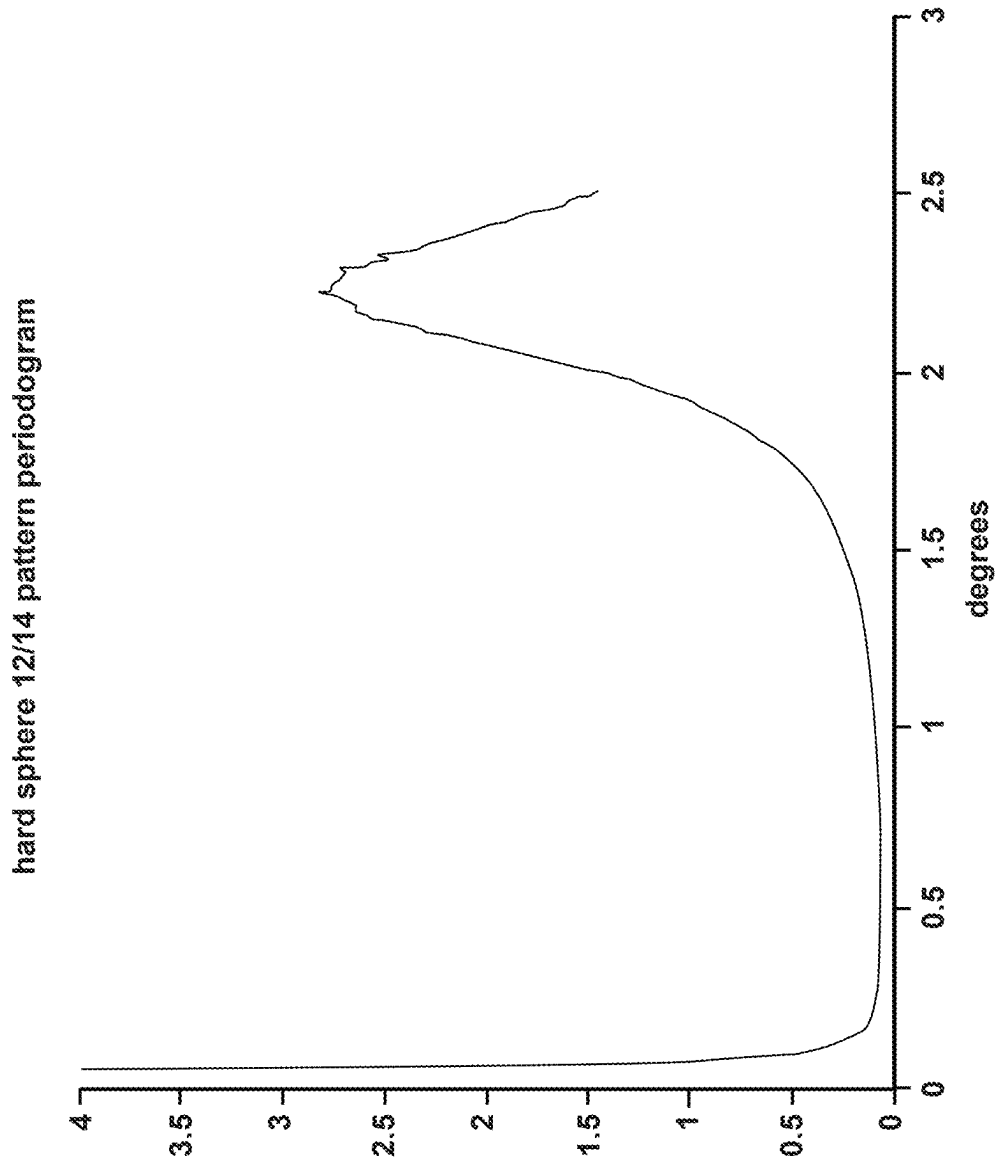


FIG. 16B

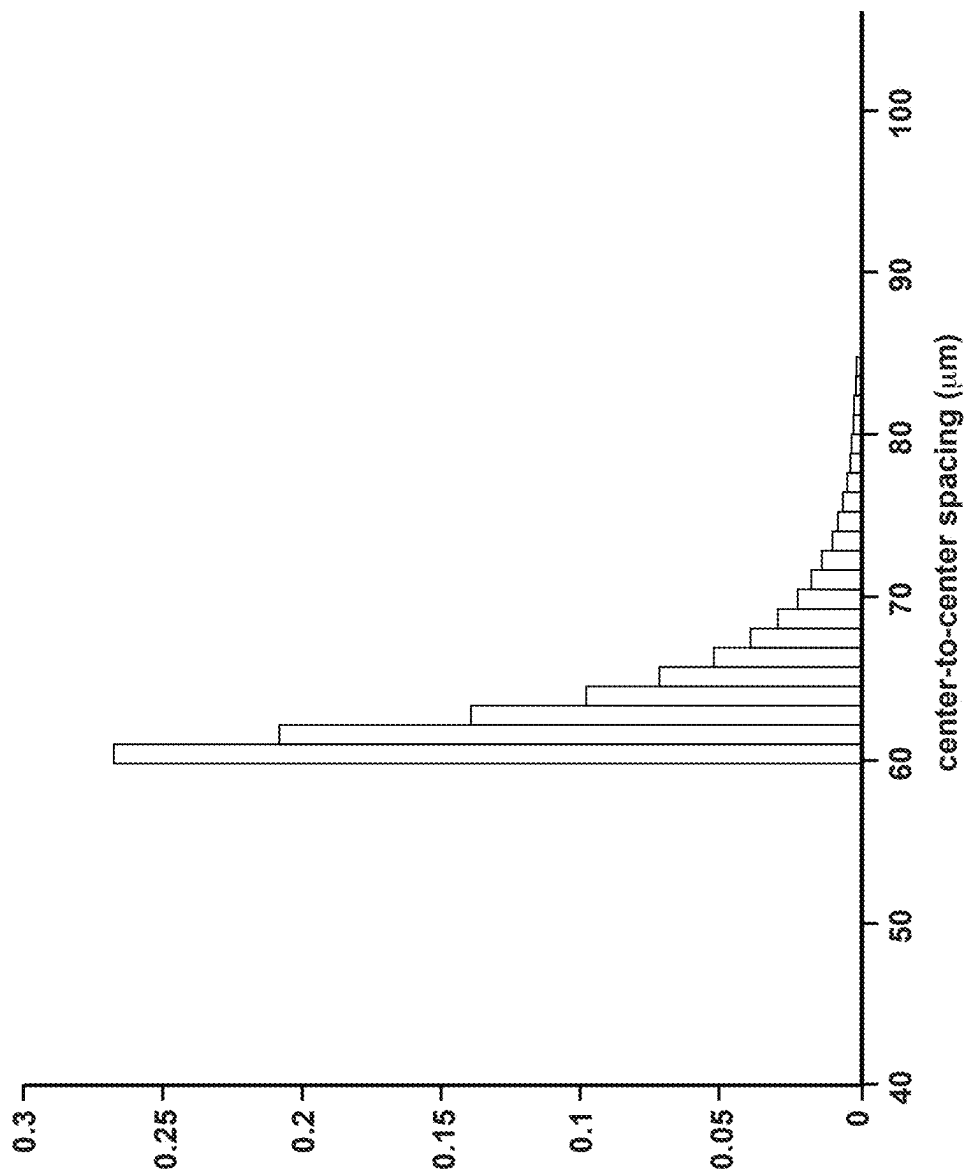


FIG. 17A

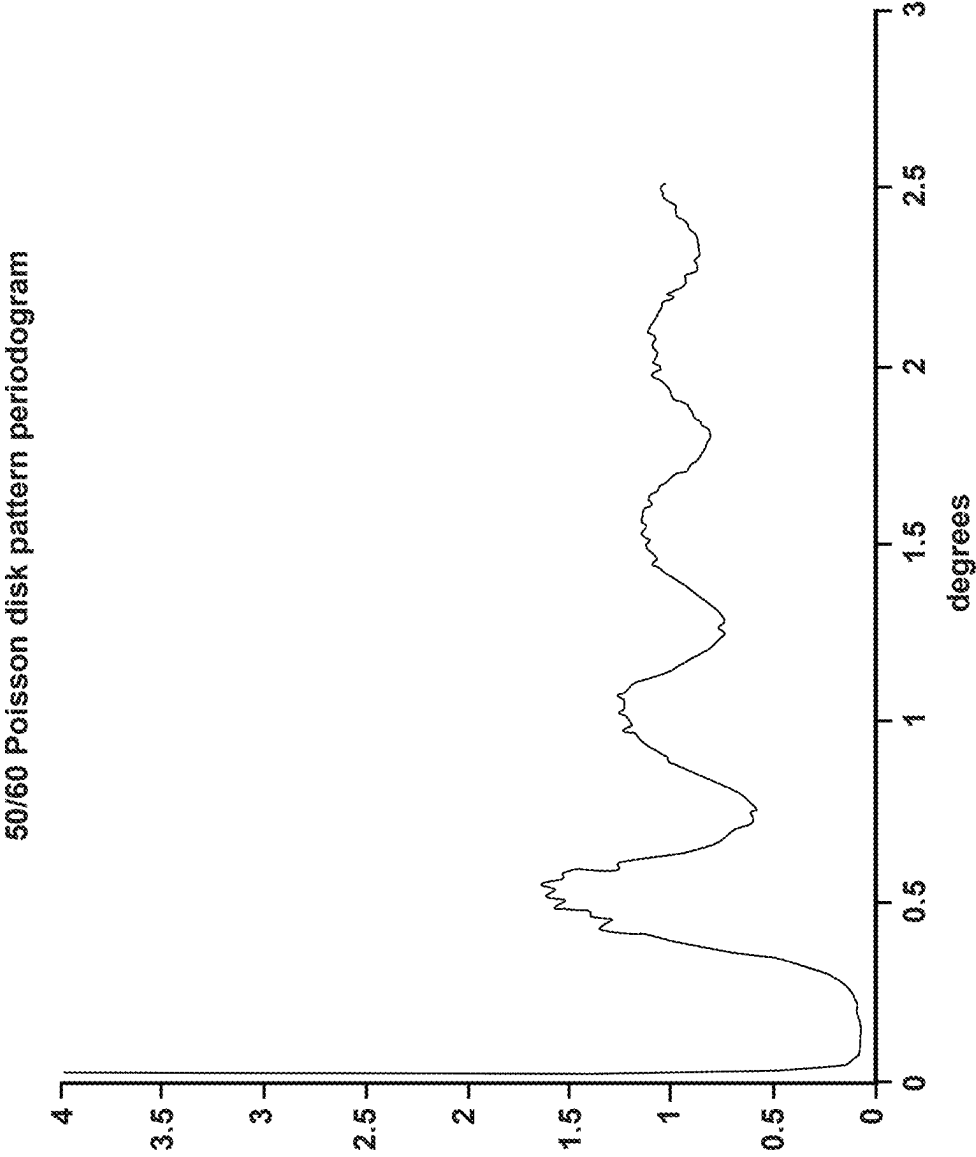


FIG. 17B

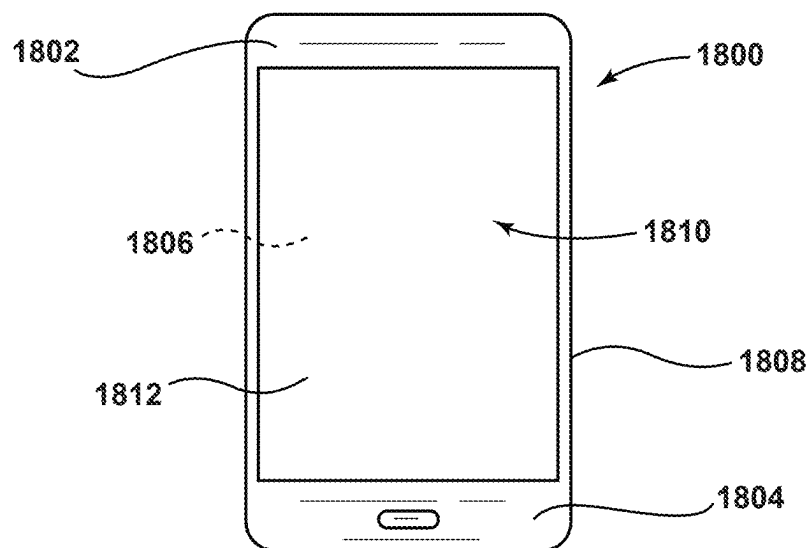


FIG. 18A

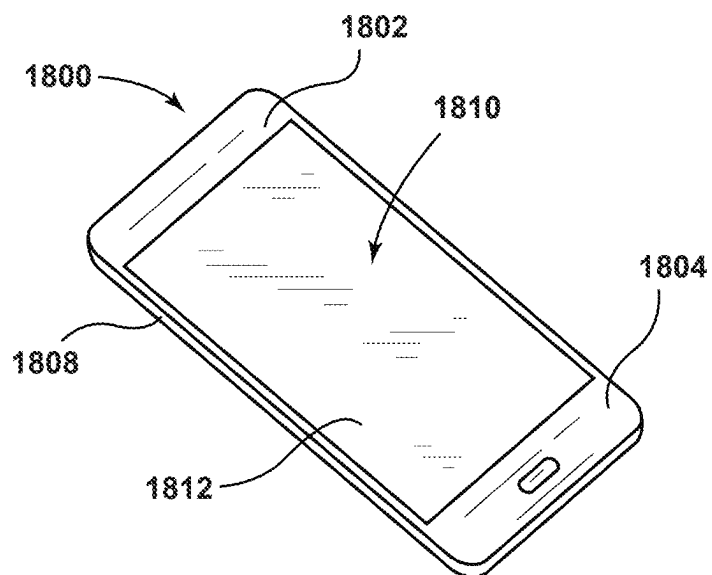


FIG. 18B

Ex. 5A, measured apparent hardness and modulus

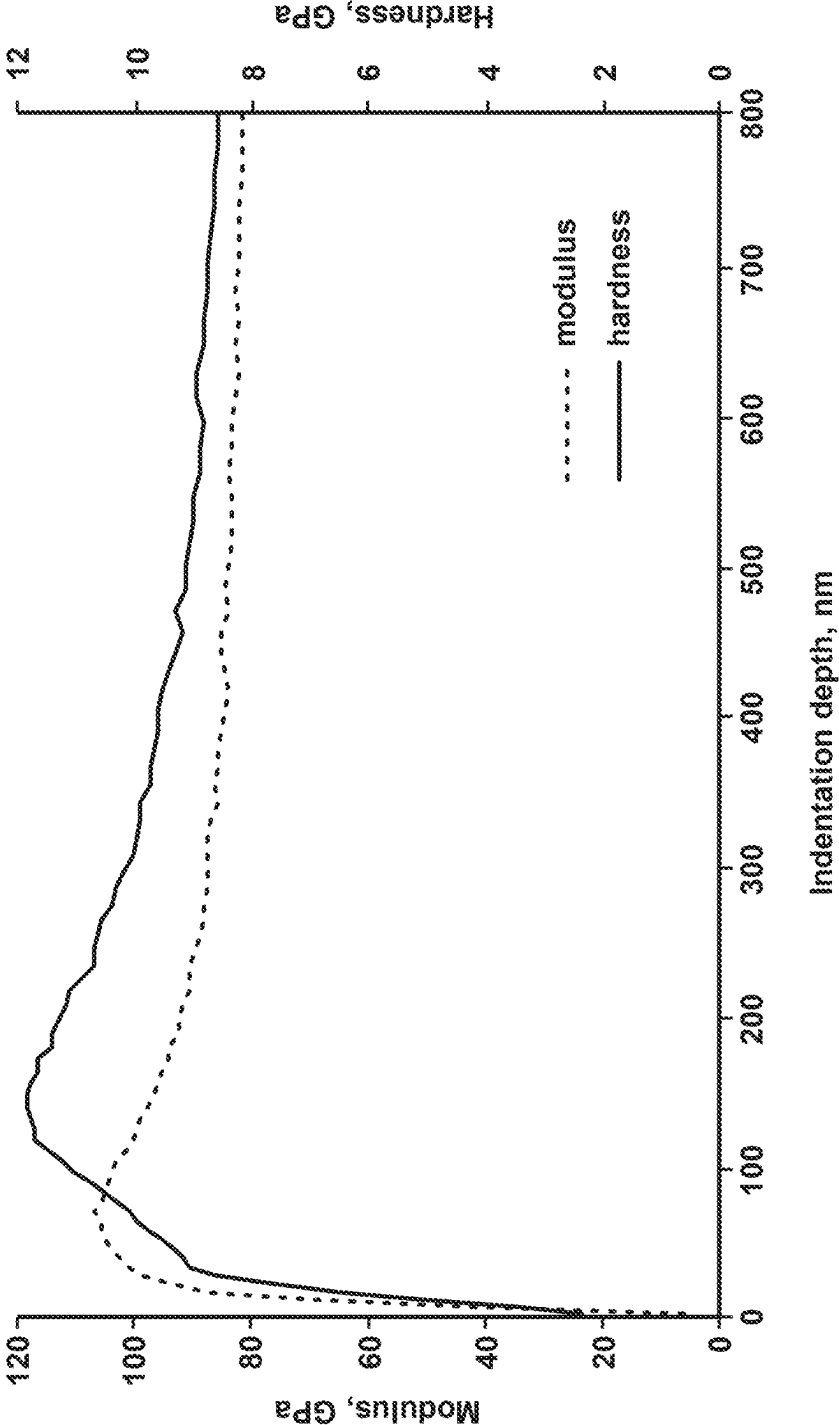


FIG. 19

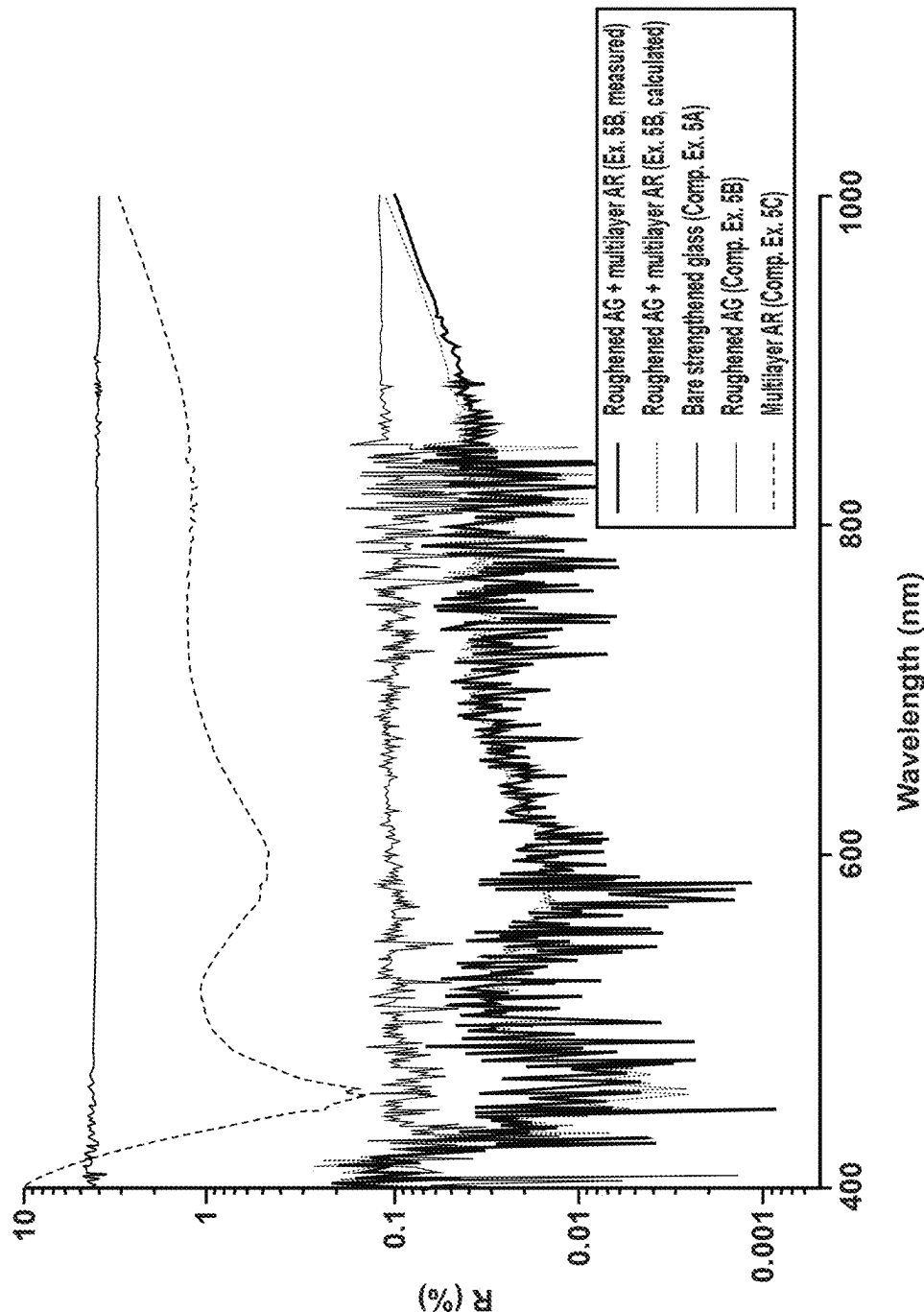


FIG. 20A

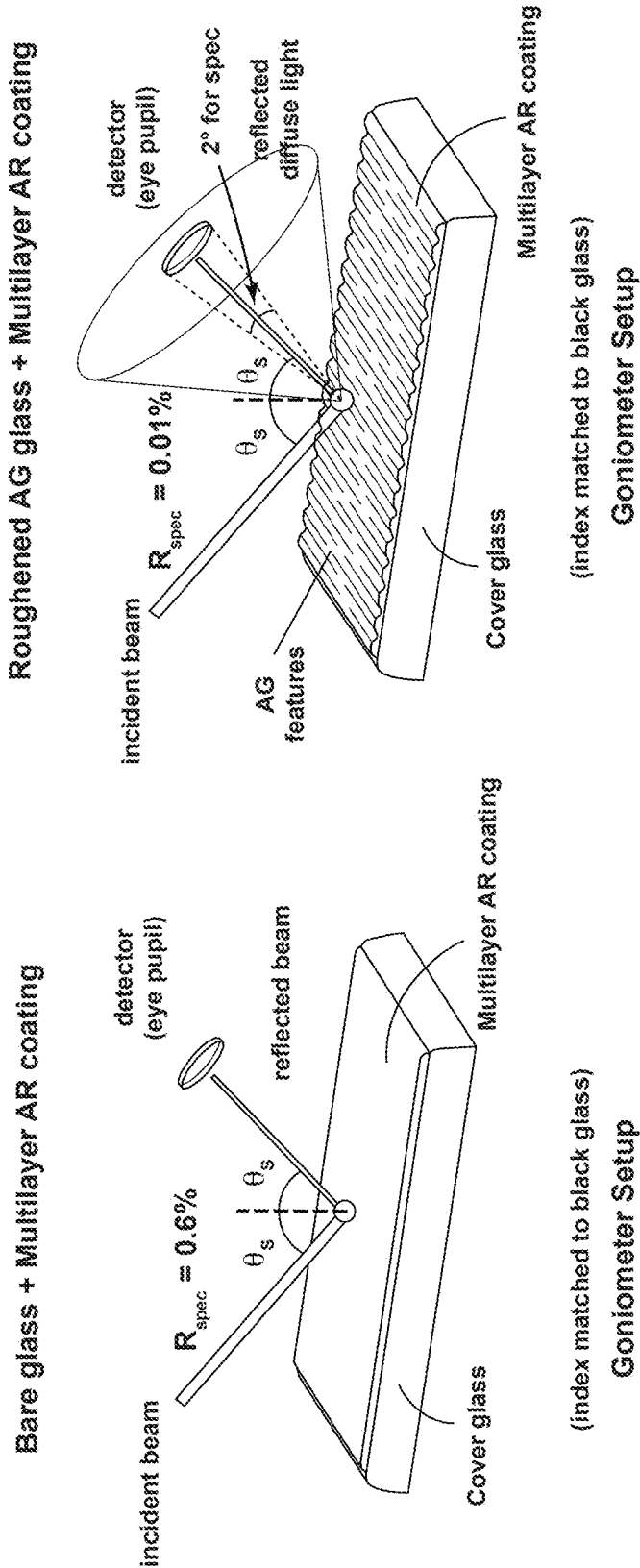


FIG. 20B

FIG. 20C

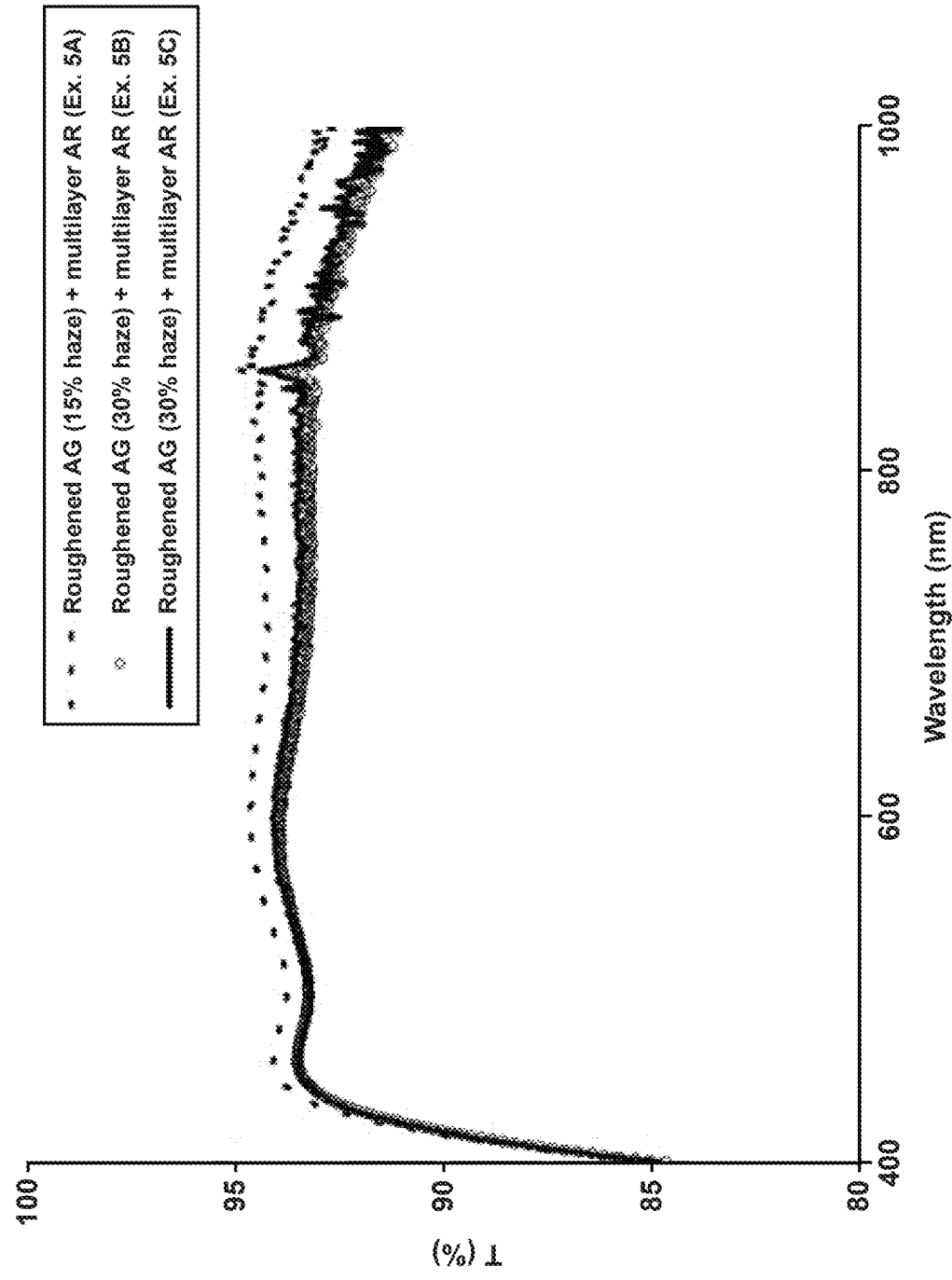


FIG. 21

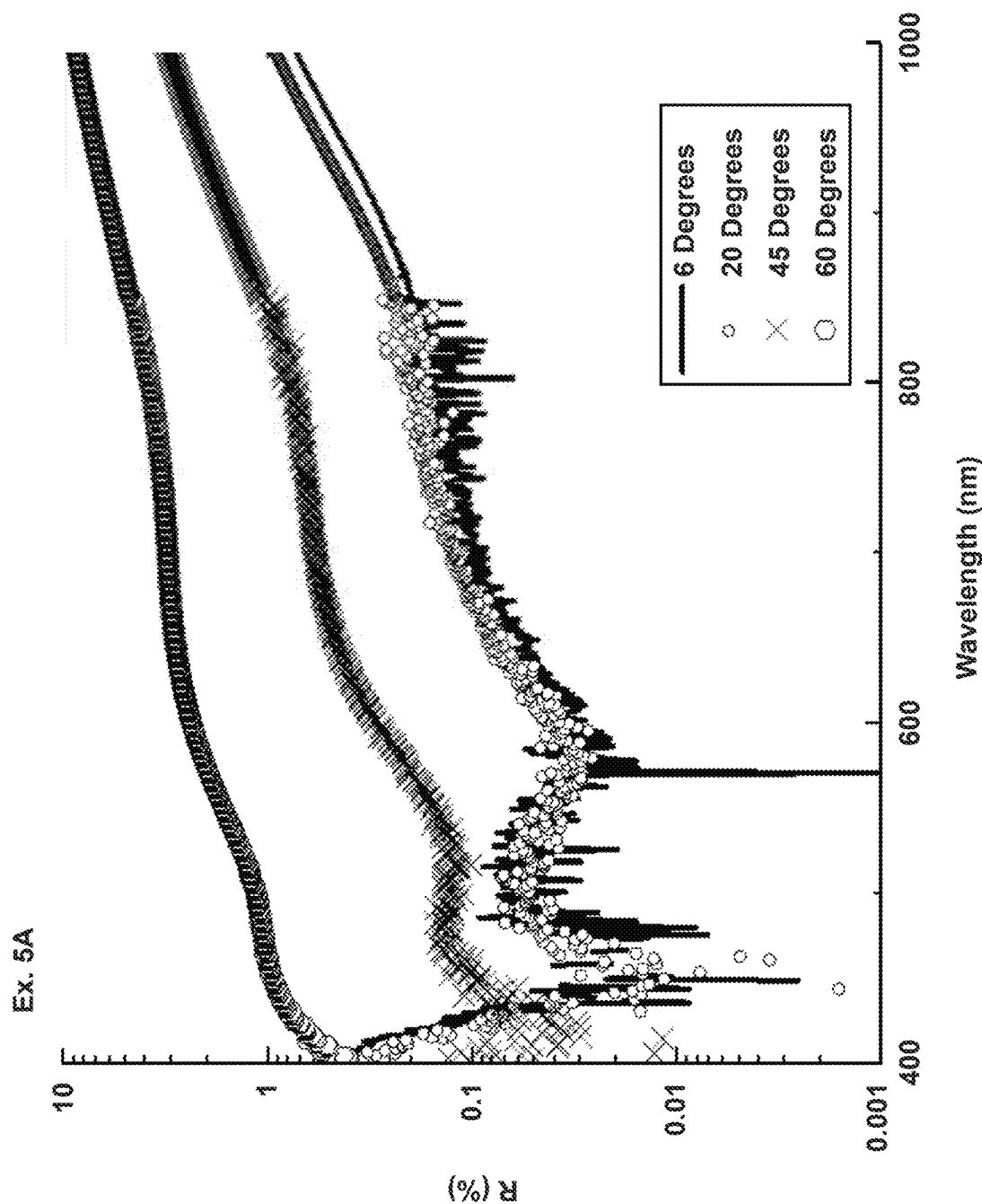


FIG. 22A

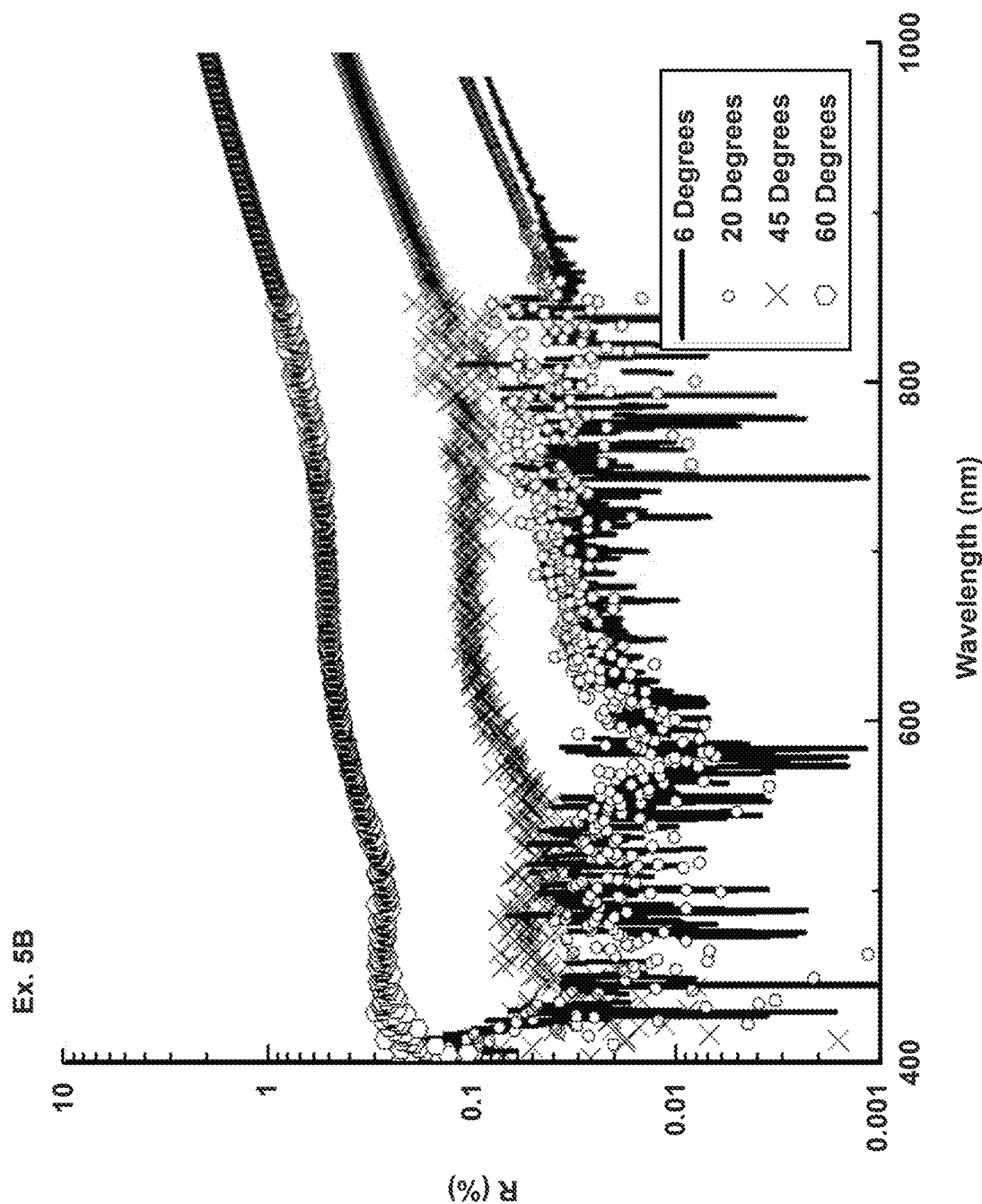


FIG. 22B

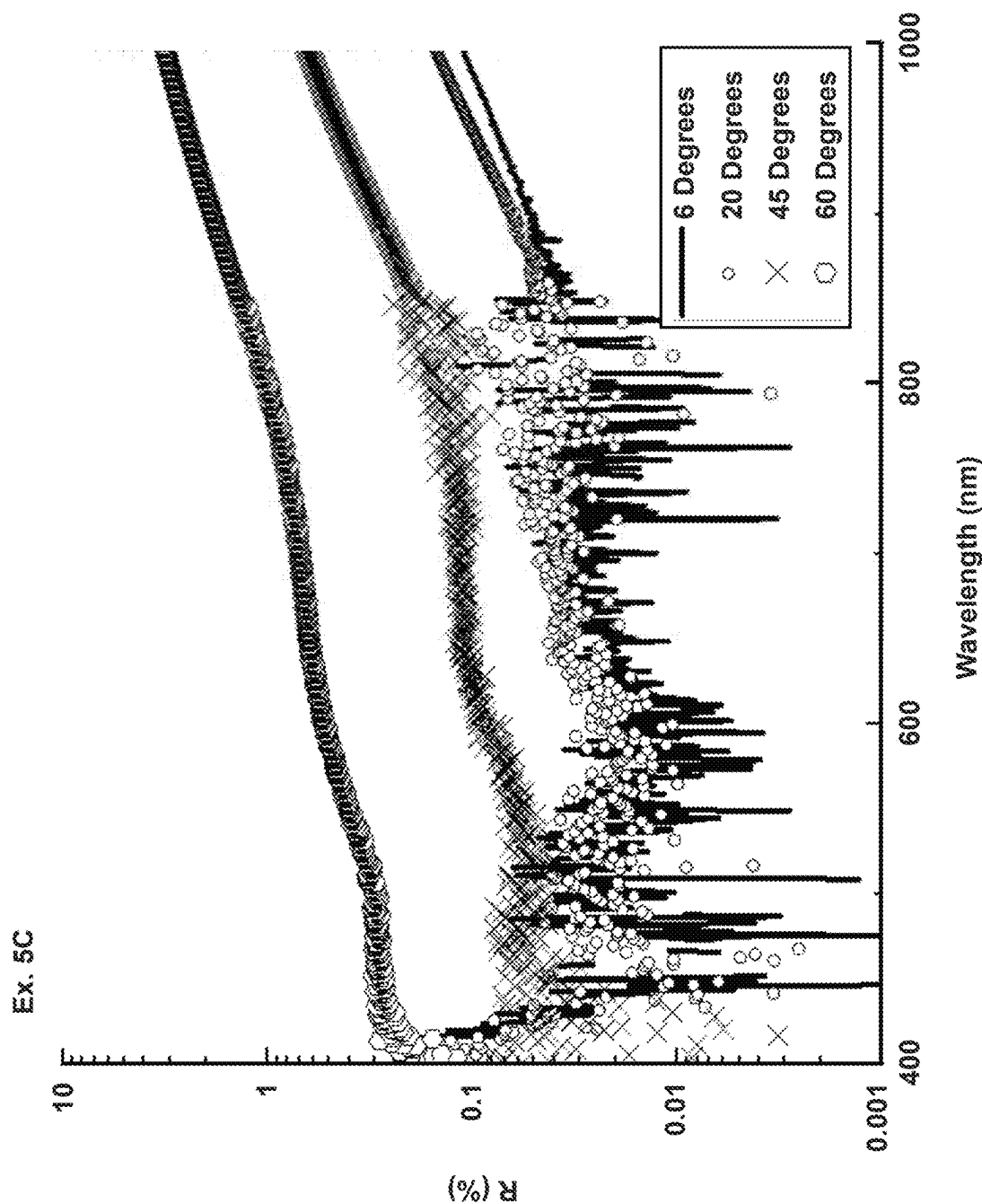


FIG. 22C

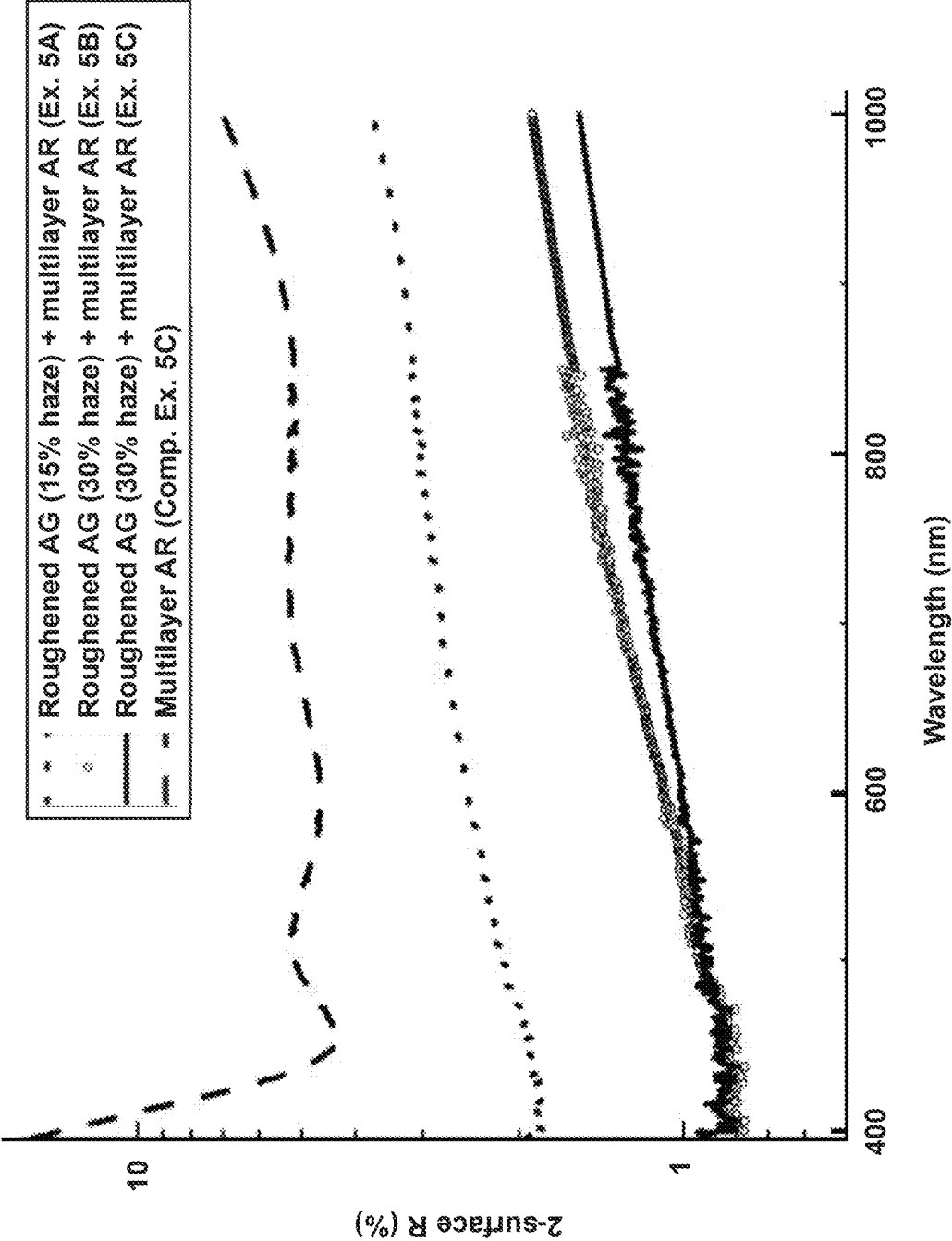


FIG. 23

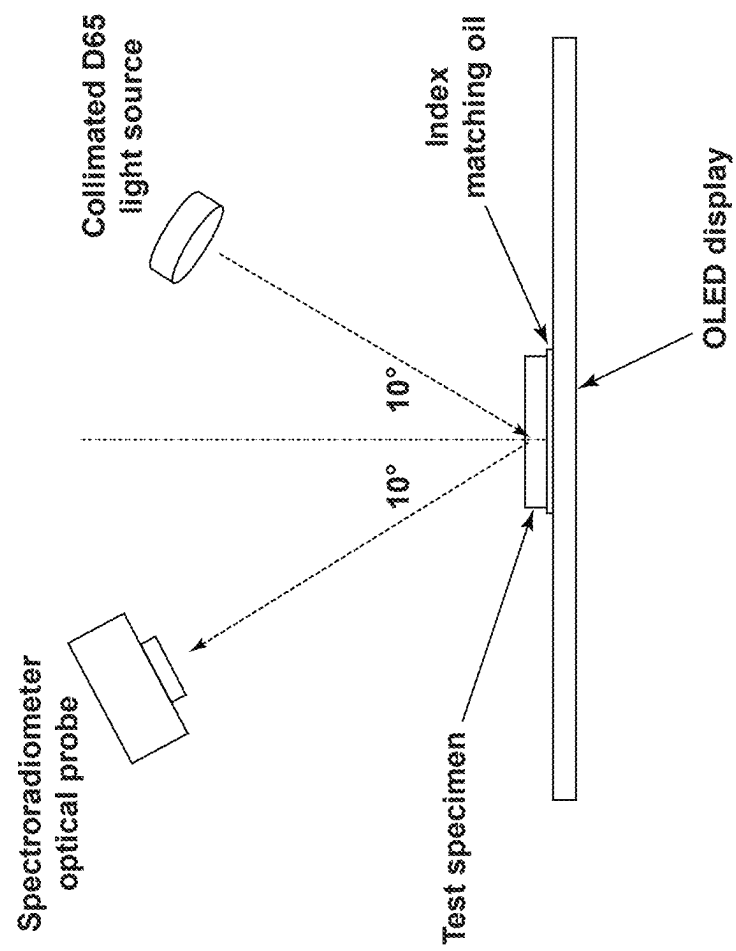


FIG. 24

**DISPLAY ARTICLES WITH ANTIGLARE
SURFACES AND THIN, DURABLE
ANTIREFLECTION COATINGS**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This patent application is a continuation of U.S. patent application Ser. No. 18/615,430, filed on Mar. 25, 2024, which is a continuation of U.S. patent application Ser. No. 17/370,350, filed on Jul. 8, 2021, now U.S. Pat. No. 11,971,519 granted on Apr. 30, 2024, which claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/049,843, entitled “DISPLAY ARTICLES WITH DIFFRACTIVE, ANTIGLARE SURFACES AND METHODS OF MAKING THE SAME,” filed Jul. 9, 2020, the entire disclosure of which is incorporated herein by reference.

[0002] The present application relates to, but does not claim priority to, commonly owned and assigned U.S. patent application Ser. No. 17/369,279 (D31977), entitled “TEXTURED REGION TO REDUCE SPECULAR REFLECTANCE INCLUDING A LOW REFRACTIVE INDEX SUBSTRATE WITH HIGHER ELEVATED SURFACES AND LOWER ELEVATED SURFACES AND A HIGH REFRACTIVE INDEX MATERIAL DISPOSED ON THE LOWER ELEVATED SURFACES” and filed on Jul. 7, 2021; U.S. patent application Ser. No. 17/369,301 (D31038/32632), entitled “ANTI-GLARE SUBSTRATE FOR A DISPLAY ARTICLE INCLUDING A TEXTURED REGION WITH PRIMARY SURFACE FEATURES AND SECONDARY SURFACE FEATURES IMPARTING A SURFACE ROUGHNESS THAT INCREASES SURFACE SCATTERING” and filed on Jul. 7, 2021; U.S. patent application Ser. No. 17/369,315 (D32630/32632), entitled “TEXTURED REGION OF A SUBSTRATE TO REDUCE SPECULAR REFLECTANCE INCORPORATING SURFACE FEATURES WITH AN ELLIPTICAL PERIMETER OR SEGMENTS THEREOF, AND METHOD OF MAKING THE SAME” and filed on Jul. 7, 2021; and U.S. patent application Ser. No. 17/370,311 (D32647), entitled “DISPLAY ARTICLES WITH DIFFRACTIVE, ANTIGLARE SURFACES AND THIN, DURABLE ANTIREFLECTION COATINGS” and filed on Jul. 8, 2021. The entire disclosures of each of the foregoing U.S. patent applications, publications and patent documents are incorporated herein by reference.

FIELD

[0003] The present disclosure generally relates to display articles with textured, antiglare surfaces (e.g., diffractive, roughened, and other textured morphologies) with thin, durable antireflective coatings, particularly display articles comprising a substrate with one or more primary surfaces with a textured surface region having antiglare characteristics and an antireflective coating thereon.

BACKGROUND

[0004] Antiglare surfaces are often used in display devices such as LCD screens, tablets, smartphones, OLEDs and touch screens to avoid or reduce specular reflection of ambient light. In many display devices, these antiglare surfaces are formed by providing a level of roughness to one or more surfaces of the glass and/or films on the glass to

spread and scatter incident light. Antiglare surfaces in the form of a roughened glass surface are often used on the front surfaces of these display devices to reduce the apparent visibility of external reflections from the display and improve readability of the display under differing lighting conditions.

[0005] Conventional approaches for imparting antiglare properties on glass substrates employed in display articles have been successful to a certain degree. Typically, these approaches have been used to employ a randomized surface roughness profile in the surfaces of the substrate and/or within films on such substrates. These approaches, however, have resulted in glass substrates and display articles with high haze and/or high display sparkle. The high haze levels can reduce display contrast by scattering high-angle light towards the user, making black levels appear ‘washed out’. The high display sparkle levels are unwanted random variations in pixel brightness resulting in a grainy appearance that shifts with viewing angle.

[0006] In addition, cover articles are often used to protect devices within electronic products, to provide a user interface for input and/or display, and/or for many other functions. Such products include mobile devices, for example smart phones, smart watches, mp3 players and computer tablets. These display articles may also benefit from some transparency, scratch-resistance, abrasion resistance or a combination thereof. These applications often demand scratch-resistance and strong optical performance characteristics, in terms of maximum light transmittance and minimum reflectance. Furthermore, for some cover applications it is beneficial that the color exhibited or perceived, in reflection and/or transmission, does not change appreciably as the viewing angle is changed. In display applications, this is because, if the color in reflection or transmission changes with viewing angle to an appreciable degree, the user of the product will perceive a change in color or brightness of the display, which can diminish the perceived quality of the display. In other applications, changes in color may negatively impact the aesthetic appearance or other functional aspects of the device.

[0007] These display articles are often used in applications (e.g., mobile devices) with packaging constraints. In particular, many of these applications can significantly benefit from reductions in overall thickness, even reductions of a few percent. In addition, many of the applications that employ such display and non-display articles benefit from low manufacturing cost, e.g., through the minimization of raw material costs, minimization of process complexity and yield improvements. Smaller packaging with optical and mechanical property performance attributes comparable to existing display and non-display articles can also serve the desire for reduced manufacturing cost (e.g., through less raw material costs, through reductions in the number of layers in an anti-reflective structure, etc.).

[0008] The optical performance of cover display articles can be improved by using various anti-reflective coatings; however, known anti-reflective coatings are susceptible to wear or abrasion. Such abrasion can compromise any optical performance improvements achieved by the anti-reflective coating. Abrasion damage can include reciprocating sliding contact from counter face objects (e.g., fingers). In addition, abrasion damage can generate heat, which can degrade chemical bonds in the film materials and cause flaking and other types of damage to the cover glass. Since abrasion

damage is often experienced over a longer term than the single events that cause scratches, the coating materials disposed experiencing abrasion damage can also oxidize, which further degrades the durability of the coating.

[0009] In view of these considerations, there is a need for display articles and substrates with improved antiglare characteristics that include, but are not limited to, the suppression of specular reflection, low sparkle and low distinctness of image. Further, there is a need for these same display articles to also possess abrasion-resistance and antireflection optical properties.

SUMMARY

[0010] According to an aspect of the disclosure, a display article is provided that includes: a substrate comprising a thickness and a primary surface; a textured surface region defined by the primary surface; and an antireflective coating disposed on the textured surface region defined by the primary surface of the substrate. The textured surface region comprises a plurality of structural features and an average texture height (R_{text}) from 50 nm to 300 nm. The substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, and a transmittance haze of less than 40% from an incident angle of 0° from normal. Further, the antireflective coating comprises a plurality of alternating high refractive index and low refractive index layers. Each of the low refractive index layers comprises a refractive index of less than or equal to 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8. In addition, the article exhibits a first-surface average photopic specular reflectance (% R) of less than 0.3% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0011] According to another aspect of the disclosure, a display article is provided that includes: a substrate comprising a thickness and a primary surface; a textured surface region defined by the primary surface; and an antireflective coating disposed on the textured surface region defined by the primary surface of the substrate. The textured surface region comprises a plurality of structural features and an average texture height (R_{text}) from 50 nm to 300 nm. The substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, and a transmittance haze of less than 40% at an incident angle of 0° from normal. Further, the antireflective coating comprises a total physical thickness from 200 nm to 500 nm and a plurality of alternating high refractive index and low refractive index layers, wherein the antireflective coating comprises a total of three (3) to nine (9) layers. Each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8. In addition, each high refractive index layer comprises one of Si_3N_4 , SiN_x and SiO_xN_y . In addition, the article exhibits a first-surface average photopic specular reflectance (% R) of less than 0.3% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0012] According to a further aspect of the disclosure, a display article is provided that includes: a substrate comprising a thickness and a primary surface; a roughened surface region defined by the primary surface; and an antireflective coating disposed on the roughened surface

region defined by the primary surface of the substrate. The roughened surface region comprises a plurality of structural features and an average surface roughness (R_q) from 20 nm to 2000 nm root-mean-square (RMS) variation. The substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, and a transmittance haze of less than 40% from an incident angle of 0° from normal. Further, the antireflective coating comprises a total physical thickness from 200 nm to 500 nm and a plurality of alternating high refractive index and low refractive index layers, wherein the antireflective coating comprises a total of three (3) to nine (9) layers. Each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8. In addition, each high refractive index layer comprises one of Si_3N_4 , SiN_x and SiO_xN_y . In addition, the article exhibits a first-surface average photopic specular reflectance (% R) of less than 1% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0013] Additional features and advantages will be set forth in the detailed description which follows, and will be readily apparent to those skilled in the art from that description or recognized by practicing the embodiments as described herein, including the detailed description which follows, the claims, as well as the appended drawings.

[0014] It is to be understood that both the foregoing general description and the following detailed description are merely exemplary, and are intended to provide an overview or framework to understanding the nature and character of the disclosure as it is claimed.

[0015] The accompanying drawings are included to provide a further understanding of principles of the disclosure, and are incorporated in, and constitute a part of, this specification. The drawings illustrate one or more embodiment(s) and, together with the description, explain, by way of example, principles and operation of the disclosure. It is to be understood that various features of the disclosure disclosed in this specification and in the drawings can be used in any and all combinations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] These and other features, aspects and advantages of the present disclosure are better understood when the following detailed description of the disclosure is read with reference to the accompanying drawings, in which:

[0017] FIG. 1A is a cross-sectional, schematic view of a display article, according to an embodiment of the disclosure.

[0018] FIG. 1B is a cross-sectional, schematic view of a display article, according to an embodiment of the disclosure.

[0019] FIG. 1C is a cross-sectional, schematic view of a display article with an antireflective coating, according to an embodiment of the disclosure.

[0020] FIG. 1D is a cross-sectional, schematic view of a display article with an antireflective coating, according to an embodiment of the disclosure.

[0021] FIG. 1E is a cross-sectional, schematic view of a display article with an antireflective coating, according to an embodiment of the disclosure.

[0022] FIG. 2 is a cross-sectional, schematic view of a diffractive antiglare structure, according to an embodiment of the disclosure.

[0023] FIGS. 3A and 3B are plots of diffraction efficiency in reflection and transmission, respectively, as a function of structure depth for the diffractive antiglare structure depicted in FIG. 2, according to embodiments of the disclosure.

[0024] FIGS. 4A-4C are plots of diffraction efficiency in reflection as a function of structure depth for the diffractive antiglare structure depicted in FIG. 2 at fill fractions of 15%, 30% and 70%, respectively, according to embodiments of the disclosure.

[0025] FIG. 5 is a plot of diffraction efficiency in reflection as a function of structure depth for the diffractive antiglare surface depicted in FIG. 2 with varying incident light wavelengths, according to embodiments of the disclosure.

[0026] FIG. 6 is a schematic flow chart of a method of making a display article, according to an embodiment of the disclosure.

[0027] FIGS. 7A-7D are optical micrographs of diffractive surface regions of substrates employed in display articles, according to embodiments of the disclosure.

[0028] FIG. 8 is a plot of etch depth as a function of etch time to form two structural features of diffractive surface regions of substrates employed in display articles, according to embodiments of the disclosure.

[0029] FIGS. 9A and 9B are plots of distinctness of image (DOI) as a function of etch depth for structural features having different sizes and fill fractions, as part of diffractive surface regions of substrates employed in display articles, according to embodiments of the disclosure.

[0030] FIGS. 10A and 10B are plots of pixel power deviation (PPD_{140}) and haze as a function of etch depth for structural features having different sizes and fill fractions, as part of diffractive surface regions of substrates employed in display articles, according to embodiments of the disclosure.

[0031] FIGS. 10C and 10D are plots of pixel power deviation (PPD_{140}) and haze as a function of etch depth for structural features having different sizes and fill fractions, as part of diffractive surface regions of substrates employed in display articles, according to embodiments of the disclosure.

[0032] FIG. 11A is an optical image and surface height distribution bar of a diffractive surface region with a first set of structural features having a depth of about 150 nm and a fill fraction of about 50%, according to an embodiment of the disclosure.

[0033] FIG. 11B is an angular plot with amplitude of reflectance vs. reflectance angle in degrees for three articles having differing diffractive surface regions, according to embodiments of the disclosure.

[0034] FIG. 12 is an angular plot with amplitude of reflectance vs. reflectance angle in degrees for a display article having a diffractive surface region, according to an embodiment of the disclosure.

[0035] FIGS. 13A and 13B are optical images of a display article with a diffractive surface region, before and after removal of a mask and etchant employed in a method of making a display article, according to an embodiment of the disclosure.

[0036] FIG. 14 is a plot of distinctness of image (DOI) as a function of etch depth for structural features of a diffractive surface region of a display article, according to an embodiment of the disclosure.

[0037] FIG. 15 is a schematic plot of seven structural features of a diffractive surface region, as arranged in a hexagonal pattern to depict hexagonality (H), according to an embodiment of the disclosure.

[0038] FIGS. 16A and 16B are nearest neighbor distribution and pattern periodogram plots, respectively, of a diffractive surface region of a display article, according to an embodiment of the disclosure.

[0039] FIGS. 17A and 17B are nearest neighbor distribution and pattern periodogram plots, respectively, of a diffractive surface region of a display article, according to an embodiment of the disclosure.

[0040] FIG. 18A is a plan view of an exemplary electronic device incorporating any of the articles disclosed herein.

[0041] FIG. 18B is a perspective view of the exemplary electronic device of FIG. 18A.

[0042] FIG. 19 is a plot of elastic modulus and hardness vs. indentation depth of a display article, according to an embodiment of the disclosure.

[0043] FIG. 20A is a plot of first-surface specular reflectance vs. wavelength of display articles according to embodiments of the disclosure and comparative display articles.

[0044] FIGS. 20B and 20C are schematic illustrations of specular reflectance measurements of display articles according to embodiments of the disclosure and comparative display articles.

[0045] FIG. 21 is a plot of two-surface total transmittance vs. wavelength of display articles, according to embodiments of the disclosure.

[0046] FIGS. 22A-22C are plots of first-surface specular reflectance vs. wavelength for various angles of incidence, according to embodiments of the disclosure.

[0047] FIG. 23 is a plot of two-surface specular reflectance vs. wavelength for a degree of incidence of 6°, according to embodiments of the disclosure.

[0048] FIG. 24 is a schematic of an optical setup for measuring optical properties of the display articles of the disclosure.

DETAILED DESCRIPTION

[0049] In the following detailed description, for purposes of explanation and not limitation, example embodiments disclosing specific details are set forth to provide a thorough understanding of various principles of the present disclosure. However, it will be apparent to one having ordinary skill in the art, having had the benefit of the present disclosure, that the present disclosure may be practiced in other embodiments that depart from the specific details disclosed herein. Moreover, descriptions of well-known devices, methods and materials may be omitted so as not to obscure the description of various principles of the present disclosure. Finally, wherever applicable, like reference numerals refer to like elements.

[0050] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0051] Directional terms as used herein—for example “up,” “down,” “right,” “left,” “front,” “back,” “top,” “bottom”—are made only with reference to the figures as drawn and are not intended to imply absolute orientation.

[0052] Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is in no way intended that an order be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps or operational flow; plain meaning derived from grammatical organization or punctuation; the number or type of embodiments described in the specification.

[0053] As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to a “component” includes aspects having two or more such components, unless the context clearly indicates otherwise.

[0054] As used herein, an “average texture height (R_{text})” is a characteristic of the structural features of the textured surface region of a primary surface of a substrate of the display articles of the disclosure and reported in units of nanometers (nm). Further, for a textured surface region that comprises a roughened surface region (e.g., as produced through an etching and/or sandblasting process), R_{text} is defined as the average surface roughness (R_q) of the roughened surface region and can be reported in units of root-mean-squared (RMS) nanometers (nm). For a textured surface region that comprises a diffractive surface region, as described in this disclosure, R_{text} is defined as the average difference in height between the two heights or depths of the structural features (e.g., pillars, holes, etc.) associated with the diffractive surface region.

[0055] As used herein, the “Berkovich Indenter Hardness Test” includes measuring the hardness of a material on a surface thereof by indenting the surface with a diamond Berkovich indenter. The Berkovich Indenter Hardness Test includes indenting the air-side surface **61** of an antireflective coating **60** of a display article **100** of the disclosure (see FIGS. 1C-1E and corresponding description) with the diamond Berkovich indenter to form an indent to an indentation depth in the range from about 50 nm to about 1000 nm (or the entire thickness of the anti-reflective coating or layer, whichever is less) and measuring the hardness from this indentation at various points along the entire indentation depth range, along a specified segment of this indentation depth (e.g., in the depth range from about 100 nm to about 500 nm), or at a particular indentation depth (e.g., at a depth of 100 nm, at a depth of 500 nm, etc.) generally using the methods set forth in Oliver, W. C.; Pharr, G. M., “An improved technique for determining hardness and elastic modulus using load and displacement sensing indentation experiments”, *J. Mater. Res.*, Vol. 7, No. 6, 1992, 1564-1583; and Oliver, W. C. and Pharr, G. M., “Measurement of Hardness and Elastic Modulus by Instrument Indentation: Advances in Understanding and Refinements to Methodology”, *J. Mater. Res.*, Vol. 19, No. 1, 2004, 3-20. Further, when hardness is measured over an indentation depth range (e.g., in the depth range from about 100 nm to about 500 nm), the results can be reported as a maximum hardness

within the specified range, wherein the maximum is selected from the measurements taken at each depth within that range. As used herein, “hardness” and “maximum hardness” both refer to as-measured hardness values, not averages of hardness values. Similarly, when hardness is measured at an indentation depth, the value of the hardness obtained from the Berkovich Indenter Hardness Test is given for that particular indentation depth.

[0056] Aspects of the disclosure generally pertain to display articles with a combination of antireflective (AR) and antiglare (AG) optical properties, along with mechanical strength and wear resistance. These display articles advantageously possess lower first surface specular reflectance levels (e.g., as low as 0.015%) as compared to articles solely with AR or AG properties and characteristics, which also facilitates higher display contrast ratios, color gamut and neutral reflected color levels. More particularly, the display articles have one or more textured, AG substrate surfaces (e.g., diffractive, roughened, and other textured morphologies) with a thin, durable multilayer AR coating. Further, as the AG textured surface region of these articles can scatter light in both transmission and reflection, it can also reduce the appearance of buried reflections in the display. Moreover, these display articles and substrates employ a textured surface region with antiglare properties, such as low pixel power deviation (PPD_{140}) and low transmitted haze. These display articles also include an AR coating with a plurality of alternating high refractive index and low refractive index layers which, in some embodiments, can enable the article to exhibit a maximum hardness of 8 GPa or greater, as measured by a Berkovich Indenter Hardness Test along an indentation depth of 50 nm or greater. The textured surface regions can enable the display articles employing them to exhibit a PPD_{140} of less than 5%, and a transmitted haze of less than 40%.

[0057] Aspects of the disclosure also generally pertain to display articles with diffractive, antiglare surfaces and methods of making the same, particularly display articles comprising a substrate with one or more primary surfaces with a diffractive surface region and antiglare characteristics. In general, the display articles and substrates of the disclosure employ an engineered diffractive surface region with antiglare properties, such as low distinctness of image (DOI), low pixel power deviation (PPD_{140}) and low transmitted haze. Each of these antiglare properties is desirable for display applications, and conventional approaches have not achieved this combination of antiglare properties. The diffractive surface regions, according to aspects of the disclosure, possess structural features, such as holes and/or pillars, with a diameter of less than 100 μm , a pitch of less than 125 μm and a fill fraction of 40 to 55%. The diffractive surface regions, as including these holes and/or pillars, can enable the display articles employing them to exhibit a first-surface reflectance DOI of less than 80%, a PPD_{140} of less than 4%, and a transmitted haze of less than 20%. Further, these properties are achievable without the additional presence of any antireflective coating structures over the diffractive surface region. Further, the diffractive surface region, in some embodiments, can have a multimodal distribution (e.g., a bimodal distribution) of surface heights with a height and/or depth of from 120 to 200 nm, which can reduce specular reflectance through diffractive interference.

[0058] The display articles of the disclosure, as including a diffractive surface region, offer several advantages over

display articles with conventional approaches to achieving antireflective characteristics. For example, the display articles of the disclosure can suppress specular reflectance by a factor of 10× or more using diffractive light scattering, while also achieving a combination of low haze, low sparkle and high mechanical durability. The high mechanical durability is associated with the relatively low aspect ratio of the structural features of the diffractive surface region. In addition, some display articles according to the disclosure employ a diffractive surface region and a multilayer antireflective coating structure to achieve specular reductions of greater than 20×, 50× or even 100×. Another advantage of the display articles of the disclosure is that the planar step-like and semi-planar morphology, together with the controlled structure depths of less than 1 micron, or less than 250 nm, of the diffractive surface region allows them to be easily fabricated with much lower consumption of glass material and etching chemicals (such as HF) compared to conventional etched, antiglare glass substrates, leading to less environmental waste and potential cost benefits. Various processes can be employed to create these structures (e.g., organic mask and etching, organic mask and vapor deposition, organic mask and liquid phase deposited oxide), which can aid in maintaining low manufacturing costs. A further advantage of these display articles is that they can exhibit a combination of antiglare, optical properties not achievable from conventional antiglare approaches. For example, the display articles of the disclosure, as incorporating a diffractive surface region, have achieved a DOI of less than 80%, a PPD₁₄₀ of less than 2% and a haze of less than 5%.

[0059] Referring to FIGS. 1A and 1B, a display article 100 is depicted as including a substrate 10 with a plurality of primary surfaces 12 and 14, and a thickness 13. The primary surface 12 of the substrate 10 also includes a diffractive surface region 30a. Hence, the primary surface 12 has a diffractive surface region 30a defined thereon such that the diffractive surface region 30a is formed from or otherwise part of the substrate 10, as shown in FIG. 1A. In some implementations (not shown), a diffractive surface region 30a can also be defined by the primary surface 14 of the substrate 10. Further, in some implementations, the diffractive surface region 30a is defined by both of primary surfaces 12 and 14. As also depicted in FIG. 1A, the diffractive surface region 30a includes a plurality of structural features 20 that comprises a plurality of different heights and/or depths in a multimodal distribution.

[0060] Referring again to the diffractive surface region 30a shown in FIGS. 1A and 1B, it can describe a surface comprised of circular pillars or holes (i.e., the structural features 20) with a fixed set of a few possible discrete height or depth levels above or below the primary surface 12 of the substrate 10. Each feature 20, in some instances, can have a planar top characterized by a single height or depth. These features 20 may have the same or a finite number of different diameters. The placement of the features 20 can be random, but this is by design or otherwise purposely engineered and not a random function of the manufacturing process. The manufacturing process can use some form of precision, engineered masking to produce the designed feature shapes exactly (to within some tolerance). As used herein, a diffractive surface region 30a with a plurality of structural features 20 (as shown in exemplary form in FIGS. 1A and 1B) with a plurality of different heights in a “multimodal distribution” means that diffractive surface region 30a con-

sists primarily of two (e.g., a bimodal distribution), three (e.g., a trimodal distribution), four, five or more distinct and purposely engineered average or primary heights, and each of these average or primary heights consists of distributions whose widths are less than or comparable to the vertical separation between the heights. Referring again to FIGS. 1A and 1B, the exemplary diffractive surface region 30a comprises a plurality of heights and/or depths in a bimodal distribution. Further, as used herein, a “bimodal distribution” means that the diffractive surface region 30a consists primarily of two distinct and purposely engineered average or primary heights, and each of these average or primary heights consists of distributions whose widths are less than or comparable to the vertical separation between the heights.

[0061] In embodiments, the plurality of structural features 20 includes pillars and/or holes and these pillars and/or holes make up a multimodal distribution of surface heights and/or depths. According to some implementations, the diffractive surface region 30a can comprise a two-dimensional array of circular, square, hexagonal, polygonal, or irregular structural features 20. Further, these structural features 20 can be arranged in an ordered or a semi-ordered array—essentially, any of various array schemes that are reproducibly fabricated and do not depend on manufacturing process randomness for their function. As such, in some embodiments of the display article 100 depicted in FIGS. 1A and 1B, the diffractive surface region 30a comprises a plurality of structural features 20 that comprises a plurality of different heights in a multimodal distribution and is distributed across the surface region 30a in a semi-ordered or an ordered array.

[0062] Referring again to the display article 100 of FIGS. 1A and 1B, the substrate 10 can exhibit a sparkle of less than 4%, as measured by PPD₁₄₀ at an incident angle of 0° from normal. The substrate 10 can also exhibit a DOI of less than 80% at an incident angle of 20° from normal. Further, the substrate 10 of the display article 100 can exhibit a transmittance haze of less than 20% from an incident angle of 0° from normal. Embodiments of the display article 100 depicted in FIGS. 1A and 1B can also exhibit a combination of these optical properties.

[0063] According to some implementations of the display article 100 depicted in FIGS. 1A and 1B, the multimodal distribution of the diffractive surface region 30a further includes a first portion of structural features 22a, 22a' at a first average height 24a (or, interchangeably referred to as a first average depth 24a) and a second portion of structural features 22b, 22b' at a second average height 24b (or, interchangeably referred to as a first average depth 24b). According to embodiments of the display article 100, the first portion of structural features 22a, 22a' can be pillars 22a, as shown in exemplary form in FIG. 1A. According to embodiments of the display article 100, the first portion of structural features 22a, 22a' can be holes 22a', as shown in exemplary form in FIG. 1B. According to some of these embodiments, from a structural standpoint, the second portion of structural features 22b, 22b' can be a set of ligaments 22b (e.g., as shown in FIG. 1A situated between the pillars 22a) or mesas 22b' (e.g., as shown in FIG. 1B situated between the holes 22a') between the first portion of structural features 22a, 22a'. According to implementations, the ligaments 22b can be the ligaments, matrixes or other comparable structure between the first portion of the structural features 22a, 22a', when the first portion is configured as pillars 22a. Further, the mesas 22b' can be mesas, pla-

teaus, matrixes or other comparable structure between the first portion of the structural features **22a**, **22a'**, when the first portion is configured as holes **22a'**. It should also be understood, however, that the second portion of structural features **22b**, **22b'**, like the first portion of structural features **22a**, **22a'**, includes a distribution of surface heights with a second average height **24b**. Further, the first average height **24a** of the pillars **22a** or the first average depth **24a** of the holes **22a'**, depending on the configuration of the diffractive surface region **30a**, can range from about 25 nm to about 300 nm, from about 50 nm to about 250 nm, from about 75 nm to about 225 nm, from about 100 nm to about 200 nm, from about 120 nm to about 180 nm, or from about 130 nm to about 170 nm.

[0064] As noted earlier, the display article **100** depicted in FIGS. **1A** and **1B** includes a substrate **10** with a diffractive surface region **30a** that can include a plurality of structural features **20** of different heights in a multimodal distribution. This multimodal distribution can have a plurality of surface height modes, e.g., the distribution may be bimodal (e.g., with a first portion of structural features **22a**, **22a'** and a second portion of structural features **22b**, **22b'**), tri-modal, four-modal, five-modal, etc. In embodiments, the diffractive surface region **30a** is configured such that each of these modes is characterized by a distinct peak of surface height vs. area fraction within the distribution of surface heights. These peaks may be distinguished by a decrease in area fraction of at least 20%, at least 50% or at least 80% from the peak surface height value between the distinct peaks associated with each of the modes. Further, the peaks of each of the modes may have a varying width, and the area fraction does not need to drop to zero between the peaks of the distribution. In some embodiments, however, the area fraction for heights in between each of the peaks on a surface height vs. area chart may drop to zero or close to zero.

[0065] Referring again to the display article **100** depicted in FIGS. **1A** and **1B**, the multimodal distribution of the diffractive surface region **30a** can be further characterized such that the difference between the first average height **24a** and the second average height **24b** is from about 25 nm to about 300 nm, from about 50 nm to about 250 nm, from about 75 nm to about 225 nm, from about 100 nm to about 200 nm, from about 120 nm to about 180 nm, or from about 130 nm to about 170 nm. For example, the difference between the first average height **24a** and the second average height **24b** can be about 25 nm, 50 nm, 75 nm, 100 nm, 125 nm, 150 nm, 175 nm, 200 nm, 225 nm, 250 nm, 275 nm, 300 nm, and all height differences between the foregoing levels. In some embodiments, the difference between the first average height **24a** and the second average height **24b** may be in a range that corresponds to about $\frac{1}{4}$ of the wavelength of visible light in air, or odd multiples of the $\frac{1}{4}$ of the wavelength of visible light.

[0066] Referring again to the display article **100** depicted in FIGS. **1A** and **1B**, embodiments are configured such that the diffractive surface region **30a** includes a first planar region **21a** that corresponds to the first average height **24a** of the first portion of structural features **22a**, **22a'** and a second planar region **21b** that corresponds to the second average height **24b** of the second portion of structural features **22b**, **22b'**. That is, each of the first and second planar regions **21a** and **21b**, respectively, is planar in the sense that these regions have close to the same surface height (i.e.,

within one mode of the surface height vs. area fraction distribution of the surface heights of the plurality of structural features **20**). Further, each of these planar regions **21a** and **21b** can be characterized by a surface height variation (or roughness) within the planar region of less than 50 nm root-mean-square (RMS) variation, less than 20 nm RMS, less than 10 nm RMS, less than 5 nm RMS, less than 2 nm RMS, or less than 1 nm RMS. For example, each of these planar regions **21a** and **21b** can be characterized by a surface height variation from 0.1 nm RMS to 50 nm RMS, from 0.1 nm RMS to 20 nm RMS, from 0.1 nm RMS to 10 nm RMS, or from 0.1 nm RMS to 1 nm RMS. Further, according to some embodiments, the planar regions **21a** and **21b** can also include individual sub-regions or domains (not shown) having an average area within each domain of greater than $5 \mu\text{m}^2$, greater than $10 \mu\text{m}^2$, greater than $20 \mu\text{m}^2$, greater than $50 \mu\text{m}^2$, or greater than $100 \mu\text{m}^2$.

[0067] According to some embodiments of the display article **100** depicted in FIGS. **1A** and **1B**, as noted earlier, the diffractive surface region **30a** can include two or more planar regions (e.g., first and second planar regions **21a** and **21b**). Further, each of these planar regions (e.g., plateaus, mesas, etc.) can be substantially planar, meaning that more than 50%, more than 80%, or more than 90% of the diffractive surface region **30a** is planar. In some embodiments, the total surface area of the first and second planar regions **21a** and **21b** is at least 50%, at least 60%, at least 70%, at least 80%, or at least 90% of a total surface area of the diffractive surface region **30a**.

[0068] According to implementations of the display article **100** depicted in FIGS. **1A** and **1B**, the article is characterized by a low level of sparkle. In general, the diffractive surface region **30a** associated with the substrate **10** of these articles can produce light deflections leading to an image artifact called “sparkle”. Display “sparkle” or “dazzle” is a generally undesirable side effect that can occur when introducing antiglare or light scattering surfaces into a pixelated display system such as, for example, an LCD, an OLED, touch screens, or the like, and differs in type and origin from the type of “sparkle” or “speckle” that has been observed and characterized in projection or laser systems. Sparkle is associated with a very fine grainy appearance of the display and may appear to have a shift in the pattern of the grains with changing viewing angle of the display. Display sparkle may be manifested as bright and dark or colored spots at approximately the pixel-level size scale.

[0069] As used herein, the terms “pixel power deviation” and “PPD₁₄₀” refer to the quantitative measurement for display sparkle. Further, as used herein, the term “sparkle” is used interchangeably with “pixel power deviation” and “PPD₁₄₀.” PPD₁₄₀ is calculated by image analysis of display pixels according to the following procedure. A grid box is drawn around each LCD pixel. The total power within each grid box is then calculated from charge-coupled device (CCD) camera data and assigned as the total power for each pixel. The total power for each LCD pixel thus becomes an array of numbers, for which the mean and standard deviation may be calculated. The PPD₁₄₀ value is defined as the standard deviation of total power per pixel divided by the mean power per pixel (times 100). The total power collected from each LCD pixel by the eye simulator camera is measured and the standard deviation of total pixel power (PPD₁₄₀) is calculated across the measurement area, which typically comprises about 30x30 LCD pixels.

[0070] The details of a measurement system and image processing calculation that are used to obtain PPD_{140} values are described in U.S. Pat. No. 9,411,180 entitled “Apparatus and Method for Determining Sparkle,” the salient portions of which that are related to PPD measurements are incorporated by reference herein in their entirety. Further, unless otherwise noted, the SMS-1000 system (Display-Messtechnik & Systeme GmbH & Co. KG) is employed to generate and evaluate the PPD_{140} measurements of this disclosure. The PPD_{140} measurement system includes: a pixelated source comprising a plurality of pixels (e.g., a Lenovo Z50 140 ppi laptop), wherein each of the plurality of pixels has referenced indices i and j ; and an imaging system optically disposed along an optical path originating from the pixelated source. The imaging system comprises: an imaging device disposed along the optical path and having a pixelated sensitive area comprising a second plurality of pixels, wherein each of the second plurality of pixels are referenced with indices n and n ; and a diaphragm disposed on the optical path between the pixelated source and the imaging device, wherein the diaphragm has an adjustable collection angle for an image originating in the pixelated source. The image processing calculation includes: acquiring a pixelated image of the transparent sample, the pixelated image comprising a plurality of pixels; determining boundaries between adjacent pixels in the pixelated image; integrating within the boundaries to obtain an integrated energy for each source pixel in the pixelated image; and calculating a standard deviation of the integrated energy for each source pixel, wherein the standard deviation is the power per pixel dispersion. As used herein, all “ PPD_{140} ” and “sparkle” values, attributes and limits are calculated and evaluated with a test setup employing a display device having a pixel density of 140 pixels per inch (PPI) (also referred herein as “ PPD_{140} ”).

[0071] According to some implementations of the display article **100** shown in FIGS. 1A and 1B, the substrate **10** exhibits a sparkle of less than 4%, 3.5%, 3%, 2.5%, 2%, 1.5%, 1%, 0.5%, and all sparkle threshold values between the foregoing levels, as measured by PPD_{140} at an incident angle of 0° from normal with a 140 ppi LCD display. For example, the substrate **10** can exhibit a sparkle of 3.5%, 3.25%, 3%, 2.75%, 2.5%, 2.25%, 2%, 1.75%, 1.5%, 1.25%, 1%, 0.75%, 0.5%, and all sparkle values between the foregoing levels, as measured by PPD_{140} at an incident angle of 0° from normal.

[0072] Referring again to the display article **100** depicted in FIGS. 1A and 1B, the article can also be configured for optimal antiglare performance, as manifested by low distinctness of image (DOI) values. As used herein, “DOI” is equal to $100 \cdot (R_s - R_{0.3^\circ}) / R_s$, where R_s is the specular reflectance flux measured from incident light (at 20° from normal) directed onto the diffractive surface region **30a** of the display article **100** of the disclosure and $R_{0.3^\circ}$ is the reflectance flux measured from the same incident light at 0.3° from the specular reflectance flux, R_s . Unless otherwise noted, the DOI values and measurements reported in this disclosure are obtained according to the ASTM D5767-18, entitled “Standard Test Method for Instrumental Measurement of Distinctness-of-Image (DOI) Gloss of Coated Surfaces using a Rhopoint IQ Gloss Haze & DOI Meter” (Rhopoint Instruments Ltd.). According to some embodiments of the display article **100** shown in FIGS. 1A and 1B, the substrate **10** exhibits a DOI of less than 90%, 85%, 80%, 75%, 70%,

65%, 60%, 55%, 50%, 45%, 40%, and all DOI threshold values between the foregoing levels, as measured at an incident angle of 20° from normal. For example, the substrate **10** can exhibit a DOI of 87.5%, 85%, 82.5%, 80%, 77.5%, 75%, 72.5%, 70%, 67.5%, 65%, 62.5%, 60%, 57.5%, 55%, 52.5%, 50%, 47.5%, 45%, 42.5%, 40%, and all DOI values between the foregoing levels, as measured at an incident angle of 20° from normal.

[0073] As used herein, the terms “transmission haze” and “haze” refer to the percentage of transmitted light scattered outside an angular cone of about $\pm 2.5^\circ$ in accordance with ASTM D1003, entitled “Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics,” the contents of which are incorporated by reference herein in their entirety. For an optically smooth surface, transmission haze is generally close to zero. According to implementations of the display article **100** depicted in FIGS. 1A and 1B, the articles can be characterized by a haze of less than 20%. According to implementations of the display article **100** shown in FIGS. 1A and 1B, the substrate **10** can also exhibit a transmittance haze of less than 20%, 15%, 10%, 5%, and all haze threshold values between the foregoing levels, as measured at an incident angle of 0° from normal. As an example, the substrate **10** can exhibit a transmittance haze of 17%, 16%, 15%, 14%, 13%, 12%, 11%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, and all haze values between the foregoing levels, as measured at an incident angle of 0° from normal.

[0074] According to some implementations of the display article **100** depicted in FIGS. 1A and 1B, the diffractive surface region **30a** of the substrate **10** can effect a reduction in specular reflectance (R_s) and absolute specular reflectance (% R). As used herein, “specular reflectance (R_s)” is defined as the total reflected light from the first surface of the substrate **10** (e.g., primary surface **12**) within a cone of angles of $\pm 0.1^\circ$, or as the peak intensity of light within this specular angular range. Further, the Rhopoint IQ meter reports an R_s value in gloss units that is normalized to a maximum of 100 for a flat glass having an index of 1.567 illuminated at 20° angle of incidence and no back-surface reflectance. As such, a glass is known to have a first-surface absolute specular reflectance (% R) of 4.91%, R_s values in gloss units (GU) reported by the Rhopoint IQ meter can be converted to an absolute specular reflectance (% R) by multiplying by a factor of 4.91/100. As such, implementations of the display article **100** are configured such that they exhibit a reduction in the specular reflectance (R_s) or absolute specular reflectance (% R) by a factor of 2, a factor of 4, a factor of 5, or a factor of 10 or more as compared to the same surface of a substrate, but lacking the diffractive surface region **30a**. In embodiments, the substrate **10** (e.g., a glass composition having a refractive index of about 1.51), as including the diffractive surface region **30a**, can exhibit a first-surface absolute specular reflectance (% R) of less than 2%, less than 1.5%, less than 1%, less than 0.8%, less than 0.6%, less than 0.5%, less than 0.4%, or even less than 0.25%, as measured at an incident angle of 20° from normal, at wavelengths between 450 nm and 650 nm.

[0075] Referring now to FIG. 1C, a display article **100** is depicted with the same features and attributes as the display article **100** shown in FIGS. 1A and 1B, with like-numbered elements having substantially the same functions and structures. Further, the display article **100** depicted in FIG. 1C employs an antireflective coating **60** disposed on the primary

surface **12** of the substrate **10**, to further improve the antiglare effect of the display article **100**. In embodiments, the display article **100**, as depicted in FIG. 1C with an antireflective coating **60**, is particularly beneficial for end use applications requiring transmittance and some degree of transmitted scattering (e.g., for display devices that include features that result in subsurface reflectance such as touch sensor layers, thin film transistor layers, indium-tin-oxide layers, and other subsurface features). In embodiments, the antireflective coating **60** is configured according to a multilayer antireflective coating or film structure, as prepared to effect antiglare, optical properties for the article employing it. As an example, the antireflective coating **60** includes a plurality of alternating low refractive index and high refractive index layers **62** and **64**, respectively. In some implementations, the antireflective coating **60** can have a total of three (3) to nine (9) layers, e.g., 3 layers, 4 layers, 5 layers, 6 layers, 7 layers, 8 layers or 9 layers in total. Further, each of the low refractive index layers **62** comprises a refractive index of less than or equal to about 1.8 and about the same or greater than a refractive index of the substrate **10**; and each of the high refractive index layers **64** comprises a refractive index that is greater than 1.8. In this configuration, the display article **100** can exhibit a first-surface absolute specular reflectance (% R) of less than 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, or even less than 0.1%, as measured at an incident angle of 0-20° from normal. Further, according to some embodiments, the display article **100** can exhibit a first-surface absolute specular reflectance (% R) of less than 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, or even less than 0.1%, as measured at an incident angle of 0-20° from normal at one or more wavelengths between 450 nm and 650 nm.

[0076] The antiglare effects afforded by the addition of the antireflective coating **60** in the display article **100** shown in FIG. 1C should be approximately additive in a proportional sense. That is, the diffractive surface region **30a** of a display article **100** shown in FIGS. 1A and 1B can lower the specular reflectance of the primary surface **12** of the substrate **10** by a factor of 10, and the addition of the antireflective coating **60** can further lower the specular reflectance by a factor of 10, leading to a reduction in the specular reflectance of the display article **100** of FIG. 1C of about a factor of 100. As such, it is believed that the display article **100**, as configured according to FIG. 1C, can, according to some embodiments, exhibit a first-surface absolute specular reflectance (% R) of less than 0.1%, less than 0.08%, less than 0.06%, less than 0.05%, less than 0.04%, or even less than 0.025%, as measured at an incident angle of 0-20° from normal at one or more wavelengths between 450 nm and 650 nm.

[0077] In an exemplary embodiment, as shown in FIG. 1C, a low refractive index layer **62** of the antireflective coating **60** is disposed directly on the diffractive surface region **30a** of the primary surface **12** of the substrate **10**. Further, according to the embodiment shown in FIG. 1C, a low refractive index layer **62** is disposed as the top-most layer of the display article **100** and the antireflective coating **60** has a total of five (5) layers, i.e., a multilayer structure with five (5) low and high refractive index layers: **62/64/62/64/62**. Suitable materials for the low refractive index layer **62** include: SiO₂, Al₂O₃, GeO₂, SiO_x, AlO_xN_y, SiO_xN_y, SiAl_xO_xN_y, MgO and MgAl₂O₄. Suitable materials for the high refractive index layer **64** include: Al₂O₃, AlO_xN_y, SiO_xN_y, SiAl_xO_xN_y, AlN, SiN_x, Si₃N₄, Nb₂O₅, Ta₂O₅, HfO₂,

TiO₂, ZrO₂, Y₂O₃, and diamond-like carbon. Further, the thicknesses of each of the low and high refractive index layers **62** and **64** can range from about 1 nm to about 250 nm and the total thickness of the antireflective coating **60** can range from about 5 nm to 3000 nm, 5 nm to 2500 nm, 5 nm to 2000 nm, from 5 nm to 1500 nm, from 5 nm to 1000 nm, from 5 nm to 750 nm, from 5 nm to 500 nm, from 5 nm to 450 nm, from 5 nm to 400 nm, from 5 nm to 350 nm, from 5 nm to 300 nm, from 5 nm to 275 nm, from 5 nm to 260 nm, from 5 nm to 250 nm, from 100 nm to 500 nm, from 100 nm to 400 nm, from 100 nm to 350 nm, from 100 nm to 300 nm, from 100 nm to 275 nm, from 100 nm to 250 nm, from 200 nm to 500 nm, from 200 nm to 400 nm, from 200 nm to 350 nm, from 200 nm to 300 nm, from 200 nm to 275 nm, from 200 nm to 250 nm, from 250 nm to 350 nm, from 250 nm to 340 nm, and all thickness values within the foregoing ranges.

[0078] According to some embodiments of the display article **100** depicted in FIG. 1C, the antireflection coating **60** can be configured with a scratch resistant layer. In some embodiments, the scratch resistant layer can be the farthest, middlemost or lowest high refractive index layer **64** from the substrate **10**. In some implementations, the scratch resistant layer is the thickest high refractive index layer **64** in the antireflection coating **60** and may also comprise one of Si₃N₄, SiN_x and SiO_xN_y. The scratch resistant layer can also have a physical thickness from 50 nm to 2000 nm, from 50 nm to 1000 nm, from 50 nm to 500 nm, from 50 nm to 400 nm, from 50 nm to 300 nm, from 50 nm to 200 nm, 50 nm to 150 nm, 75 nm to 175 nm, 100 nm to 160 nm, and all thickness values within the foregoing ranges.

[0079] According to an implementation of the display article **100** depicted in FIG. 1C, the structure of the antireflection coating **60** can be configured with a design as listed below in Table 1A. In this implementation, the antireflection coating **60** has a total thickness of 260.5 nm, and employs a scratch resistant layer with a thickness of 105.9 nm and a SiO_xN_y composition.

TABLE 1A

Layer	Material	Refractive Index (@ 550 nm)	Extinction Coefficient (@ 550 nm)	Thickness (nm)
Air		1.000	0	
1	SiO ₂	1.461	0	85.9
2	SiO _x N _y	1.997	3.6e-4	105.9
3	SiO ₂	1.461	0	22.3
4	SiO _x N _y	1.997	3.6e-4	21.4
5	SiO ₂	1.461	0	25.00
Substrate	ion-exchanged strengthened glass	1.510		
Total Thickness				260.5

[0080] According to another implementation of the display article **100** depicted in FIG. 1C, the structure of the antireflection coating **60** can be configured with a design as listed below in Table 1B. In this implementation, the antireflection coating **60** has a total thickness of 338.4 nm, and employs a scratch resistant layer with a thickness of 158.5 nm and a SiN_x composition.

TABLE 1B

Layer	Material	Refractive Index (@ 550 nm)	Extinction Coefficient (@ 550 nm)	Thickness (nm)
Air		1.000	0	
1	SiO ₂	1.471	0	94.6
2	SiN _x	2.028	0.00057	158.5
3	SiO ₂	1.456	0	37.2
4	SiN _x	2.041	0.00057	23.1
5	SiO ₂	1.449	0	25.0
Substrate	ion-exchanged, strengthened glass	1.508		
Total Thickness				338.4

[0081] According to a further implementation of the display article **100** depicted in FIG. 1C, the structure of the antireflection coating **60** can be configured with a design as listed below in Table 1C. In this implementation, the anti-reflection coating **60** has a total thickness of 301.48 nm, and employs a scratch resistant layer with a thickness of 135.0 nm and a SiN_x composition.

TABLE 1C

Layer	Material	Refractive Index (@ 550 nm)	Thickness (nm)
Air		1	
1	SiO ₂	1.477	87.7
2	SiN _x	2.041	135.0
3	SiO ₂	1.477	38.6
4	SiN _x	2.041	15.2
5	SiO ₂	1.477	25.0
Substrate	ion-exchanged, strengthened glass	1.506	
Total thickness			301.48

[0082] According to a further implementation of the display article **100** depicted in FIG. 1C, the structure of the antireflection coating **60** can be configured with a design as listed below in Table 1D. In this implementation, the anti-reflection coating **60** has a total thickness of 295.5 nm, and employs a scratch resistant layer with a thickness of 128.0 nm and a SiN_x composition.

TABLE 1D

Layer	Material	Refractive Index (@ 550 nm)	Thickness (nm)
Air		1	
1	SiO ₂	1.465	90.4
2	SiN _x	2.043	128.0
3	SiO ₂	1.465	37.8
4	SiN _x	2.043	14.4
5	SiO ₂	1.465	25.0
Substrate	ion-exchanged, strengthened glass	1.506	
Total thickness			295.5

[0083] Referring now to FIG. 1D, a display article **100** is depicted with the same features and attributes as the display article **100** shown in FIG. 1C, with like-numbered elements having substantially the same functions, structures and properties (unless otherwise noted in the following sections). The display article **100** depicted in FIG. 1D employs an antireflective coating **60** that is the same as, or substantially similar to the coating **60** employed in the article **100** in FIG. 1C, as disposed on the primary surface **12** of the substrate **10** to further improve the antiglare effect of the display article **100**. However, the display article **100** depicted in FIG. 1D employs a roughened surface region **30b** defined by the primary surface **12** of the substrate **10**, as opposed to the diffractive surface region **30a** employed by the article **100** shown in FIG. 1C. Hence, the antireflective coating **60** of the display article **100** shown in FIG. 1D is disposed on the roughened surface region **30b**.

[0084] Referring again to FIG. 1D, the roughened surface region **30b** comprises a plurality of structural features **20**. In some aspects, the roughened surface region **30b** is a surface with randomly or semi-randomly formed structural features **20** that are formed through various etching and/or sandblasting processes directed at the primary surface **12** of the substrate **10**, as understood by those of ordinary skill in the art. According to some embodiments, the majority of the structural features **20** of the roughened surface region **30b** have lateral etched feature dimensions (i.e., X-Y dimensions) that range from 1 μm to 125 μm, 1 μm to 100 μm, 1 μm to 75 μm, 1 μm to 50 μm, 1 μm to 40 μm, 1 μm to 30 μm, 5 μm to 125 μm, 5 μm to 100 μm, 5 μm to 75 μm, 5 μm to 60 μm, 5 μm to 50 μm, 5 μm to 40 μm, 5 μm to 30 μm, 10 μm to 60 μm, 10 μm to 100 μm, and lateral dimensions within the foregoing ranges.

[0085] In embodiments of the display article **100** shown in FIG. 1D, the roughened surface region **30b** possesses an average surface roughness (R_q) from 20 nm to 2000 nm RMS variation. According to further implementations, the roughened surface region **30b** possesses an average surface roughness (R_q) in RMS variation from 10 nm to 2500 nm, 10 nm to 2000 nm, 10 nm to 1500 nm, 20 nm to 2500 nm, 20 nm to 2000 nm, 20 nm to 1500 nm, 50 nm to 2500 nm, 50 nm to 2000 nm, 50 nm to 1500 nm, 50 nm to 1000 nm, 50 nm to 500 nm, 50 nm to 250 nm, 100 nm to 2500 nm, 100 nm to 2000 nm, 100 nm to 1500 nm, 100 nm to 1000 nm, 100 nm to 500 nm, 100 nm to 250 nm, and all surface roughness values between the foregoing ranges.

[0086] In some implementations of the display article **100** shown in FIG. 1D, the roughened surface region **30b** can be described such that its structural features **20** have a first average height and a second average height. The first average height corresponds to the average height of the peaks of the roughened surface region **30b** and the second average height corresponds to the depth of the troughs between the peaks. In such configurations, the difference between the first and second average heights of the roughened surface region **30b** can range from 10 nm to 500 nm, 10 nm to 250 nm, 25 nm to 500 nm, 25 nm to 250 nm, 50 nm to 500 nm, 50 nm to 250 nm, 50 nm to 150 nm, 100 nm to 200 nm, 120 nm to 200 nm, and all height differences between the foregoing ranges.

[0087] Referring again to the display article **100** shown in FIG. 1D, in this configuration, the display article **100** can exhibit a first-surface average specular (or average photopic specular) reflectance (% R) of less than 1%, 0.9%, 0.8%,

0.7%, 0.6%, 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, or even less than 0.1%, as measured at any incident angle from about 5° to 20° from normal. Further, according to some embodiments, the display article **100** can exhibit a first-surface average specular (or average photopic specular) reflectance (% R) of less than 1%, 0.9%, 0.8%, 0.7%, 0.6%, 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, or even less than 0.1%, as measured at an incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0088] The display article **100** shown in FIG. 1D can also exhibit sparkle of less than 5%, as measured by pixel power deviation (PPD₁₄₀) at an incident angle of 0° from normal, according to some embodiments. In some implementations, the display article **100** shown in exemplary form in FIG. 1D can exhibit sparkle levels of less than 5%, 4.5%, 4%, 3.5%, 3%, 2.5%, 2%, 1.5%, 1%, or 0.5%, as measured by pixel power deviation (PPD₁₄₀) at an incident angle of 0° from normal. In some aspects, the display article **100** of FIG. 1D exhibits sparkle levels from 5% to 0.1%, from 5% to 0.5%, from 5% to 1%, and all sparkle levels in the foregoing ranges.

[0089] Referring again to the display article **100** of FIG. 1D, the article can exhibit transmittance haze values of less than 40%, according to embodiments. In some implementations, the display article **100** shown in exemplary form in FIG. 1D can exhibit transmittance haze levels of less than 40%, 35%, 30%, 25%, 20%, 15%, 10%, or 5%. In some embodiments, the display article **100** of FIG. 1D can exhibit transmittance haze levels from 50% to 1%, 50% to 5%, 40% to 1%, 40% to 5%, 30% to 1%, 30% to 5%, 20% to 1%, 20% to 5%, and all transmittance haze values in the foregoing ranges.

[0090] The display article **100** shown in FIG. 1D can also exhibit distinctness of image (DOI) levels of 80% or less, as measured at an incident angle of 20° from normal. In some implementations, the display article **100** shown in exemplary form in FIG. 1D can exhibit DOI levels of less than 99%, 95%, 90%, 85%, 80%, 75%, 70%, 65%, 60%, or 55%, as measured at an incident angle of 20° from normal. In some aspects, the display article **100** of FIG. 1D exhibits DOI levels from 99% to 40%, from 99% to 50%, from 95% to 40%, from 95% to 50%, from 90% to 40%, from 90% to 50%, from 85% to 40%, from 85% to 50%, from 80% to 40%, and from 80% to 50%, and all DOI levels in the foregoing ranges.

[0091] The display article **100** shown in FIG. 1D can also be characterized by various color characteristics. In some embodiments, these display articles **100** exhibit ambient contrast ratios of up to 60, up to 50, or up to 40, as measured with 1000 lux. In some embodiments, these display articles **100** exhibit ambient contrast ratios of up to 15, up to 12.5, up to 10, up to 7.5, up to 5, or up to 2.5, as measured with 100 incident lux. These display articles **100** depicted in FIG. 1D can exhibit color gamut levels of at least 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, or even higher levels at a direct contrast ratio of 100 lux. In some embodiments, these display articles **100** can exhibit color gamut levels of at least 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, or even higher levels at an ambient contrast ratio of 1000 lux.

[0092] According to some implementations, the display article **100** shown in FIG. 1D can exhibit a high degree of color neutrality. In some aspects, the display article **100** can exhibit a first surface reflected color ($\sqrt{a^2+b^2}$) of less

than 0.5, 0.25, 0.1, or 0.05, at each incident angle of 6° and 20° from normal. In some embodiments, the display article **100** can exhibit a first surface reflected color ($\sqrt{a^2+b^2}$) of less than 5, 4, 3, 2, or 1, at all incident angles from 0° to 60° from normal. According to some implementations, the display article **100** can exhibit a two-surface (i.e., through both primary surfaces **12**, **14** of the substrate **10** and one antireflective coating **60** disposed on one of surfaces **12**, **14**) transmitted color ($\sqrt{a^2+b^2}$) of less than 2, 1.5, 1, 0.9, 0.8, 0.7, 0.6, or 0.5 at 0° incidence using a 2° acceptance angle.

[0093] Referring to the display articles **100** depicted in FIGS. 1C and 1D, each can exhibit a maximum hardness of 8 GPa or greater as measured on the air-side surface **61** of the antireflection coating **60** by a Berkovich Indenter Hardness Test along an indentation depth of 50 nm or greater. In some implementations, the display article **100** can exhibit a maximum hardness level of 8 GPa, 9 GPa, 10 GPa, 11 GPa, 12 GPa, 13 GPa, 14 GPa, 15 GPa, 16 GPa, 17 GPa, 18 GPa, 19 GPa, 20 GPa, or higher, as measured with a Berkovich Indenter Hardness Test along an indentation depth of 50 nm or greater, or 100 nm or greater.

[0094] Referring now to FIG. 1E, a display article **100** is depicted with the same attributes as the display article **100** shown in FIGS. 1C and 1D, with like-numbered elements having substantially the same functions, structures and properties (unless otherwise noted in the following sections). The display article **100** depicted in FIG. 1E employs an antireflective coating **60** that is the same as, or substantially similar to the coating **60** employed in the article **100** in FIGS. 1C and 1D, as disposed on the primary surface **12** of the substrate **10** to further improve the antiglare effect of the display article **100**. Further, the display article **100** depicted in FIG. 1E employs a textured surface region **31** defined by the primary surface **12** of the substrate **10**. As such, the antireflective coating **60** of the display article **100** shown in FIG. 1D is disposed on the textured surface region **31**. More specifically, the textured surface region **31** is inclusive of the diffractive surface region **30a** (see FIGS. 1A-1C), the roughened surface region **30b** (see FIG. 1D), and other textured or roughened primary surfaces of substrates **10** employed in the display articles **100** of the disclosure.

[0095] In embodiments of the display article **100** shown in FIG. 1E, the textured surface region **31** possesses an average texture height (R_{text}) from 20 nm to 2000 nm. According to further implementations, the roughened surface region **30b** possesses an average texture height (R_{text}) from 10 nm to 2500 nm, 10 nm to 2000 nm, 10 nm to 1500 nm, 20 nm to 2500 nm, 20 nm to 2000 nm, 20 nm to 1500 nm, 50 nm to 2500 nm, 50 nm to 2000 nm, 50 nm to 1500 nm, 50 nm to 1000 nm, 50 nm to 500 nm, 50 nm to 250 nm, 100 nm to 2500 nm, 100 nm to 2000 nm, 100 nm to 1500 nm, 100 nm to 1000 nm, 100 nm to 500 nm, 100 nm to 250 nm, and all texture height values between the foregoing ranges. Further, for a textured surface region **31** that comprises a roughened surface region **30b** (e.g., as produced through an etching and/or sandblasting process), R_{text} can be defined as the average surface roughness (R_q) of the structural features **20** of the roughened surface region **30b** and is reported in units of root-mean-squared (RMS) nanometers. For a textured surface region **31** that comprises a diffractive surface region **30a**, as described in this disclosure, R_{text} is defined as the average difference in height between the two heights or depths of the structural features **20** (e.g., pillars, holes, etc.) associated with the diffractive surface region.

[0096] Referring again to FIGS. 1A-1E, the substrate **10** of the display article **100** can be configured with a multi-component glass composition having about 40 mol % to 80 mol % silica and a balance of one or more other constituents, e.g., alumina, calcium oxide, sodium oxide, boron oxide, etc. In some implementations, the bulk composition of the substrate **10** is selected from the group consisting of aluminosilicate glass, a borosilicate glass and a phosphosilicate glass. In other implementations, the bulk composition of the substrate **10** is selected from the group consisting of aluminosilicate glass, a borosilicate glass, a phosphosilicate glass, a soda lime glass, an alkali aluminosilicate glass, and an alkali aluminoborosilicate glass. In further implementations, the substrate **10** is a glass-based substrate, including, but not limited to, glass-ceramic materials that comprise a glass component at about 90% or greater by weight and a ceramic component. In other implementations of the display article **100**, the substrate **10** can be a polymer material, with durability and mechanical properties suitable for the development and retention of the diffractive surface region **30a**.

[0097] In one embodiment of the display article **100** depicted in FIGS. 1A-1E, the substrate **10** has a bulk composition that comprises an alkali aluminosilicate glass that comprises alumina, at least one alkali metal and, in some embodiments, greater than 50 mol % SiO_2 , in other embodiments, at least 58 mol % SiO_2 , and in still other embodiments, at least 60 mol % SiO_2 , wherein the ratio $(\text{Al}_2\text{O}_3 \text{ (mol \%)} + \text{B}_2\text{O}_3 \text{ (mol \%)}) / \Sigma \text{ alkali metal modifiers (mol \%)} > 1$, where the modifiers are alkali metal oxides. This glass, in particular embodiments, comprises, consists essentially of, or consists of: about 58 mol % to about 72 mol % SiO_2 ; about 9 mol % to about 17 mol % Al_2O_3 ; about 2 mol % to about 12 mol % B_2O_3 ; about 8 mol % to about 16 mol % Na_2O ; and 0 mol % to about 4 mol % K_2O , wherein the ratio $(\text{Al}_2\text{O}_3 \text{ (mol \%)} + \text{B}_2\text{O}_3 \text{ (mol \%)}) / \Sigma \text{ alkali metal modifiers (mol \%)} > 1$, where the modifiers are alkali metal oxides.

[0098] In another embodiment of the display article **100**, as shown in FIGS. 1A-1E, the substrate **10** has a bulk composition that comprises an alkali aluminosilicate glass comprising, consisting essentially of, or consisting of: about 61 mol % to about 75 mol % SiO_2 ; about 7 mol % to about 15 mol % Al_2O_3 ; 0 mol % to about 12 mol % B_2O_3 ; about 9 mol % to about 21 mol % Na_2O ; 0 mol % to about 4 mol % K_2O ; 0 mol % to about 7 mol % MgO ; and 0 mol % to about 3 mol % CaO .

[0099] In yet another embodiment, the substrate **10** has a bulk composition that comprises an alkali aluminosilicate glass comprising, consisting essentially of, or consisting of: about 60 mol % to about 70 mol % SiO_2 ; about 6 mol % to about 14 mol % Al_2O_3 ; 0 mol % to about 15 mol % B_2O_3 ; 0 mol % to about 15 mol % Li_2O ; 0 mol % to about 20 mol % Na_2O ; 0 mol % to about 10 mol % K_2O ; 0 mol % to about 8 mol % MgO ; 0 mol % to about 10 mol % CaO ; 0 mol % to about 5 mol % ZrO_2 ; 0 mol % to about 1 mol % SnO_2 ; 0 mol % to about 1 mol % CeO_2 ; less than about 50 ppm As_2O_3 ; and less than about 50 ppm Sb_2O_3 ; wherein 12 mol % $\leq \text{Li}_2\text{O} + \text{Na}_2\text{O} + \text{K}_2\text{O} \leq 20$ mol % and 0 mol % $\leq \text{MgO} + \text{Ca} \leq 10$ mol %.

[0100] In still another embodiment, the substrate **10** has a bulk composition that comprises an alkali aluminosilicate glass comprising, consisting essentially of, or consisting of: about 64 mol % to about 68 mol % SiO_2 ; about 12 mol % to about 16 mol % Na_2O ; about 8 mol % to about 12 mol % Al_2O_3 ; 0 mol % to about 3 mol % B_2O_3 ; about 2 mol % to

about 5 mol % K_2O ; about 4 mol % to about 6 mol % MgO ; and 0 mol % to about 5 mol % CaO , wherein: 66 mol % $\leq \text{SiO}_2 + \text{B}_2\text{O}_3 + \text{CaO} \leq 69$ mol %; $\text{Na}_2\text{O} + \text{K}_2\text{O} + \text{B}_2\text{O}_3 + \text{MgO} + \text{CaO} + \text{SrO} > 10$ mol %; 5 mol % $\leq \text{MgO} + \text{CaO} + \text{SrO} \leq 8$ mol %; $(\text{Na}_2\text{O} + \text{B}_2\text{O}_3) - \text{Al}_2\text{O}_3 \leq 2$ mol %; 2 mol % $\leq \text{Na}_2\text{O} - \text{Al}_2\text{O}_3 \leq 6$ mol %; and 4 mol % $\leq (\text{Na}_2\text{O} + \text{K}_2\text{O}) - \text{Al}_2\text{O}_3 \leq 10$ mol %.

[0101] In other embodiments, the substrate **10** has a bulk composition that comprises SiO_2 , Al_2O_3 , P_2O_5 , and at least one alkali metal oxide (R_2O), wherein $0.75 > [(\text{P}_2\text{O}_5 \text{ (mol \%)} + \text{R}_2\text{O (mol \%)}) / \text{M}_2\text{O}_3 \text{ (mol \%)}] \leq 1.2$, where $\text{M}_2\text{O}_3 = \text{Al}_2\text{O}_3 + \text{B}_2\text{O}_3$. In some embodiments, $[(\text{P}_2\text{O}_5 \text{ (mol \%)} + \text{R}_2\text{O (mol \%)}) / \text{M}_2\text{O}_3 \text{ (mol \%)}] = 1$ and, in some embodiments, the glass does not include B_2O_3 and $\text{M}_2\text{O}_3 = \text{Al}_2\text{O}_3$. The substrate **10** comprises, in some embodiments: about 40 to about 70 mol % SiO_2 ; 0 to about 28 mol % B_2O_3 ; about 0 to about 28 mol % Al_2O_3 ; about 1 to about 14 mol % P_2O_5 ; and about 12 to about 16 mol % R_2O . In some embodiments, the glass substrate comprises: about 40 to about 64 mol % SiO_2 ; 0 to about 8 mol % B_2O_3 ; about 16 to about 28 mol % Al_2O_3 ; about 2 to about 12 mol % P_2O_5 ; and about 12 to about 16 mol % R_2O . The substrate **10** may further comprise at least one alkaline earth metal oxide such as, but not limited to, MgO or CaO .

[0102] In some embodiments, the substrate **10** has a bulk composition that is substantially free of lithium; i.e., the glass comprises less than 1 mol % Li_2O and, in other embodiments, less than 0.1 mol % Li_2O and, in other embodiments, 0.01 mol % Li_2O , and in still other embodiments, 0 mol % Li_2O . In some embodiments, such glasses are free of at least one of arsenic, antimony, and barium; i.e., the glass comprises less than 1 mol % and, in other embodiments, less than 0.1 mol %, and in still other embodiments, 0 mol % of As_2O_3 , Sb_2O_3 , and/or BaO .

[0103] In other embodiments of the display article **100** depicted in FIGS. 1A-1E, the substrate **10** has a bulk composition that comprises, consists essentially of or consists of a glass composition, such as Corning® Eagle XG® glass, Corning® Gorilla® glass, Corning® Gorilla® Glass 2, Corning® Gorilla® Glass 3, Corning® Gorilla® Glass 4 or Corning® Gorilla® Glass 5.

[0104] According to other embodiments, the substrate **10** of the display article **100** depicted in FIGS. 1A-1E can possess an ion-exchangeable glass composition that is strengthened by either chemical or thermal means that are known in the art. In one embodiment, the substrate **10** is chemically strengthened by ion exchange. In this process, metal ions at or near a primary surface **12** and/or primary surface **14** of the substrate **10** are exchanged for larger metal ions having the same valence as the metal ions in the glass substrate. The exchange is generally carried out by contacting the substrate **10** with an ion exchange medium such as, for example, a molten salt bath that contains the larger metal ion. The metal ions are typically monovalent metal ions such as, for example, alkali metal ions. In one non-limiting example, chemical strengthening of a substrate **10** that contains sodium ions by ion exchange is accomplished by immersing the substrate **10** in an ion exchange bath comprising a molten potassium salt such as potassium nitrate (KNO_3) or the like. In one particular embodiment, the ions in the surface layer of the substrate **10** and the larger ions are monovalent alkali metal cations, such as Li^+ (when present in the glass), Na^+ , K^+ , Rb^+ , and Cs^+ . Alternatively, monovalent cations in the surface layer of the substrate **10** may

be replaced with monovalent cations other than alkali metal cations, such as Ag^+ or the like.

[0105] In these embodiments of the display article 100 depicted in FIGS. 1A-1E, the replacement of small metal ions by larger metal ions in the ion exchange process creates a compressive stress region 50 in the substrate 10 that extends from the primary surface 12 to a depth 52 (referred to as the “depth of layer”) that is under compressive stress. It should also be understood that a compressive stress region can be formed in the glass substrate that extends from the primary surface 14 to a depth (not shown in FIGS. 1A-1E) that is comparable in nature to the compressive stress region 50. More particularly, this compressive stress at the primary surface 14 of the glass substrate is balanced by a tensile stress (also referred to as “central tension”) within the interior of the glass substrate. In some embodiments, the primary surface 12 of the substrate 10 described herein, when strengthened by ion exchange, has a compressive stress of at least 350 MPa, and the region under compressive stress extends to a depth 52, i.e., depth of layer, of at least 15 μm below the primary surface 12.

[0106] Ion exchange processes are typically carried out by immersing the substrate 10 in a molten salt bath containing the larger ions to be exchanged with the smaller ions in the glass. It will be appreciated by those skilled in the art that parameters for the ion exchange process, including, but not limited to, bath composition and temperature, immersion time, the number of immersions of the glass in a salt bath (or baths), use of multiple salt baths, additional steps such as annealing, washing, and the like, are generally determined by the composition of the glass and the desired depth of layer and compressive stress of the glass as a result of the strengthening operation. By way of example, ion exchange of alkali metal-containing glasses may be achieved by immersion in at least one molten bath containing a salt such as, but not limited to, nitrates, sulfates, and chlorides of the larger alkali metal ion. The temperature of the molten salt bath typically is in a range from about 380° C. up to about 450° C., while immersion times range from about 15 minutes up to about 16 hours. However, temperatures and immersion times different from those described above may also be used. Such ion exchange treatments, when employed with a substrate 10 having an alkali aluminosilicate glass composition, result in a compressive stress region 50 having a depth 52 (depth of layer) ranging from about 10 μm up to at least 50 μm with a compressive stress ranging from about 200 MPa up to about 800 MPa, and a central tension of less than about 100 MPa.

[0107] As the etching processes that can be employed to create the diffractive surface region 30a, roughened surface region 30b and textured surface region 31 of the display article 100 shown in FIGS. 1A-1E, according to some embodiments, can remove alkali metal ions from the substrate 10 that would otherwise be replaced by a larger alkali metal ion during an ion exchange process, a preference exists for developing a compressive stress region 50 in the display article 100 after the formation and development of the diffractive surface region 30a, roughened surface region 30b and textured surface region 31. In other embodiments, a compressive stress region 50 can be developed in the substrate 10 prior to development of the diffractive surface region 30a, roughened surface region 30b and textured surface region 31 to a depth 52 sufficient to account for some loss in the depth of layer in the region 50 associated with the

various treatments associated with forming these regions, as outlined below. Alternatively, the diffractive surface region 30a, roughened surface region 30b and textured surface region 31 may be created by an additive or coating process rather than a substrate etching process, in which case it may be desirable to develop the compressive stress region 50 before this additive or coating process.

[0108] According to another implementation of the display article 100 depicted in FIGS. 1A-1E, the article can further include an easy-to-clean (ETC) coating (not shown) disposed over the diffractive surface region 30a, roughened surface region 30b or textured surface region 31. In most embodiments, the ETC coating is deposited over any of the regions 30a, 30b and 31 such that its surface morphology generally mirrors the underlying morphology of the regions 30a, 30b and 31. In one embodiment, the display article 100 further includes a smudge-resistant fluorine-based ETC coating, deposited on at least a portion of the regions 30a, 30b and 31. In embodiments, the ETC coating comprises at least one amphiphobic substance having fluorine termination groups to provide the regions 30a, 30b and 31 with amphiphobicity (i.e., hydrophobicity and oleophobicity, or lack of affinity for both oils and water), thus minimizing wetting of the surface by water and/or oils. The fluorine termination groups of the ETC coating are less polar than a surface having—OH termination groups, and therefore minimize hydrogen (i.e., Van der Waals) bonding between particles and liquids. For fingerprint oils and debris associated with fingerprints, bonding—and adhesion—is minimized. Consequently, mass transport of fingerprint oils and debris from the finger of a person to the ETC coating is minimized. In one embodiment, the ETC coating is formed by exchanging the hydrogen found in terminal OH groups on the diffractive surface region 30a of the display article 100 with a fluorine-based moiety such as, for example, a fluorine-containing monomer (e.g., a fluorosilane), to form a glass having terminal fluorinated groups.

[0109] In another embodiment, an ETC coating of the display article 100 depicted in FIGS. 1A-1E comprises a self-assembled monolayer of a fluorine-terminating molecular chain. In yet another embodiment, the ETC coating comprises a thin, fluoro-polymeric coating and, in yet another embodiment, the ETC coating comprises silica soot particles that have been treated to have pendent fluorocarbon groups attached to the soot particles. Such ETC coatings can be applied to the diffractive surface region 30a of the display article 100 by dipping, vapor coating, spraying, application with a roller, or other suitable method known in the art. After the ETC coating has been applied, it may be “cured” at temperatures ranging from about 25° C. up to about 150° C. and, in another embodiment, at temperatures ranging from about 40° C. up to about 100° C. Curing times can range from about 1 up to about 4 hours, and may be carried out in an atmosphere containing 40-95% moisture. After curing, the display article 100 with an ETC coating can be solvent rinsed to remove any unbound coating and air-dried prior to use.

[0110] Referring again to the display article 100 depicted in FIGS. 1A-1C, embodiments are configured in which the diffractive surface region 30a includes a plurality of structural features 20 with a first portion of structural features 22a, 22a' and a second portion of structural features 22b, 22b'. Further, the first portion of structural features 22a (e.g., pillars), 22a' (e.g., holes) can be defined by a pitch 42a of

less than 125 μm , and the second portion of structural features **22b**, **22b'** can be defined by a pitch **42b** (see FIG. 1A) substantially the same or different from the pitch **42a**. According to some implementations, the second portion of structural features **22b**, **22b'** (e.g., ligaments, mesas) can be defined by a pitch **42b** of less than 125 μm , with the pitch **42a** substantially the same as or different from the pitch **42b**. Further, as used herein, the pitch **42a** and **42b** of the first and second portion of structural features **22a**, **22a'** and **22b**, **22b'** are pitch values of these features, as would be commonly understood by those of ordinary skill in the field of the disclosure. As such, the pitch **42a** and/or **42b** of the first and second portion of structural features **22a**, **22a'** and **22b**, **22b'**, respectively, can be less than 125 μm , 110 μm , 100 μm , 90 μm , 80 μm , 70 μm , 60 μm , 50 μm , 40 μm , 30 μm , 20 μm and all pitch values between these upper thresholds of the pitch value. In embodiments, the pitch values **42a**, **42b** may have a lower threshold, e.g., the pitch values may be larger than about 2 microns, larger than 5 microns, or larger than 10 microns.

[0111] Once again referring to the display article **100** depicted in FIGS. 1A-1C, embodiments are configured such that the first or the second portion of structural features **22a**, **22a'** (e.g., pillars or holes) or **22b**, **22b'** (e.g., ligaments or mesas), respectively, is at a fill fraction from about 30% to 70%, with the other portion (i.e., **22a**, **22a'** or **22b**, **22b'**) at a fill fraction of 100% minus the fill fraction of the first or second portions **22a**, **22a'** or **22b**, **22b'**. Hence, the first portion of structural features **22a**, **22a'** or second portion of structural features **22b**, **22b'** of the diffractive surface region **30a** can be configured at a fill fraction from about 30% to 70%, from about 35% to 65%, from about 40% to 60%, or from about 40% to 55%. For example, the first or second portions **22a**, **22a'** or **22b**, **22b'** can be configured within the diffractive surface region **30a** such that they have a fill fraction of 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, and all fill fraction levels between the foregoing values.

[0112] Still referring to the display article **100** depicted in FIGS. 1A-1C, the diffractive surface region **30a** is arranged within the primary surface **12** of the substrate **10** such that each of the plurality of structural features **20**, e.g., the first and second portions **22a**, **22a'** and **22b**, **22b'**, has a diameter **32a**, **32b** of less than 100 μm . Further, as used herein, the diameter **32a**, **32b** of the first and second portion of structural features **22a**, **22a'** and **22b**, **22b'** represents the effective diameter or longest width dimension of these features at their full width height maximum (FWHM), as would be commonly understood by those of ordinary skill in the field of the disclosure. As such, the diameter **32a**, **32b** of the structural features of the first and second portions **22a**, **22a'** and **22b**, **22b'** can be less than 100 μm , 90 μm , 80 μm , 70 μm , 60 μm , 50 μm , 40 μm , 30 μm , 20 μm , 10 μm , and all diameter values less than the foregoing diameters.

[0113] According to some embodiments of the display article **100** depicted in FIGS. 1A-1C, the diffractive surface region **30a** is configured such that each of the structural features **20** has an aspect ratio of more than 10. Unless otherwise noted, the aspect ratio of each of the structural features **20**, e.g., the first and second portion of structural features **22a**, **22a'** and **22b**, **22b'**, is given by the average diameter **32a**, **32b** divided by the respective average height **24a**, **24b**. In some implementations, the aspect ratio of the structural features **20** of the diffractive surface region **30a** is

more than 10, more than 20, more than 50, or more than 100. For example, a first portion of structural features **22a**, **22a'** with an average diameter **32a** of 20 μm and an average height **24a** of 0.2 μm corresponds to an aspect ratio of 100. More generally, the diffractive surface region **30a**, as characterized by these aspect ratios, is substantially flat or planar, at least as viewed under ambient lighting without any magnification aids.

[0114] According to some implementations of the display article shown in FIGS. 1A-1C, the structural features **20** of the diffractive surface region **30a** can be configured according to a period **47** to effect antiglare properties. In some implementations of the display article **100**, the structural features **20** of the diffractive surface region **30a** are configured with a period **47** that ranges from 1 μm to 200 μm , from 5 μm to 200 μm , from 5 μm to 150 μm , from 5 μm to 100 μm , from 5 μm to 50 μm , from 5 μm to 30 μm , from 20 μm to 150 μm , from 20 μm to 100 μm , from 10 μm to 30 μm , from 10 μm to 20 μm , and all period values between the foregoing ranges. For example, the structural features **20** can be configured with a relatively large period **47** on the order of about 100 μm (e.g., from about 70 to 200 μm) for end use applications of the display article **100** that benefit from maximizing a scattered light component near 0.30, such as in the case where the application has a particular DOI target. Such DOI targets may require a scattered light component at or near 0.30 from the specular reflection direction, which can be enhanced by relatively large structural features **20**. For end use applications for the display article **100** in which the DOI requirements are not as strict, smaller structural features **20** may be desirable, e.g., with a period **47** that ranges from about 5 to 30 μm and, according to some implementations, is semi-randomized to minimize color and/or Moiré artifacts. The level and type of feature randomization in the X-Y dimension can be very important to achieving low PPD while also minimizing other display artifacts such as Moiré or color banding. Put another way, traditional, perfectly ordered grating-like structures are not preferred for the articles of the disclosure. Structural features **20** that range from about 5 to 30 μm can affect DOI, and can be designed to achieve DOI values, e.g., below 90 or below 80, so it is not essential to have larger structural features **20** to have some effect on DOI. The randomization of the structural features **20** at the period **47** can be characterized by a variation in nearest-neighbor distances over a distribution range which may be limited to 1.2, 1.3, 1.4 or 1.5 $\times$ a minimum distance or an average distance. These smaller periods **47** can still effectively suppress specular reflection, but are believed to be more desirable to reduce artifacts such as sparkle (i.e., as measured by PPD_{140}) in display cover applications. In some embodiments, the period **47** can be configured to range from 5 μm to 30 μm , from 10 μm to 30 μm , and from 10 μm to 20 μm , which can lower DOI to less than 90, or even less than 80, while still maintaining desirable, low sparkle levels.

[0115] Referring now to FIG. 2, a cross-sectional, schematic view of a diffractive, antiglare structure is provided. As shown schematically in FIG. 2, a diffractive optical model can illustrate the principles of the diffractive surface region **30a**. Optical modeling calculations were performed using Gsolver©, a commercially available software package. The Gsolver© package employs a rigorous, coupled-wave analysis to find solutions for Maxwell's equations, which can govern the optical performance of a diffraction

grating. In the context of developing a diffractive surface region **30a**, the Gsolver© software can be applied to a linear, rectangular diffractive antiglare structure as shown in FIG. 2. Note that the rectangular nature of the diffractive antiglare structure shown in FIG. 2 departs from the structural features **20** of the diffractive surface region **30a** in the sense that the structural features **20** can comprise pillars, holes, mesas, plateaus, polygons, and other discrete, non-rectangular shapes. Nevertheless, the calculations conducted on the schematic shown in FIG. 2 demonstrate trends and provide data that is useful in the definition of the structural features **20** of the diffractive surface region **30a** that is consistent with the principles of this disclosure. Notably, as shown in FIG. 2, the as-modeled diffractive antiglare structure has a period of 20 μm and is integrated within the surface of a glass substrate having a refractive index of 1.518 at 550 nm. Further, as shown in FIG. 2, incident light at a wavelength of 550 nm is directed against the diffractive antiglare structure at an incident angle of 20 degrees. For optimal antiglare properties, it is generally desirable to suppress the zero order reflected light (i.e., specular light).

[0116] Referring now to FIGS. 3A and 3B, plots of diffraction efficiency in reflection and transmission, respectively, as a function of structure depth for the diffractive antiglare structure depicted in FIG. 2 are provided. These plots were developed from the Gsolver© software. As shown in FIG. 3A, the diffractive antiglare structure has a period of 20 μm and structure depths ranging from 0 to 0.5 μm are modeled. Further, as shown in FIG. 3A, plots of the zero through the fifth order are shown (i.e., $m=0$ to 5). Notably, the zero order curve is the amplitude of the specular reflectance and, as shown in the figure, is suppressed close to 0% at a structure depth of 0.14 to 0.15 μm . In addition, the specular reflectance can be suppressed by a factor of 10 compared to flat glass over a structure depth range of about 0.12 to 0.17 μm , which corresponds to about $\frac{1}{4}$ of the wavelength of light in air. A second minimum in specular reflectance is observed at the $\frac{3}{4}$ wavelength structure depth. As shown in FIG. 3B, the amplitude of the transmitted diffracted orders for the same structure as shown in FIGS. 2 and 3A is provided. For display applications, it can be desirable to maximize the zero order, i.e., the specular transmission. Specular transmission is greater than 78% of the transmittance of flat glass at a structure depth of ~ 0.15 μm , which is also the preferred depth for minimization of specular reflectance. The total transmission (considering all angles) remains close to the flat glass value, and the majority of scattered light in transmission is within 10 degrees or 5 degrees of the specular direction for preferred embodiments.

[0117] Referring now to FIGS. 4A-4C, plots of diffraction efficiency in reflection as a function of structure depth for the diffractive antiglare structure depicted in FIG. 2 at fill fractions of 15%, 30% and 70%, respectively, are provided. Notably, the diffractive structures with these fill fractions at 15%, 30% and 70% do not allow the zero order (specular) reflectance to drop to zero. This indicates that the preferred fill fraction for simple bimodal height, single material structure such as this one is close to 50%, or in the range of 35% to 65%, as modeled and depicted in FIG. 2.

[0118] Referring now to FIG. 5, a plot of diffraction efficiency in reflection as a function of structure depth for the diffractive antiglare structure depicted in FIG. 2 with varying incident light wavelengths is provided. In particular, this figure illustrates the effect of incident light wavelengths on

the suppression of specular reflection, particularly at the zero order. Although the optimal structure depth varies with wavelength, as shown in FIG. 5, a single structure depth near the first $\frac{1}{4}$ wavelength minimum (at the arrow) can successfully achieve a 10 $\times$ reduction in specular reflectance for all visible light wavelengths from 450 to 650 nm. This shows that broadband suppression of specular reflectance in the visible range is viable using a basic, diffractive antiglare structure as modeled in FIG. 2, with a bimodal surface height distribution.

[0119] Referring again to FIGS. 2-5, these figures generally provide guidance for the display article **100** to be configured to minimize specular reflectance while maximizing specular transmittance. Further, these figures illustrate how the depth of the diffractive antiglare structure (e.g., as serving as the general basis for the diffractive surface region **30a**) affects specular reflectance. As noted earlier, the modeled linear diffractive antiglare structure of FIG. 2 differs from the diffractive surface region **30a** of the display articles **100** shown in FIGS. 1A-1C, as the latter contains a plurality of structural features **20**, such as pillars, holes, polygons and other discrete features disposed in an ordered or semi-ordered array. However, in view of the modeling of the basic diffractive structure shown in FIGS. 2-5, the height or depth of the structural features **20** of the diffractive surface region **30a** is preferably maintained from 50 to 250 nm, 75 to 225 nm, 100 to 200 nm, 120 to 190 nm or from 140 to 190 nm.

[0120] More generally, the two-dimensional array of structural features **20** of the diffractive surface region **30a** can be fabricated by many processes, such as optical lithography (photomask), ink jet printing, laser patterning and/or screen printing once the intended structure for the surface region **30a** has been defined. The selection of the process depends on the resolution of the structural features **20** (e.g., in terms of diameter and/or pitch) and the technical capabilities of the given process. In some embodiments, once the structural parameters of the diffractive surface region **30a** has been defined (e.g., pillars or holes, average heights, pitch, diameter, period, etc.), the design can be converted to a computer-aided design (CAD) file and then used with any of the foregoing processes to transfer it to a substrate **10** to create the 'engineered' diffractive surface region **30a**.

[0121] Referring now to FIG. 6, a schematic flow chart is provided of a method **200** of making a display article (i.e., the display article **100** shown and described in FIGS. 1A-1C). The method **200** includes: a step **202** of masking a substrate **10** comprising a thickness **13** and a primary surface **12** with a mask; a step **204** of forming a diffractive surface region **30a** within or on the primary surface **12** of the substrate **10**; and a step **206** of removing the mask from the substrate **10**. The net result of the method **200** is a display article **100**, for example, as depicted in FIGS. 1A and 1B. In other embodiments of the method **200** shown in FIG. 6, steps **202** and **206** involving a mask are not employed and step **204** is conducted to form a roughened surface region **30b** within or on the primary surface **12** of the substrate **10**. The net result of this embodiment of the method **200** is a display article **100**, for example, as depicted in FIG. 1D.

[0122] In some embodiments, the method **200** shown in FIG. 6 further includes a step **208** of forming an antireflective coating **60** over the diffractive surface region **30a** or roughened surface region **30b**, to define the display article **100** (see FIGS. 1C and 1D). In the former embodiment, the diffractive surface region **30a** comprises a plurality of struc-

tural features **20** that comprises a plurality of different heights in a multimodal distribution. Further, the multimodal distribution further comprises a first portion of structural features **22a**, **22a'** at a first average height **24a** and a second portion of structural features **22b**, **22b'** at a second average height **24b**. In addition, the substrate **10**, as fabricated according to the method **200**, exhibits a sparkle of less than 4%, as measured by PPD₁₄₀ at an incident angle of 0° from normal, a DOI of less than 80% at an incident angle of 20° from normal, and a transmittance haze of less than 20% from an incident angle of 0° from normal.

[0123] According to some implementations of the method **200** depicted in FIG. 6 of making a display article **100** (see FIGS. 1A-1C and earlier description), the step **202** of masking the substrate **10** can include one or more of screen print masking, inkjet print masking and photoresist masking. In some embodiments, the step **204** of forming the diffractive surface region **30a** includes etching the primary surface **12** of the substrate **10** through the mask to form the diffractive surface region **30a**, wherein each structural feature is a hole at a depth from 50 nm to 250 nm. The step **204**, for example, can be conducted by etching the substrate **10**, as comprising a glass composition, with an HF/HNO₃ etchant. In embodiments, the wet etching solution employed in step **204** can consist of hydrofluoric acid (HF, 49 w/w %) and nitric acid (HNO₃, 69 w/w %) combinations with 0.1-5 v/v % HF and 0.1-5 v/v % HNO₃. Typical concentrations used to achieve 100-250 nm etching depths are 0.1 v/v % HF/1 v/v % HNO₃ to 0.5 v/v % HF/1 v/v % HNO₃ solutions. For example, etching in step **204** can be carried out using a dip or spray etch process from room temperature to about 45° C. In other embodiments, the step **204** of forming the diffractive surface region **30a** can include depositing a film on the primary surface **12** of the substrate **10** through the mask (e.g., by sputtering, evaporation, or chemical vapor deposition) to form the diffractive surface region **30a**, and each structural feature is a pillar at a height from 50 nm to 250 nm. The diffractive surface region **30a** can also be formed by masking plus 'dry etching', plasma-based etching, reactive ion etching, or other vacuum-based etching methods. In some embodiments, such a film can be deposited through the mask with a liquid phase silica layer or other oxide layer, followed by mask removal and lift off.

[0124] The articles **100** disclosed herein (e.g., as shown in FIGS. 1C-1E) may be incorporated into a device article, e.g., a device article with a display (or display device articles) (e.g., consumer electronics, including mobile phones, tablets, computers, navigation systems, wearable devices (e.g., watches) and the like), augmented-reality displays, heads-up displays, glasses-based displays, architectural device articles, transportation device articles (e.g., automotive, trains, aircraft, sea craft, etc.), appliance device articles, or any device article that benefits from some transparency, scratch-resistance, abrasion resistance or a combination thereof. An exemplary device article incorporating any of the articles disclosed herein (e.g., as consistent with the articles **100** depicted in FIGS. 1C-1E) is shown in FIGS. **18A** and **18B**. Specifically, FIGS. **18A** and **18B** show a consumer electronic device **1800** including a housing **1802** having a front **1804**, a back **1806**, and side surfaces **1808**; electrical components (not shown) that are at least partially inside or entirely within the housing and including at least a controller, a memory, and a display **1810** at or adjacent to the front surface of the housing; and a cover substrate **1812** at or over

the front surface of the housing such that it is over the display. In some embodiments, the cover substrate **1812** may include any of the articles disclosed herein. In some embodiments, at least one of a portion of the housing or the cover glass comprises the articles disclosed herein.

EXAMPLES

[0125] The following examples describe various features and advantages provided by the disclosure, and are in no way intended to limit the invention and appended claims.

Example 1

[0126] Referring generally to FIGS. 7A-7D, a photomask/optical lithography process was used to develop the diffractive surface region structures shown in these images. In this case, a light-sensitive polymer (i.e., a photoresist) is exposed and developed to form three-dimensional relief images on the substrate (e.g., a substrate **10**). In general, the ideal photoresist image has the exact shape of the designed or intended pattern in the plane of the substrate, with vertical walls through the thickness of the resist (<3 μm for spin-coatable resists, <20 μm for dry film resists, and <15 m for screen-coatable photoresists). When exposed, the final resist pattern is binary with parts of the substrate covered with resist while other parts are completely uncovered. The general sequence of processing steps for a typical photolithography process, as consistent with the method **200** depicted in FIG. 6, is as follows: substrate preparation (clean and dehydration followed by an adhesion promoter for spin-coatable resist, e.g., hexamethyl disilazane (HMDS), photoresist spin coat, prebake, exposure, and development (i.e., step **202**), followed by a wet etching process to transfer the binary image onto glass (i.e., step **204**). The final step is to strip the resist after the resist pattern has been transferred into the underlying layer (i.e., step **206**). In some cases, post bake and post exposure bake steps are required to ensure resist adhesion during wet etching process.

[0127] Referring now to FIGS. 7A-7D, optical micrographs are provided of diffractive surface regions of aluminosilicate glass substrates prepared according to this example with preferred etch depths in the range of 100 to 250 nm for the holes. These diffractive surface regions were prepared according to the foregoing photomasking process, followed by an etching process. More particularly, the wet etching solution used to prepare the samples of FIGS. 7A-7D consisted of hydrofluoric acid (HF, 49 w/w %) and nitric acid (HNO₃, 69 w/w %) with 0.1-5 v/v % HF and 0.1-5 v/v % HNO₃. For this example, the concentration of the etchant employed to achieve the 100-250 nm etch depths ranged from 0.1 v/v % HF/1 v/v % HNO₃ to 0.5 v/v % HF/1 v/v % HNO₃. FIGS. 7A and 7B show two-dimensional (2D) ordered arrays of structural features of diffractive surface regions at two spatial frequencies: (a) structural features with a diameter of 12 μm with a pitch of 17 μm (FIG. 7A) (referred to as a "12-17" array); and (b) structural features with a 25 μm diameter and a pitch of 50 μm (referred to as a "25-50" array). The surfaces for the ordered arrays were designed using hexagonal packing or a square packing array with a fill fraction varying from 20 to 50%. Further, FIGS. 7C and 7D show 2D arrays of structural features of diffractive surface regions with a randomly packed structure at two spatial frequencies: (a) a 50 μm structural feature diameter with a minimum pitch of 60 μm (FIG. 7C) (referred to as a

“50-60” array); and (b) a 12 μm structural feature diameter with a minimum pitch of 14 μm (referred to as a “12-14” array).

[0128] Referring now to FIG. 8, a plot is provided of etch depth as a function of etch time to form two structural features of diffractive surface regions according to this example. In this example, structural features having a diameter of 12 μm and 50 μm are etched at various etch times and the resulting etch depths are plotted (e.g., as consistent with the structures shown in FIGS. 7D and 7C, respectively). In this example, the substrate is a glass composition and it was etched with a 49 vol % stock solution of HF. Etch depth was measured using a stylus-based profilometer (with a Z depth resolution set at $<5\text{ }\mu\text{m}$, $<200\text{ }\mu\text{m}$ scan, 3 regions with a 3 mg force) or a Bruker ContourGT-X white light interferometer (with a 20 $\times$ or 50 $\times$ objective lens, green narrow-band light, in a single field of view image). As is evident from FIG. 8, the larger structural features (i.e., those at a diameter of about 50 μm) etch faster at longer times as compared to smaller structural features (i.e., those at a diameter of about 12 μm), which allows one to engineer the diffractive surface region to have features of different sizes at different depths or heights using the same etchant and one mask.

[0129] Referring now to FIGS. 9A and 9B, plots are provided of DOI as a function of etch depth for the structural features of the diffractive surface regions depicted in FIGS. 7C and 7D above, as having different sizes and fill fractions. As shown in FIG. 9A, the DOI levels of the 50-60 array of structural features (see also FIG. 7C) at fill fractions of 36%

tively, show that PPD_{140} and haze increase as a function of etch depth. Further, the samples having a larger spatial frequency (i.e., the 50-60 array) have more of an impact on PPD_{140} with increasing etch depth (see FIG. 10A), and the samples having a higher frequency (i.e., the 12-17 array) have more of an impact on haze with increasing etch depth (see FIG. 10D).

Example 2

[0131] According to this example, an array of structural features (i.e., holes) was developed on a glass substrate with depths of 0.15 μm , 0.2 μm and 0.23 μm (Samples 950, 951, 952, respectively) according to a method consistent with the principles of the disclosure. Table 2 below lists the optical properties measured on these samples, including PPD_{140} (%), as measured in a display unit at 0°), transmissivity (%), haze (%), as measured in transmission at 0°), DOI (coupled, %, as measured in reflectance at 20°) and specular reflectance, R_s (coupled, %, as measured in reflectance at 20°). As is evident from Table 2, the sample (950) with an etch depth of 0.15 μm exhibits DOI $<80\%$, PPD_{140} $<2\%$, and a haze $<5\%$, as consistent with the diffractive surface regions consistent with the disclosure. The other samples, with depths of 0.2 and 0.23 μm , do not exhibit this combination of optical properties. This illustrates the value of preferred depth ranges in achieving targeted combinations of properties, which may vary for different preferred applications.

TABLE 2

Sample	Depth (μm)	PPD_{140}	Transmissivity	Haze	DOI (Coupled, %)	R_s (Coupled, GU)
950	0.15	1.34	92.9	4.85	76	5.9
951	0.2	1.61	92.9	8.18	93	12.5
952	0.23	1.77	92.9	9.07	97	26.1

and 50% is shown as a function of etch depth. Similarly, as shown in FIG. 9B, the DOI levels of the 12-14 array of structural features (see also FIG. 7D) at fill fractions of 20% and 50% is shown as a function of etch depth. As is evident from FIGS. 9A and 9B, both arrays of structural features demonstrate that the minimum DOI is observed at an etch depth from about 150 to 180 nm, as generally predicted or otherwise suggested by the modeling outlined earlier and shown in FIGS. 2-5.

[0130] Referring now to FIGS. 10A and 10B, plots are provided of measured PPD_{140} and haze as a function of etch depth for structural features having different sizes and fill fractions from this example. As shown in FIGS. 10A and 10B, the 50-60 array of structural features (see also FIG. 7C) at fill fractions of 36% and 50% exhibits minimum PPD_{140} and haze values at low etch depths of less than 0.2 μm . With regard to FIGS. 10C and 10D, plots are provided of measured PPD_{140} and haze as a function of etch depth for structural features having different sizes and fill fractions from this example. As shown in FIGS. 10C and 10D, the 12-17 array of structural features (see also FIG. 7A) at fill fractions of 20% and 50% also exhibits minimum PPD_{140} and haze values at low etch depths of less than 0.2 μm . More generally, FIGS. 10A-10D demonstrate that these arrays of structural features at periods of 60 μm and 17 μm , respec-

[0132] Referring now to FIG. 11A, an optical image and surface height distribution bar of the diffractive surface region of the sample with best combination of optical properties from Table 2 (Sample 950) is provided. More specifically, the structural features of this sample (Sample 950) have a depth of about 150 nm, a fill fraction of 50%, a 12 μm feature diameter/size and a minimum pitch of 14 μm . Referring now to FIG. 11B, an angular spectra plot is provided of the samples from Table 2 in this example. More particularly, the plot of FIG. 11B shows amplitude of reflectance vs. reflectance angle for Samples 950-952. It is evident from FIG. 11B that a reduction in specular reflectance is observed for Sample 950 as compared to Samples 951 and 952. Note that the Rhopoint IQ Gloss Haze & DOI Meter reports an R_s value in gloss units (GU), as listed in Table 2, that is normalized to a maximum of 100 for a flat glass having an index of 1.567 illuminated at 20° angle of incidence and no back-surface reflectance. Such glass is known to have a first-surface absolute reflectance (% R) value of 4.91%. Thus, the R_s value reported by the Rhopoint IQ Meter can be converted to an absolute specular reflectance value (% R value) by multiplying by a factor of 4.91/100. As such, Sample 950, with an R_s amplitude of ~ 6 at 0° , corresponds to a first-surface absolute specular reflectance value (% R) of $6/100 \times 4.91\% = \sim 0.295\%$.

Example 3

[0133] According to this example, samples with an array of structural features defining a diffractive surface region were made using screen printing and etching, generating pillars on a glass surface. The target pillar size/diameter was 75 μm on the screen print pattern, which after wetting with the etchant on the glass expanded closer to a diameter of about 100 μm . Further, the target pitch for these samples was 125 μm in a hexagonal pattern with a target fill fraction of 55% (the actual fill fraction was closer to 56%). The screen print pattern was made using an ink on a clean glass surface. Table 3 lists the samples made according to this example, which reflect various etch times that result in various etch depths (i.e., pillar heights), and optical properties associated with these samples (sparkle, haze, DOI and Rs). Further, as noted earlier, the first-surface specular reflectance (Rs, in gloss units (GU)) can be converted to first-surface absolute specular reflectance (% R) by multiplying them by a factor of 4.91/100. As is evident from Table 3, the optimal etch depth range in terms of the optical property measurements corresponds to $\sim 1/4$ wavelength of light in air, i.e., the samples at etch depths of 0.141 and 0.172 μm .

TABLE 3

Sample ID	Etch time (s)	Etch depth (μm)	% Sparkle @ 140 PPI	% Haze	Coupled DOI	Coupled Rs (GU)
C17-T10a-75H-E30-Bare-C	30	0.099	2.9	0.35	78.6	41.0
C17-T10a-75H-E45-Bare-C	45	0.141	4.5	0.44	49.0	24.5
C17-T10a-75H-E60-Bare-C	60	0.172	4.9	0.44	37.3	12.7
C17-T10a-75H-E75-Bare-C	75	0.203	11.1	0.58	72.7	18.7
C17-T10a-75H-E90-Bare-C	90	0.260	13.1	0.57	89.2	27.1
C17-T10a-75H-E130-Bare-C	130	0.339	16.5	0.97	92.3	40.2
C17-T10a-75H-E145-Bare-C	145	0.396	19.5	1.19	80.5	20.5
C17-T10a-75H-E160-Bare-C	160	0.420	9.9	1.35	61.7	15.3
C17-T10a-75H-E175-Bare-C	175	0.448	9.8	1.49	62.5	30.9
C17-T10a-75H-E190-Bare-C	190	0.505	25.7	1.96	79.7	20.9
C17-T10a-75H-E205-Bare-C	205	0.531	15.9	2.06	86.1	25.6

[0134] Referring now to FIG. 12, an angular spectra plot is provided for the sample in this example with an etch depth of ~ 0.172 μm (C17-T10a-75H-E60-Bare-C), with the plot depicting amplitude of reflectance vs. reflectance angle. As is evident from FIG. 12 and Table 3, the optimal etch depth for obtaining low DOI and low Rs lies in the range of 120 to 180 nm for the structural features, and the other etch depths do not exhibit this combination of optical properties.

[0135] Referring now to FIGS. 13A and 13B, optical images are provided of the diffractive surface region and structural features of the sample depicted in FIG. 12 and Table 3 with an etch depth of 0.172 μm (C17-T10a-75H-E60-Bare-C), as viewed before and after removal of the ink as part of the screen printing and etching process. As noted earlier in the description of this example, the target size/diameter of these features was about 75 μm ; however, upon expansion of the ink upon drying, the actual size of the screen-printed features (i.e., the mask) ranged from 101 to 110 μm for the specific features depicted in FIG. 13A. Also, as is evident from FIG. 13B, the regions around these ink circles were etched, thus forming the pillars of this example.

[0136] Referring now to FIG. 14, a plot is provided of DOI as a function of etch depth for the structural features of this example and detailed above in Table 3. As noted earlier, the preferred etch depth is at about $1/4$ wavelength of light

between 0.1 and 0.2 μm with regard to reductions in DOI. In addition, a second preferred etch depth in terms of DOI is located at about $3/4$ wavelength of the light at an etch depth between 0.4 and 0.5 μm . The $3/4$ wavelength etch depth can be less desirable because of its effect on transmitted light, as shown by the higher PPD_{140} values in Table 3, as well as its lesser impact on DOI as relative to the samples at the $1/4$ wavelength etch depth. These results are generally consistent with the previously described diffractive antiglare optical models.

Example 4

[0137] While the diffractive surface regions of the display articles of the disclosure employ structural features having a multimodal distribution of surface heights (e.g., a bimodal distribution of surface heights), embodiments of these diffractive surface regions employ spatial randomization of the features in the X-Y dimension. In this example, two X-Y pattern randomizations are employed for the diffractive surface regions—a “hard sphere” distribution and a “Poisson disk” distribution. The former hard sphere pattern design targeted structural features at approximate 50% fill fractions,

and samples with structural features having 12 μm and 50 μm diameters, respectively, were prepared according to this pattern. The latter Poisson disk pattern design targeted structural features at approximate 36% fill fractions. Each of these X-Y randomization schemes can also target diffractive surface regions with structural features at different fill fractions, feature depths, and combinations with antireflective coatings. Further, as is evident from this example and noted earlier, the level and type of feature randomization in the X-Y dimension can be very important to achieving low PPD while also minimizing other display artifacts such as Moiré or color banding.

[0138] X-Y pattern randomization can be defined in different ways. One method is hexagonality with FIG. 15 showing the definition of hexagonality (H). Hexagonality is a metric for quantifying locally how close a pattern is to a hexagonal lattice. For each point in the pattern, the hexagonality (H) at that point is computed using the angles of its six nearest neighbors with respect to an arbitrary axis. The mean hexagonality (H) can be defined as follows by Equation (1):

$$H = \frac{1}{6} \left| \sum_{k=1}^6 e^{6i\alpha_k} \right| \in [0, 1] \quad (1)$$

where $k=1$ to 6 for the six nearest neighbor structural features to the one structural feature and α_k represents the angles of each of the six nearest neighbor structural features to an arbitrary axis. As such, in the context of the diffractive surface region of the display articles of the disclosure, the spatial distribution of the plurality of structural features can have a mean hexagonality (H) that ranges from 0.4 to 1.0, according to embodiments of the disclosure. As shown in FIG. 15, the variables α_k represent the angles of the six nearest neighbors. On a hexagonal lattice, these six angles all differ by 60 degrees ($\pi/3$ radians); therefore, the exponents in the six summands differ by 2π radians and the six complex numbers in the summation are all the same. H is unity in that case. One may take the mean hexagonality (H) of the pattern as a global metric of deviation from a perfect hexagonal lattice, useful if the pattern is homogenous. A perfect hexagonal lattice has a mean hexagonality (H) of 1.0. A completely random, stationary Poisson distribution of points has a mean hexagonality (H) of approximately 0.36, or 36%. As such, according to embodiments of the display article 100 depicted in FIGS. 1A and 1B, the plurality of structural features 20 can be characterized with a non-random spatial distribution such that H ranges from 0.4 to 0.95, from 0.4 to 0.9, from 0.4 to 0.8, from 0.36 to 0.9, from 0.36 to 0.8, from 0.36 to 0.7, and all mean hexagonality (H) values and ranges between the foregoing ranges.

[0139] With further regard to the hard sphere randomization pattern, this pattern was employed in the formation of the structural features of the diffractive surface region depicted in FIGS. 7C and 7D. These patterns were generated using a molecular dynamics simulation tool (LAMMPS) to generate the pattern. A gas of “molecules” representing the features was initially placed on a two-dimensional hexagonal lattice to fix the fill fraction at 50%. Then the gas was heated and allowed to randomize in two dimensions. The molecules were endowed with a repulsive hard-sphere potential to maintain the minimum specified spacing. The example pattern in FIG. 7C has a mean hexagonality (H) of 49% (see also FIG. 15), indicating a large deviation from a regular lattice. The “12-14” and “50-60” arrays have pattern geometries that are the same: the 50-60 pattern is simply the 12-14 pattern scaled up by an overall factor of 50/12. This scaling does not impact fill fraction, hexagonality or feature distribution functions (aside from an obvious scaling of axes), but it does impact the optical performance of the texture. The distribution of nearest neighbor distances (defined as center-to-center spacing of features in the X-Y space) of the 12/14 pattern is shown in FIG. 16A. Its periodogram is shown in FIG. 16B. Instead of spatial frequency, the X-axis of FIG. 16B is converted to off-specular scattering angle in degrees, for an optical wavelength of 0.55 μm .

[0140] Turning to the other 50/60 array (see FIGS. 9A and 10A) has a larger distribution of nearest neighbor distances and a lower fill fraction (36%). The structural features of these samples were arranged in a Poisson disk pattern that was generated using a “dart throwing” algorithm. This algorithm ensures an absolute minimum feature spacing (60 μm in this example), but is not very efficient at filling space. The mean hexagonality (H) is lower, at 41%. The distribution of nearest neighbor distances for the 50/60 Poisson disk pattern is shown in FIG. 17A. Its periodogram is shown in FIG. 17B. Instead of spatial frequency, the X-axis of FIG.

17B is converted to off-specular scattering angle in degrees, for an optical wavelength of 0.55 μm .

Example 5A

[0141] In this example (samples denoted “Ex. 5A”), a strengthened glass substrate was prepared with a roughened surface region having a transmittance haze of about 15%, an average surface roughness (R_q) in the range of 100 nm to 300 nm and a majority of lateral etched features dimensions in the range of 5 μm to 30 μm . In this example, the roughened surface region was prepared with a two-step HF etching process. Next, the roughened surface region was coated with a five-layer AR coating consistent with the embodiment of the antireflection coating 60 described earlier in the disclosure in Table 1B.

[0142] With regard to mechanical properties, the display article of this example (Ex. 5A) was characterized with a maximum hardness of 11.8 GPa at indentation depths greater than 100 nm (specifically, @140-160 nm); a total thickness of the AR coating of less than 500 nm (specifically, 338.4 nm) and a percentage of high refractive index material of greater than 35% (specifically, 53.7%). As to its substrate surface morphology, the display article of this example was characterized with a radially averaged surface roughness power spectral density (PSD) of 80.2 nm (S_a), 96.4 nm (S_q) and average lateral feature size of $\sim 4 \mu\text{m}$, as measured and calculated through a Watershed algorithm readily understood by those with skill in the field of the disclosure.

[0143] Optical properties were measured for the display articles of this example, as given below in Table 4A.

TABLE 4A

Measured optical properties of Ex. 5A				
	6°	20°	45°	60°
<u>1st-surface Reflectance % average</u>				
R % (450-650 nm)	0.039	0.045	0.218	1.788
R % (photopic, Y)	0.042	0.045	0.196	1.758
R % (840-860 nm)	0.200	0.238	1.062	4.736
R % (930-950 nm)	0.431	0.553	2.147	7.104
<u>1st-surface Reflected Lightness and Color metrics</u>				
L*	0.38	0.40	1.77	14.16
a*	-0.07	-0.03	1.75	5.23
b*	-0.03	0.08	1.54	9.78
			0°	
<u>2-surface Transmitted light metrics, coating on 1 surface only</u>				
T % (450-650 nm)			94.28	
T % (840-860 nm)			94.62	
T % (930-950 nm)			93.71	
L*			97.74	
a*			0.02	
b*			0.62	
<u>1st-surface Contrast Ratio Metrics</u>				
Direct Contrast Ratio (100 lux)			3.6	
Ambient Contrast Ratio (1000 lux)			61	
Color Gamut (Direct Contrast Ratio)			18.9	
Color Gamut (Ambient Contrast Ratio)			88	

TABLE 4A-continued

Measured optical properties of Ex. 5A	
Other Optical metrics	
Sparkle % (PPD)	1.1
Coupled DOI (%)	89
	(estimate)
Uncoupled DOI (%)	98.8
Haze (%)	15.9

[0144] Referring now to FIG. 19, a plot is provided of elastic modulus and hardness (GPa) vs. indentation depth (nm) of a display article according to this example (Ex. 5A). As shown in FIG. 19, the measured apparent hardness and modulus was measured using a Berkovich Indenter Hardness Test, with a maximum hardness of 11.8 GPa observed at indentation depths of 140 to 160 nm.

Example 5B

[0145] In this example (samples denoted “Ex. 5B”), a strengthened glass substrate was prepared with a roughened surface region having a transmittance haze of about 30%, an average surface roughness (R_q) in the range of 100 nm to 300 nm and a majority of lateral etched features dimensions in the range of 5 μm to 30 μm . In this example, the roughened surface region was prepared with a process that includes an HF etching step followed by a sandblasting step. Next, the roughened surface region was coated with a five-layer AR coating consistent with the embodiment of the antireflection coating 60 described earlier in the disclosure in Table 1B.

[0146] With regard to mechanical properties, the display article of this example (Ex. 5B) was characterized with a maximum hardness of 11.8 GPa at indentation depths greater than 100 nm (specifically, @140-160 nm); a total thickness of the AR coating of less than 500 nm (specifically, 338.4 nm) and a percentage of high refractive index material of greater than 35% (specifically, 53.7%). As to its substrate surface morphology, the display article of this example was characterized with a radially averaged surface roughness power spectral density (PSD) of 239.6 nm (S_a), 306.1 nm (S_q) and average lateral feature size of $\sim 8 \mu\text{m}$, as measured and calculated through a Watershed algorithm readily understood by those with skill in the field of the disclosure.

[0147] Also in this example, comparative control samples were prepared with just the bare, strengthened glass substrate (denoted “Comp. Ex. 5A”); just the strengthened, glass substrate with the roughened surface region (denoted “Comp. Ex. 5B”); and just the bare, strengthened glass substrate and the foregoing antireflection coating (denoted “Comp. Ex. 5C”).

[0148] Optical properties were measured for the display articles of this example (Ex. 5B and Comp. Ex. 5C), as given below in Tables 4B and 4B1, respectively.

TABLE 4B

Measured optical properties of Ex. 5B				
	6°	20°	45°	60°
1 st -surface Reflectance % average				
R % (450-650 nm)	0.016	0.019	0.064	0.364
R % (photopic, Y)	0.018	0.022	0.066	0.416

TABLE 4B-continued

Measured optical properties of Ex. 5B				
R % (840-860 nm)	0.036	0.041	0.145	0.894
R % (930-950 nm)	0.063	0.079	0.293	1.507
1st-surface Reflected Lightness and Color metrics				
L*	0.15	0.17	0.54	3.25
a*	−0.04	−0.03	0.30	1.30
b*	0.01	0.02	0.31	1.46
0°				
2-surface Transmitted light metrics, coating on 1 surface only				
T % (photopic, Y)	93.66			
T % (840-860 nm)	93.34			
T % (930-950 nm)	92.30			
L*	97.50			
a*	−0.08			
b*	0.57			
1st-surface Contrast Ratio Metrics				
Direct Contrast Ratio (100 lux)	13.5			
Ambient Contrast Ratio (1000 lux)	60			
Color Gamut (Direct Contrast Ratio)	59.7			
Color Gamut (Ambient Contrast Ratio)	88			
Other Optical metrics				
Sparkle % (PPD)	4.5			
Coupled DOI (%)	2.7			
	(estimated)			
Uncoupled DOI (%)	55			
Transmittance Haze (%)	30			

TABLE 4B1

Measured optical properties of Comp. Ex. 5C				
	6°	20°	45°	60°
1 st -surface Reflectance % average				
R % (photopic, Y)	0.71	0.69	1.53	5.30
R % (840-860 nm)	1.37	1.50	3.02	7.48
R % (930-950 nm)	2.17	2.49	4.92	10.02
1st-surface Reflected Lightness and Color metrics				
L*	6.43	6.27	12.77	27.56
a*	-2.72	-2.54	5.27	5.27
b*	-1.16	1.48	2.95	2.83
0°				
2-surface Transmitted light metrics, coating on 1 surface only				
T % (photopic, Y)	94.51			
T % (840-860 nm)	93.89			
T % (930-950 nm)	93.15			
L*	97.84			
a*	0.06			
b*	0.46			

[0149] Referring now to FIG. 20A, a plot is provided of first-surface specular reflectance (% R) vs. visible and near-IR wavelengths (nm) of display articles according to this example (Ex. 5B) and comparative display articles (Comp. Exs. 5A-5C). As is evident from FIG. 20A, the specular reflectance of the example with a strengthened

glass substrate having a roughened surface region and a 5-layer AR coating disposed thereon is approximately an order of magnitude lower than the comparative samples with a bare, strengthened glass and the same 5-layer AR coating.

[0150] Referring now to FIGS. 20B and 20C, schematic illustrations of specular reflectance measurements of display articles according to embodiments of the disclosure and comparative display articles are depicted and can be used to interpret the data and results of FIG. 20A. As shown in FIGS. 20B and 20C, a beam of light is incident at an angle θ_i with respect to the sample normal on a comparative bare glass with a 5-layer AR coating (left) (e.g., Comp. Ex. 5C) and a display article of the disclosure having a roughened surface region and the same 5-layer AR coating (right) (e.g., Ex. 5B). The specular reflectance is defined as the light power reflected within a 2 degree cone that subtends the incident plane and is conjugate to the incident beam (see FIGS. 20B and 20C) divided by the incident power. For the comparative sample (e.g., Comp. Ex. 5C), the total reflectance ($\sim 0.7\%$) is measured within the 2 degree cone and is equal to the specular reflectance. For the display article according to the disclosure with a roughened surface region and a 5-layer AR coating (e.g., Ex. 5B), the light is scattered into a continuum of angles (shown as a cone in FIG. 2), with most light intensity falling within $\sim \pm 15^\circ$ of the specular direction. In this case, the detector only captures a portion of the total reflected light since the rest falls outside of the angular field-of-view of the detector. It is assumed that the total reflectance (integrated over all angles) of the display article surface (e.g., Ex. 5B) is the same as the comparative sample's substrate surface (e.g., Comp. Ex. 5C). This means that the specular reflectance of the surface of the example (e.g., Ex. 5B) is given by $R_{DXC} \cdot \eta$, where η is the fraction of reflected light power within the detector cone divided by the total reflected light. Also, η can be calculated by measuring the specular reflectance of an antiglare surface with no AR coating (e.g., Comp. Ex. 5B) (identical AG treatment to Ex. 5B) and the specular reflectance of a bare strengthened glass substrate surface (e.g., Comp. Ex. 5A) so that $\eta = R_{AG}/R_{GG}$. The calculated value of $R_{DXC} \cdot (R_{AG}/R_{GG})$ is shown as the short dash/long dash line in FIG. 20A for the sample of this example (Ex. 5B, calculated) and shows good agreement with the measured value of this example (Ex. 5B, measured).

Example 5C

[0151] In this example (samples denoted "Ex. 5C"), a strengthened glass substrate was prepared with a roughened surface region having a transmittance haze of about 30%, an average surface roughness (R_q) in the range of 100 nm to 300 nm and a majority of lateral etched features dimensions in the range of 5 μm to 30 μm . In this example, the roughened surface region was prepared with a process that includes an HF etching step followed by a sandblasting step. Next, the roughened surface region was coated with a five-layer AR coating consistent with the embodiment of the antireflection coating 60 described earlier in the disclosure in Table 1B.

[0152] With regard to mechanical properties, the display article of this example (Ex. 5C) was characterized with a maximum hardness of 11.8 GPa at indentation depths greater than 100 nm (specifically, @140-160 nm); a total thickness of the AR coating of less than 500 nm (specifically, 338.4 nm) and a percentage of high refractive index material

of greater than 35% (specifically, 53.7%). As to its substrate surface morphology, the display article of this example was characterized with a radially averaged surface roughness power spectral density (PSD) of 165.8 nm (S_d), 207.5 nm (S_q) and average lateral feature size of $\sim 6 \mu\text{m}$, as measured and calculated through a Watershed algorithm readily understood by those with skill in the field of the disclosure.

[0153] Optical properties were measured for the display articles of this example, as given below in Table 4C.

TABLE 4C

Measured optical properties of Ex. 5C				
	6°	20°	45°	60°
<u>1st-surface Reflectance % average</u>				
R % (450-650 nm)	0.015	0.023	0.071	0.422
R % (photopic, Y)	0.018	0.022	0.066	0.416
R % (840-860 nm)	0.041	0.043	0.188	1.361
R % (930-950 nm)	0.072	0.093	0.415	2.362
<u>1st-surface Reflected Lightness and Color metrics</u>				
L*	0.16	0.20	0.60	3.76
a*	-0.03	0.01	0.29	1.69
b*	-0.01	-0.01	0.30	2.13
0°				
<u>2-surface Transmitted light metrics, coating on 1 surface only</u>				
T % (photopic, Y)	93.72			
T % (840-860 nm)	93.70			
T % (930-950 nm)	92.43			
L*	97.52			
a*	-0.01			
b*	0.63			
<u>1st-surface Contrast Ratio Metrics</u>				
Direct Contrast Ratio (100 lux)	10.7			
Ambient Contrast Ratio (1000 lux)	59			
Color Gamut (Direct Contrast Ratio)	55.6			
Color Gamut (Ambient Contrast Ratio)	88			
<u>Other Optical metrics</u>				
Sparkle % (PPD)	3.8			
Coupled DOI (%)	25			
	(estimated)			
Uncoupled DOI (%)	84			
Haze (%)	30			

[0154] Referring now to FIG. 21, a plot is provided of two-surface total transmittance (T %) vs. wavelength (nm) of display articles in the foregoing examples (Exs. 5A-5C). As noted earlier, one primary surface of the strengthened glass substrate includes a roughened surface region and a 5-layer multilayer AR coating disposed thereon; and the other primary surface of the glass substrate is bare. The bare glass surface has a $\sim 4\%$ reflectance, limiting the maximum transmittance of these samples to about 96%. As shown in FIG. 21, visible photopic average transmittance levels were obtained for each of the samples: 94.28% (Ex. 5A); 93.65% (Ex. 5B); and 93.72% (Ex. 5C). Transmittance levels at infrared (IR) wavelengths of 850 nm are greater than 93% and at 940 nm are greater than 92%.

[0155] Referring now to FIGS. 22A-22C, plots are provided of first-surface specular reflectance (R %) vs. wavelength (nm) for various angles of incidence (6°, 20°, 45°, and

60°) for the foregoing samples (Ex. 5A, FIG. 22A; Ex. 5B, FIG. 22B; and Ex. 5C, FIG. 22C). In FIG. 22A, for Ex. 5A, the reflectance over the entire wavelength range from 425 nm to 950 nm remains below 0.4% for all angles from 0-20 degrees. The reflectance over the entire wavelength range from 425 nm to 950 nm remains below 1.1% for all angles from 0-45 degrees. In FIG. 22B, for Ex. 5B, the reflectance over the entire wavelength range from 425 nm to 950 nm remains below 0.06% for all angles from 0-20 degrees. The reflectance over the entire wavelength range from 425 nm to 950 nm remains below 0.3% for all angles from 0-45 degrees. As for FIG. 22C, for Ex. 5C, the reflectance over the entire wavelength range from 425 nm to 950 nm remains below 0.06% for all angles from 0-20 degrees. The reflectance over the entire wavelength range from 425 nm to 950 nm remains below 0.3% for all angles from 0-45 degrees.

[0156] Referring now to FIG. 23, a plot is provided of two-surface specular reflectance (R %) vs. wavelength (nm) for a degree of incidence of 6° for the foregoing samples (Exs. 5A-5C) and the comparative example with a bare, strengthened glass substrate and a 5-layer AR coating (Comp. Ex. 5C). One side of each of the sample from the foregoing examples (Exs. 5A-5C) comprises the 5-layer AR coating and the roughened surface region, and the other side is bare glass. The 2-surface reflectance of the examples (Exs. 5A-5C) is more than a factor of 3 lower than the 2-surface reflectance from the comparative example with a 5-layer AR coating alone (Comp. Ex. 5C), which is ~4% due to the flat back surface. The reduction in specular reflectance from the flat back surface of the examples (Exs. 5A-5C) is caused by angular redistribution of the transmitted and reflected light, as shown in FIGS. 20B and 20C. As a consequence of the angular redistribution of the back-scattered light, it is believed that the perception of buried reflectances from the stack of layers in the 5-layer AR coating will be significantly reduced if one of the examples (Exs. 5A-5C) is employed with a roughened surface region as compared to a bare glass substrate with the 5-layer AR coating (Comp. Ex. 5C).

[0157] With further regard to the optical properties and measurements reported in the foregoing examples, as shown in FIGS. 20A, 21, 22A-22C and 23, and Tables 4A-4C, reflectance and transmittance values are reported as polarization averages. That is, average values combining both s- and p-polarization values into a single average. Photopic averages (Y), L*, a*, and b* values were calculated from measured sample data using known methods according to the CIE 1964 standards with 10° observer and D65 illuminant. These create weighted values according to the human eye's response to visible light. Specular reflectance was measured in an angular range of +/-1 degrees using an Agilent Cary 5000 UV-Vis-NIR spectrophotometer. First-surface reflectance values were obtained by using an index matching oil to couple the back surface of the glass sample to a light absorber. Further, transmitted haze was measured by a BYK Gardner Haze-Gard. Uncoupled (2-surface) DOI was measured using a Rhopoint IQ glossmeter. The coupled DOI was estimated from the uncoupled DOI values using a relation derived from historical data. Total transmittance was measured using a Perkin-Elmer Lambda 950 spectrophotometer.

[0158] Also with regard to the optical properties in the foregoing examples, the contrast ratio (CR) of a display for white can be defined by the following in Equation (2):

$$CR = \left(\frac{L_{white\ screen} + L_{ambient\ light}}{L_{black\ screen} + L_{ambient\ light}} \right) \quad (2)$$

where $L_{white\ screen}$ and $L_{black\ screen}$ are luminance of white and black screen, respectively and $L_{ambient\ light}$ is luminance of reflected ambient light from the display. According to this definition, the highest CR can be found in the absence of the external (ambient) light. Further, CR ratio and color gamut associated with the display and relevant cover glass were measured under two different illumination conditions as follows: 1) Ambient CR (ACR): CR measurement under uniform diffused (omnidirectional) D65 white light illumination; and 2) Direct CR (DCR): CR measurement under a directional/collimated white light illumination.

ACR (Ambient Contrast Ratio) Measurement

[0159] The diffused illumination-based system employed for these measurements of the display articles of the disclosure and those in the foregoing examples mainly consists of an integrating sphere-based illumination source equipped with D65 LED light connected through an optical fiber. The ACR measurement was performed by following NISTIR 6738, a method/procedure to measure the ACR of the display, as developed by the National Institute of Standards and Technologies. The measurement was performed by coupling the untreated side of the test (glass) sample on to an OLED display with index matching oil. The brightness of the display was maintained at 380 Cd/m² during all of the contrast ratio and color gamut measurements. Instrument Systems's CAS140D spectroradiometer equipped with TOP 200 optical probe was used to measure the photopic luminance from the sample/display unit.

Dcr (Direct Contrast Ratio) Measurement

[0160] In the DCR measurements, a collimated (D65) LED light source was used as the illumination source. Referring now to FIG. 24, a schematic is provided of an optical setup for measuring the DCR of the display articles in the disclosure and the foregoing examples. As depicted in FIG. 24, each test specimen was coupled onto the surface of the OLED display through index matching oil. As also depicted in FIG. 24, Angle of Incident (AOI) of the light was set to 10° while luminance of the display and the external light source was detected by placing a spectroradiometer at the specular direction (at -10°). Intensity of the external light source was controlled by controlling the current to the LED source and/or by placing an appropriate neutral density (ND) filter in the optical path. Prior to the DCR measurement the light illuminance intensity (in lux) for each illumination condition was measured by replacing the test specimen with an illuminance meter (model A58U-223 from Konica Minolta). Similar to the ACR setup, a CAS140D spectroradiometer equipped with a TOP 200 optical probe (Instrument Systems) was used to measure the photopic luminance from the sample/display unit. The collection (cone) angle of the TOP200 optical probe was set to 1 degree. Photopic luminance of the display, while it was coupled with the test specimen, was measured while the display was loaded with black and white images, respectively. The contrast was calculated by dividing luminance of the white screen by that of the black screen. The same procedure was followed at various different illuminance

intensities by illuminating the test specimen using the calibrated light source source/ND filter setup.

Color Gamut Area (CGA) measurement

[0161] Besides contrast ratio (CR) measurements, the performance color under external illumination can be a very important property of the display articles of the disclosure. The color performance can be defined as the range of colors that a particular display device produces. The color information for a given display can be represented by specifying chromaticity coordinates for red/blue/green (RGB) in the CIE 1976 color space diagram. The area inside the triangle is proportional to the available colors for a given display illumination condition. The total area of the CIE 1976 diagram represents the full range of colors visible to the human eye. Hence, the available colors for a given display can be assessed by measuring the area of the triangle in the color space. Considering this principle, measurement of the color gamut area (CGA) of the display under variable illuminance was performed in the foregoing examples. The area of the color gamut (CGA) can be obtained by measuring the chromaticity coordinate in the color space (CIE 1976 (u', v')) for red (u'_R, v'_R), green (u'_G, v'_G) and blue (u'_B, v'_B) colors under variable illuminance intensities. Here, CGA can be calculated by measuring CIE (u', v') coordinates of the RGB color triangle for the display with the coupled test specimen while it is lit with red, green and blue images. The area of the RGB triangle (CGA) is proportional to the number of colors emitted through the display/test specimen unit. The area of the triangle was measured by using the Herons formula given by Equation (3):

$$CGA = \frac{1}{4} \sqrt{(a^2 + b^2 + c^2)^2 - 2(a^4 + b^4 + c^4)} \quad (3)$$

where a, b and c can be found according to the following Equations (3A), (3B) and (3C), respectively:

$$a = \sqrt{(v'_R - v'_B)^2 + (u'_R - u'_B)^2} \quad (3A)$$

$$b = \sqrt{(v'_R - v'_G)^2 + (u'_R - u'_G)^2} \quad (3B)$$

$$c = \sqrt{(v'_B - v'_G)^2 + (u'_B - u'_G)^2} \quad (3C)$$

The same procedure was applied to measure the CGA under various illumination intensities and illumination conditions. The CGA under diffused illumination was performed by using the calibrated integrating sphere setup, whereas the DCR setup was used to measure CGA under direct/collimated illumination. The largest CGA can be found in the absence of external illumination. Therefore, for better understanding, the CGA at any given illuminance, is presented as a percentage with respect to its original value without any external illuminance.

[0162] As outlined herein, a first aspect of the disclosure pertains to a display article. The display article comprises: a substrate comprising a thickness and a primary surface; and the primary surface having defined thereon a diffractive surface region. The diffractive surface region comprises a plurality of structural features that comprises a plurality of different heights in a multimodal distribution. Further, the substrate exhibits a sparkle of less than 4%, as measured by pixel power deviation (PPD₁₄₀) at an incident angle of 0°

from normal, a distinctness of image (DOI) of less than 80% at an incident angle of 20° from normal, and a transmittance haze of less than 20% from an incident angle of 0° from normal.

[0163] According to a second aspect, the first aspect is provided, wherein the plurality of structural features further comprises a first portion of structural features having a first average height and a second portion of structural features having a second average height.

[0164] According to a third aspect, the second aspect is provided, wherein the first average height and the second average height have a difference of from 50 nm to 250 nm.

[0165] According to a fourth aspect, the second aspect is provided, wherein the first average height and the second average height have a difference of from 120 nm to 200 nm.

[0166] According to a fifth aspect, any one of the second through fourth aspects is provided, wherein the diffractive surface region further comprises a first planar region corresponding to the first average height and a second planar region corresponding to the second average height. Further, each of the first planar region and the second planar region comprises a root-mean-square (RMS) height variation of less than 50 nm.

[0167] According to a sixth aspect, the fifth aspect is provided, wherein the first and second planar regions have a total surface area that is at least 50% of a total surface area of the diffractive surface region.

[0168] According to a seventh aspect, any one of the first through sixth aspects is provided, wherein the substrate comprises a glass composition.

[0169] According to an eighth aspect, any one of the first through seventh aspects is provided, wherein the substrate further exhibits a sparkle of less than 2%, as measured by PPD₁₄₀ at an incident angle of 0° from normal.

[0170] According to a ninth aspect, any one of the first through eighth aspects is provided, wherein the substrate further exhibits a transmittance haze of less than 5% from an incident angle of 0° from normal.

[0171] According to a tenth aspect, any one of the first through ninth aspects is provided, wherein the substrate further exhibits a first-surface absolute specular reflectance (% R) of less than 1% at an incident angle of 20° from normal.

[0172] According to an eleventh aspect, any one of the first through tenth aspects is provided, further comprising an antireflective coating disposed on the primary surface of the substrate. The antireflective coating comprises a plurality of alternating high refractive index and low refractive index layers. Each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index that is greater than 1.8. Further, the article exhibits a first-surface absolute specular reflectance (% R) of less than 0.1% at an incident angle of 20° from normal.

[0173] A twelfth aspect of the disclosure pertains to a display article. The display article comprises: a substrate comprising a thickness and a primary surface; and the primary surface having defined thereon a diffractive surface region. The diffractive surface region comprises a plurality of structural features comprising a pitch of less than 125 μm and a fill fraction from 30% to 70%, each structural feature comprising a diameter of less than 100 μm. Further, the substrate exhibits a sparkle of less than 4%, as measured by pixel power deviation (PPD) at an incident angle of 0° from

normal, a distinctness of image (DOI) of less than 80% at an incident angle of 20° from normal, and a transmittance haze of less than 20% from an incident angle of 0° from normal.

[0174] According to a thirteenth aspect, the twelfth aspect is provided, wherein the substrate comprises a glass composition.

[0175] According to a fourteenth aspect, the twelfth or the thirteenth aspect is provided, wherein the substrate further exhibits a sparkle of less than 2%, as measured by PPD₁₄₀ at an incident angle of 0° from normal.

[0176] According to a fifteenth aspect, any one of the twelfth through fourteenth aspects is provided, wherein the substrate further exhibits a transmittance haze of less than 5% from an incident angle of 0° from normal.

[0177] According to a sixteenth aspect, any one of the twelfth through fifteenth aspects is provided, wherein the substrate further exhibits a first-surface absolute specular reflectance (% R) of less than 1% at an incident angle of 20° from normal.

[0178] According to a seventeenth aspect, any one of the twelfth through sixteenth aspects is provided, wherein a first portion of the plurality of structural features comprises pillars having a first average height from 50 nm to 250 nm.

[0179] According to an eighteenth aspect, the seventeenth aspect is provided, wherein the plurality of structural features further comprises a plurality of heights in a multimodal distribution. The multimodal distribution further comprises the first portion of structural features at the first average height and a second portion of structural features at a second average height. Further, the first portion of the distribution has a first fill fraction from 30 to 70% and the second portion of the distribution has a second fill fraction of 100% minus the first fill fraction.

[0180] According to a nineteenth aspect, any one of the twelfth through sixteenth aspects is provided, wherein a first portion of the plurality of structural features comprises holes having a first average depth from 50 nm to 250 nm.

[0181] According to a twentieth aspect, the nineteenth aspect is provided, wherein the plurality of structural features further comprises a plurality of depths in a multimodal distribution. The multimodal distribution further comprises the first portion of structural features at the first average depth and a second portion of structural features at a second average depth. Further, the first portion of the distribution has a first fill fraction from 30 to 70% and the second portion of the distribution has a second fill fraction of 100% minus the first fill fraction.

[0182] According to a twenty-first aspect, any one of the twelfth through twentieth aspects is provided, wherein the fill fraction is from 40% to 55%.

[0183] According to a twenty-second aspect, any one of the twelfth through twenty-first aspects is provided, wherein each structural feature further comprises an aspect ratio of more than 10.

[0184] According to a twenty-third aspect, any one of the twelfth through twenty-second aspects is provided, wherein the plurality of structural features further comprises a period of from m to 100 μm.

[0185] According to a twenty-fourth aspect, any one of the twelfth through sixteenth aspects is provided, further comprising an antireflective coating disposed on the primary surface of the substrate. The antireflective coating comprises a plurality of alternating high refractive index and low refractive index layers. Each of the low refractive index

layers comprises a refractive index of less than or equal to about 1.8 and about the same or greater than a refractive index of the substrate. Each of the high refractive index layers comprises a refractive index that is greater than 1.8. Further, the article exhibits a first-surface absolute specular reflectance (% R) of less than 0.1% at an incident angle of 20° from normal.

[0186] A twenty-fifth aspect of the disclosure pertains to a display article. The display article comprises: a substrate comprising a thickness and a primary surface; and the primary surface having defined thereon a diffractive surface region. The diffractive surface region comprises a plurality of structural features comprising a pitch of less than 125 μm, and a fill fraction from 30% to 70%. Further, each structural feature comprises a height or depth of from 50 nm to 250 nm.

[0187] According to a twenty-sixth aspect, the twenty-fifth aspect is provided, wherein the plurality of structural features further comprises a non-random spatial distribution.

[0188] According to a twenty-seventh aspect, the twenty-fifth or twenty-sixth aspect is provided, wherein the substrate exhibits a sparkle of less than 4%, as measured by pixel power deviation (PPD₁₄₀) at an incident angle of 0° from normal, a distinctness of image (DOI) of less than 80% at an incident angle of 20° from normal, and a transmittance haze of less than 20% from an incident angle of 0° from normal.

[0189] According to a twenty-eighth aspect, any one of the twenty-fifth through twenty-seventh aspects is provided, wherein the non-random spatial distribution of the plurality of structural features comprises a mean hexagonality (H) of from greater than 0.4 to less than 1.0, wherein the hexagonality (H) of one structural feature is given by Equation (1):

$$H = \frac{1}{6} \left| \sum_{k=1}^6 e^{6i\alpha_k} \right| \in [0, 1] \quad (1)$$

where k=1 to 6 for the six nearest neighbor structural features to the one structural feature and α_k represents the angles of each of the six nearest neighbor structural features to an arbitrary axis.

[0190] According to a twenty-ninth aspect, any one of the twenty-fifth through twenty-eighth aspects is provided, wherein a first portion of the plurality of structural features comprises pillars having a first average height from 50 nm to 250 nm.

[0191] According to a thirtieth aspect, the twenty-ninth aspect is provided, wherein the height of each pillar is from 120 nm to 180 nm.

[0192] According to a thirty-first aspect, any one of the twenty-fifth through twenty-eighth aspects is provided, wherein a first portion of the plurality of structural features comprises holes having a first average depth from 50 nm to 250 nm.

[0193] According to a thirty-second aspect, the thirty-first aspect is provided, wherein the depth of each hole is from 120 nm to 180 nm.

[0194] According to a thirty-third aspect, any one of the twenty-fifth through thirty-second aspects is provided, further comprising an antireflective coating disposed on the primary surface of the substrate. The antireflective coating comprises a plurality of alternating high refractive index and low refractive index layers. Each of the low refractive index

layers comprises a refractive index of less than or equal to about 1.8 and about the same or greater than a refractive index of the substrate. Each of the high refractive index layers comprises a refractive index that is greater than 1.8. Further, the article exhibits a first-surface absolute specular reflectance (% R_s) of less than 0.1% at an incident angle of 20° from normal.

[0195] A thirty-fourth aspect of the disclosure pertains to a method of making a display article. The method comprises: masking a substrate comprising a thickness and a primary surface with a mask; forming a diffractive surface region within the primary surface of the substrate; and removing the mask from the substrate. The diffractive surface region comprises a plurality of structural features that comprises a plurality of different heights in a multimodal distribution. The multimodal distribution further comprises a first portion of structural features at a first average height and a second portion of structural features at a second average height. Further, the substrate exhibits a sparkle of less than 4%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, a distinctness of image (DOI) of less than 80% at an incident angle of 20° from normal, and a transmittance haze of less than 20% from an incident angle of 0° from normal.

[0196] According to a thirty-fifth aspect, the thirty-fourth aspect is provided, wherein the step of masking comprises one or more of screen print masking, inkjet print masking, photoresist masking.

[0197] According to a thirty-sixth aspect, the thirty-fourth or thirty-fifth aspect is provided, wherein the step of forming comprises etching the primary surface of the substrate through the mask to form the diffractive surface region, and wherein each structural feature is a hole comprising a depth from 50 nm to 250 nm.

[0198] According to a thirty-seventh aspect, the thirty-fourth or thirty-fifth aspect is provided, wherein the step of forming comprises depositing a film on the primary surface of the substrate through the mask to form the diffractive surface region, and wherein each structural feature is a pillar comprising a height from 50 nm to 250 nm.

[0199] A thirty-eighth aspect of the present disclosure pertains to a display article. The display article includes: a substrate comprising a thickness and a primary surface; a textured surface region defined by the primary surface; and an antireflective coating disposed on the textured surface region defined by the primary surface of the substrate. The textured surface region comprises a plurality of structural features and an average texture height (R_{text}) from 50 nm to 300 nm. The substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, and a transmittance haze of less than 40% from an incident angle of 0° from normal. The antireflective coating comprises a plurality of alternating high refractive index and low refractive index layers. Each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8. Further, the article exhibits a first-surface average photopic specular reflectance (% R) of less than 0.3% at any incident angle from about 5° to 200 from normal at wavelengths from 450 nm to 650 nm.

[0200] According to a thirty-ninth aspect, the thirty-eighth aspect is provided, wherein the coated article exhibits a maximum hardness of 8 GPa or greater as measured on an

air-side surface of the antireflective coating by a Berkovich Indenter Hardness Test along an indentation depth of 50 nm or greater.

[0201] According to a fortieth aspect, the thirty-eighth aspect is provided, wherein the plurality of structural features further comprises a first average height and a second average height.

[0202] According to a forty-first aspect, the fortieth aspect is provided, wherein the first average height and the second average height have a difference of from 50 nm to 250 nm.

[0203] According to a forty-second aspect, the fortieth aspect is provided, wherein the first average height and the second average height have a difference of from 120 nm to 200 nm.

[0204] According to a forty-third aspect, any one of the thirty-eighth through forty-second aspects is provided, wherein the first-surface average photopic specular reflectance (% R) is less than 0.1% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0205] According to a forty-fourth aspect, any one of the thirty-eighth through forty-third aspects is provided, wherein the sparkle is less than 3%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal.

[0206] According to a forty-fifth aspect, any one of the thirty-eighth through forty-fourth aspects is provided, wherein the substrate comprises a glass substrate or a glass-ceramic substrate.

[0207] According to a forty-sixth aspect, any one of the thirty-eighth through forty-fifth aspects is provided, wherein the substrate further exhibits a distinctness of image (DOI) of less than 85% at an incident angle of 20° from normal.

[0208] According to a forty-seventh aspect, any one of the thirty-eighth through forty-sixth aspects is provided, wherein the transmittance haze is less than 30% from an incident angle of 0° from normal.

[0209] A forty-eighth aspect of the present disclosure pertains to a consumer electronic product. The consumer electronic product includes: a housing comprising a front surface, a back surface and side surfaces; electrical components at least partially within the housing, the electrical components comprising a controller, a memory, and a display, the display at or adjacent to the front surface of the housing; and a cover substrate disposed over the display. At least one portion of the housing or the cover substrate comprises the article of any one of the thirty-eighth through forty-seventh aspects.

[0210] A forty-ninth aspect of the present disclosure pertains to a display article. The display articles includes: a substrate comprising a thickness and a primary surface; a textured surface region defined by the primary surface; and an antireflective coating disposed on the textured surface region defined by the primary surface of the substrate. The textured surface region comprises a plurality of structural features and an average texture height (R_{text}) from 50 nm to 300 nm. The substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, and a transmittance haze of less than 40% from an incident angle of 0° from normal. The antireflective coating comprises a total physical thickness from 200 nm to 500 nm and a plurality of alternating high refractive index layers and low refractive index layers, wherein the antireflective coating comprises a total of three

(3) to nine (9) layers. Each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8. Each high refractive index layer comprises one of Si_3N_4 , SiN_x and SiO_xN_y . Further, the article exhibits a first-surface average photopic specular reflectance (% R) of less than 0.3% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0211] According to a fiftieth aspect, the forty-ninth aspect is provided, wherein one of the low refractive index layers is disposed directly on the textured region defined by the primary surface of the substrate.

[0212] According to a fifty-first aspect, the forty-ninth or fiftieth aspect is provided, wherein each low refractive index layer comprises SiO_2 or SiO_x .

[0213] According to a fifty-second aspect, any one of the forty-ninth through fifty-first aspects is provided, wherein the antireflective coating further comprises a scratch resistant layer which is the thickest high refractive index layer, wherein the scratch resistant layer comprises one of Si_3N_4 , SiN_x and SiO_xN_y , having a physical thickness from 50 nm to 200 nm, and further wherein the plurality of alternating high refractive index and low refractive index layers ranges from three (3) to six (6) layers.

[0214] According to a fifty-third aspect, the fifty-second aspect is provided, wherein the total physical thickness of the antireflective coating is from 200 nm to 350 nm and the physical thickness of the scratch resistant layer is from about 75 nm to 175 nm.

[0215] According to a fifty-fourth aspect, the fifty-second aspect is provided, wherein the total physical thickness of the antireflective coating is from 250 nm to 340 nm and the physical thickness of the scratch resistant layer is from 100 nm to 160 nm.

[0216] According to a fifty-fifth aspect, any one of the forty-ninth through fifth-fourth aspects is provided, wherein the plurality of structural features further comprises a first average height and a second average height.

[0217] According to a fifty-sixth aspect, any one of the forty-ninth through fifty-fifth aspects is provided, wherein the coated article exhibits a maximum hardness of 8 GPa or greater as measured on an air-side surface of the antireflective coating by a Berkovich Indenter Hardness Test along an indentation depth of 50 nm or greater.

[0218] According to a fifty-seventh aspect, the fifty-fifth or fifty-sixth aspect is provided, wherein the first average height and the second average height have a difference of from 120 nm to 200 nm.

[0219] According to a fifty-eighth aspect, any one of the forty-ninth through fifty-seventh aspects is provided, wherein the first-surface average photopic specular reflectance (% R) is less than 0.1% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0220] According to a fifty-ninth aspect, any one of the forty-ninth through fifty-eighth aspects is provided, wherein the sparkle is less than 3%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal.

[0221] According to a sixtieth aspect, any one of the forty-ninth through fifty-ninth aspects is provided, wherein the substrate comprises a glass substrate or a glass-ceramic substrate.

[0222] According to a sixty-first aspect, any one of the forty-ninth through sixtieth aspects is provided, wherein the substrate further exhibits a distinctness of image (DOI) of less than 80% at an incident angle of 20° from normal.

[0223] According to a sixty-second aspect, any one of the forty-ninth through sixty-first aspects is provided, wherein the transmittance haze is less than 30% from an incident angle of 0° from normal.

[0224] A sixty-third aspect of the present disclosure pertains to a consumer electronic product. The consumer electronic product includes: a housing comprising a front surface, a back surface and side surfaces; electrical components at least partially within the housing, the electrical components comprising a controller, a memory, and a display, the display at or adjacent to the front surface of the housing; and a cover substrate disposed over the display. At least one portion of the housing or the cover substrate comprises the article of any one of the forty-ninth through sixty-second aspects.

[0225] A sixty-fourth aspect of the present disclosure pertains to a display article. The display articles includes: a substrate comprising a thickness and a primary surface; a roughened surface region defined by the primary surface; and an antireflective coating disposed on the roughened surface region of the primary surface of the substrate. The roughened surface region comprises a plurality of structural features and an average surface roughness (R_q) from 20 nm to 2000 nm root-mean-square (RMS) variation. The substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal, and a transmittance haze of less than 40% from an incident angle of 0° from normal. The antireflective coating comprises a total physical thickness from 200 nm to 500 nm and a plurality of alternating high refractive index and low refractive index layers, wherein the antireflective coating has a total of three (3) to nine (9) layers. Each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8. Each high refractive index layer comprises one of Si_3N_4 , SiN_x and SiO_xN_y . Further, the article exhibits a first-surface average photopic specular reflectance (% R) of less than 1% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0226] According to a sixty-fifth aspect, the sixty-fourth aspect is provided, wherein one of the low refractive index layers is disposed directly on the roughened surface region defined by the primary surface of the substrate.

[0227] According to a sixty-sixth aspect, the sixty-fourth or sixty-fifth aspect is provided, wherein each low refractive index layer comprises SiO_2 or SiO_x .

[0228] According to a sixty-seventh aspect, any one of the sixty-fourth through sixty-sixth aspects is provided, wherein the substrate comprises a glass substrate or a glass-ceramic substrate.

[0229] According to a sixty-eighth aspect, any one of the sixty-fourth through sixty-seventh aspects is provided, wherein the roughened surface region comprises an average surface roughness (R_q) from 50 nm to 250 nm root-mean-square (RMS) variation.

[0230] According to a sixty-ninth aspect, any one of the sixty-fourth through sixty-eighth aspects is provided, wherein the first-surface average photopic specular reflectance is less than 3%, as measured by pixel power deviation (PPD_{140}) at an incident angle of 0° from normal.

tance (% R) is less than 0.1% at any incident angle from about 5° to 20° from normal at wavelengths from 450 nm to 650 nm.

[0231] According to a seventieth aspect, any one of the sixty-fourth through sixty-ninth aspects is provided, wherein the substrate further exhibits a distinctness of image (DOI) of less than 80% at an incident angle of 20° from normal.

[0232] According to a seventy-first aspect, any one of the sixty-fourth through seventieth aspects is provided, wherein the article exhibits a first surface reflected color ($\sqrt[3]{(a^{*2}+b^{*2})}$) of <0.5 at each incident angle of 6° and 200 from normal.

[0233] According to a seventy-second aspect, any one of the sixty-fourth through seventieth aspects is provided, wherein the article exhibits a two-surface transmitted color ($\sqrt[3]{(a^{*2}+b^{*2})}$) of <1 at an incident angle of 0° using a 2 degree acceptance angle.

[0234] A seventy-third aspect of the present disclosure pertains to a consumer electronic product. The consumer electronic product includes: a housing comprising a front surface, a back surface and side surfaces; electrical components at least partially within the housing, the electrical components comprising a controller, a memory, and a display, the display at or adjacent to the front surface of the housing; and a cover substrate disposed over the display. At least one portion of the housing or the cover substrate comprises the article of any one of the sixty-fourth through seventy-second aspects. Many variations and modifications may be made to the above-described embodiments of the disclosure without departing substantially from the spirit and various principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

What is claimed is:

1. A display article, comprising:

a substrate comprising a thickness and a primary surface; one of a diffractive surface region and a roughened surface region defined by the primary surface; and

an antireflective coating disposed on the textured surface region or the diffractive surface region defined by the primary surface of the substrate,

wherein the substrate exhibits a sparkle of less than 5%, as measured by pixel power deviation (PPD₁₄₀) at an incident angle of 0° from normal, and a transmittance haze of less than 40% from an incident angle of 0° from normal, wherein:

the antireflective coating comprises a plurality of alternating high refractive index and low refractive index layers,

each of the low refractive index layers comprises a refractive index of less than or equal to about 1.8, and each of the high refractive index layers comprises a refractive index of greater than 1.8,

if the primary surface defines the diffractive surface region, the diffractive surface region exhibits a mean hexagonality (H) in a range from 0.36 to 0.8, the hexagonality for each structural feature of the diffractive surface region being computed as

$$H = \frac{1}{6} \left| \sum_{k=1}^6 e^{6i\alpha_k} \right| \in [0, 1]$$

where α_k represents the angles of the six nearest structural features to one of the structural features with respect to an arbitrary axis, and

the display article exhibits a reflected color ($\sqrt[3]{(a^{*2}+b^{*2})}$) from the antireflective coating that is less than 5 for all incident angles from 0° to 60°, as measured with a D65 illuminant.

2. The display article according to claim 1, wherein the article exhibits a first-surface average photopic specular reflectance (% R) of less than 0.3% at any incident angle from about 5° to 200 from normal at wavelengths from 450 nm to 650 nm.

3. The display article according to claim 1, wherein the mean hexagonality is in a range from 0.36 to 0.7.

4. The display article according to claim 1, wherein, when the display article is disposed over a display, the display article exhibits an ambient contrast ratio that is up to 15 as measured with 100 incidence lux with a D65 illuminant.

5. The display article according to claim 4, wherein, when the display article is disposed over the display, the display article exhibits color a gamut level of at least 15 for a direct contrast ratio measured with 100 incidence lux with a D65 illuminant.

6. The display article according to claim 1, wherein the article exhibits a first-surface average photopic specular reflectance (% R) of less than 0.1% at any incident angle from about 5° to 200 from normal at wavelengths from 450 nm to 650 nm.

7. The display article according to claim 1, wherein the coated article exhibits a maximum hardness of 8 GPa or greater as measured on an air-side surface of the antireflective coating by a Berkovich Indenter Hardness Test along an indentation depth of 50 nm or greater.

8. The display article according to claim 1, wherein:

the antireflective coating comprises a total physical thickness from 200 nm to 500 nm, and

the antireflective coating comprises a total of three (3) to nine (9) layers.

9. The display article according to claim 1, wherein the article comprises the diffractive surface region, the diffractive surface region comprising a plurality of structural features.

10. The display article of claim 9, wherein the plurality of structural features comprise a plurality of different heights in a multimodal distribution.

11. The display article according to claim 10, wherein the plurality of structural features further comprises first portion at a first average height and a second portion at a second average height, wherein the first average height and the second average height have a difference of from 50 nm to 250 nm.

12. The display article of claim 11, wherein the first portion comprises a fill fraction that is from 30% to 70% of the diffractive surface region, wherein the second portion comprises a second fill fraction that is 100% minus the fill fraction.

13. The display article of claim 12, wherein the fill fraction is one of 40%, 45%, 50%, 55%, and 60%.

14. The display article of claim **1**, wherein the article comprises the roughened surface region, the roughened surface region having structural features, wherein a majority of the structural features have a lateral dimension in a range from 1 μm to 125 μm .

15. The display article of claim **14**, wherein the roughened surface region exhibits an average surface roughness (R_q) in RMS variation from 10 nm to 2500 nm.

16. The display article of claim **15**, wherein the display article exhibits a distinctness of image of 80% or less, as measured at an incident angle of 20° from normal.

17. The display article of claim **1**, wherein the display article exhibits a transmittance haze of less than 20%, as measured at an incidence angle of 0° from normal.

18. The display article of claim **1**, wherein the display article exhibits a transmittance haze of less than 5%, as measured at an incidence angle of 0° from normal.

19. The display article of claim **1**, wherein the display article exhibits a sparkle of less than 2%, as measured by PPD_{140} at an incident angle of 0° from normal.

20. The display article of claim **1**, wherein the primary surface defines the diffractive surface region comprising a plurality of pillars with heights ranging from 120 nm to 180 nm.

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